Namibia University of Science and Technology · Faculty of Computing and Informatics

ABI820S · Advanced Business Intelligence and Analytics · Group project · Domain: Health services

Green Basket Health Services (GBHS): one trusted view of patient flow, resource utilisation and disease surveillance

End-to-end business intelligence solution: Kimball data warehouse with SCD Type 2, reproducible ELT with 44 validated data-quality rules and 22 reconciliation checks, Power BI semantic model with dynamic row-level and object-level security, OLAP and four advanced analytics techniques, accessible dashboards, performance optimisation and health-data governance.

Client (scenario)	Green Basket Health Services (Pty) Ltd – health-services subsidiary of Green Basket cc: 5 hospitals and 5 clinics in 9 Namibian regions
Scenario / domain	Health services: patient flow, resource utilisation, disease surveillance (synthetic and public data only)
Submission	28 October 2026 (Week 16, as stated in the brief header) · Final report, version 1.0
Data cut-off	30 September 2026 (epi week 2026-W39), plus a day-2 incremental load of 1 October 2026
Platform	PostgreSQL 16 · Python 3.11+ · Power BI Desktop (PBIP: TMDL + PBIR)

Team role	Report sections owned	Name	Student number	Signature
Member 1 – Project lead and business analyst	3.1, 3.9, presentation			
Member 2 – Data warehouse architect	3.2, data dictionary			
Member 3 – ETL, data quality and governance	3.3, 3.8			
Member 4 – Power BI developer	3.4, 3.6			
Member 5 – Analytics and performance lead	3.5, 3.7			

Declaration. We declare that this report and the accompanying artefacts are our own group work and that every source is acknowledged in APA style. **Ethical and legal use of data:** all patient-level data are synthetic (seeded generator, no real person), the external data are public (NSA 2023 census, NASA POWER), and no personal information about real people is collected, stored or published (§3.1 and §3.8). **Use of generative AI:** an AI assistant (Anthropic Claude) was used to draft code, documentation and analysis. The team executed, verified and edited every artefact and takes full responsibility for the content. Each member can explain the sections they own. This use is disclosed in line with the course's academic-integrity rules; the team confirms it complies with the NUST policy in force.

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Executive summary

The problem. Green Basket Health Services (GBHS) is the health-services subsidiary of Green Basket cc. It runs five hospitals (Windhoek, Oshakati, Rundu, Katima Mulilo and Walvis Bay) and five clinics in nine Namibian regions. In 2025 its emergency departments (EDs) recorded 81,612 attendances, 14 % more than in 2024. Its data sit in nine disconnected places:

- the patient administration system (PAS): attendances, inpatient spells and the patient index;
- paper-based midnight ward returns and the nursing e-roster;
- weekly IDSR disease returns;
- a master-data workbook;
- two external sources: the Namibia Statistics Agency (NSA) 2023 census and NASA POWER weather.

Management could not answer three questions with confidence: why ED waits rose, whether beds and staff are used well, and whether a malaria surge is coming. Profiling the raw extracts found:

- 2,966 duplicate attendance rows;
- 16,477 timestamps in day-first format;
- 30 spellings of 9 notifiable conditions;
- 16,223 superseded spell images;
- occupied-bed counts keyed ten times too high.

The solution. We built the components below. One command (`python run_all.py`, about 5 minutes) reproduces the whole build.

- **A Kimball warehouse in PostgreSQL 16.** It has 8 fact tables of four types: a transaction fact, an accumulating snapshot, periodic snapshots and a factless coverage fact. They share 13 conformed dimensions, including a complete date dimension with ISO epidemiological weeks and Namibian public holidays. Two dimensions are snowflaked (Care Unit › Facility › Region; Diagnosis › Condition). Wards and patients keep SCD Type 2 history.
- **An ELT pipeline with 44 data-quality rules and 22 reconciliation checks.** The generator recorded every defect it injected, so we could test the rules. They caught **134,964 of 134,992 injected defects** (recall 99.98 %) with **zero false positives**.
- **A Power BI semantic model.** It has 108 DAX measures, dynamic row-level security, static row-level security and object-level security, plus an incremental-refresh policy. The report has six pages, including a drill-through page and a tooltip page.
- **Four advanced analytics techniques**, each validated on held-out data or known ground truth: malaria forecasting, a 30-day readmission risk model, outbreak detection and an analysis of what drives ED waiting times.

78.8 % → 66.7 %	**+6.3 %**	**14,571 → 17,514**	**37 %**
ED patients seen within their triage target, 2022 → 2026 YTD (target 80 %)	longer door-to-doctor time for each extra arrival per doctor-shift (95 % CI 6.2–6.4 %)	confirmed malaria cases, epi year 2025 → 2026 to week 39 (2024: 3,071)	of 30-day readmissions found by calling the top 20 % of patients by predicted risk

What the data says.

1. **ED performance collapsed in the malaria seasons, and staffing per arrival explains most of it.**
   - The share of patients seen within the South African Triage Scale (SATS) target held at 77–79 % in 2022–2024. It fell to 67.5 % in 2025 and 66.7 % in 2026 to date. Over the same period, patients who left without being seen (LWBS) almost doubled (3.3 % → 6.2 %).
   - Holding triage category, facility and time constant, waits rise **6.3 % per extra arrival per doctor-shift** and **3.6 % per 10 points of hospital bed occupancy** (access block).
   - In January–June 2026, one extra day-shift doctor at Rundu and Katima Mulilo coincided with a rise in patients seen on time: 43.9 % → 53.5 % at Rundu and 36.1 % → 48.8 % at Katima Mulilo. This happened despite 16–23 % more attendances (§3.5).
2. **Beds are running hot.**
   - Bed occupancy rose from 76.8 % (2022) to 84.8 % (2026), just under the 85 % ceiling.
   - Katima Mulilo's medical ward ran at 108.5 % in 2025.
   - Only about 27 % of patients are discharged before noon. This leaves too few morning beds for the afternoon ED peak.
3. **Malaria returned at scale.**
   - Confirmed cases rose almost five-fold from 2024 to 2025, and 2026 is already 20 % above 2025. The north-east carries 87 % of the load: Rundu, Katima Mulilo and Nkurenkuru.
   - Rainfall 4–8 weeks earlier is a significant lead indicator (SARIMAX coefficient 0.405, $p < 0.001$).
   - The historical-limits alert fired in the first four epi weeks of 2025 and 2026, 2–6 weeks before each seasonal peak (2025-W07, 2026-W05). At Rundu it fired as early as 2024-W49.
   - The best forecast (Holt-Winters) reaches 23.8 % WAPE over a 52-week rolling-origin test, and 19.4 % one week ahead.
4. **Readmissions sit above the 12 % ceiling** (13.0–13.6 %). Patients who left against medical advice have twice the odds of readmission, and chronic conditions 1.7 times the odds. The risk model is modest (AUC 0.66), but it is well calibrated and similar across sexes. It is useful for prioritising follow-up calls, not for decisions about individual patients.

Recommendations.

1. **Roster to the load.** From January to May, add one ED doctor per day shift at Rundu, Katima Mulilo and Nkurenkuru. Trigger the uplift from the malaria early warning: the historical-limits alert plus the forecast (§3.5, §3.7).
2. **Fight access block.**
   - Make *discharged before noon* a ward KPI, with a target of 40 %.
   - Keep the occupancy ceiling of 85 %.
   - Prioritise the Katima Mulilo medical ward, whose complement grew from 18 to 22 beds in March 2026.
3. **Call the top 20 % risk patients within 72 hours of discharge.** That is about 3,600 patients a year at 2025 volumes, and the calls would reach 37 % of all 30-day readmissions.
4. **Use the warehouse as the system of record.**

- Run the 44 data-quality rules weekly under named stewards.
- Publish the Power BI app with facility-level RLS.
- Give the MoHSS surveillance unit aggregate-only access (object-level security and small-cell suppression) (§3.8).

How to read this report.

- Sections 3.1–3.9 follow the brief, item by item.
- Section 4 maps every required deliverable to its file, cross-checks each requirement of the brief in a compliance matrix, and self-assesses against the rubric.
- Every results table is generated from the warehouse by the build scripts, and every code listing is copied verbatim from the submitted files, so the PDF cannot drift from the code.
- Confidence labels: we state a figure plainly when it is computed from our data. We write *likely* or *uncertain* when a statement depends on assumptions or external facts that could not be verified.

Assumptions and scope

Table 1. Assumptions (each is tested or stated in the section shown)

#	Assumption	Consequence / where handled
A1	GBHS is a teaching scenario. All patient-level and operational data (PAS, ward returns, e-roster, IDSR returns, master data) are **synthetic**. They come from a seeded generator (`01_data/generate_synthetic_data.py`, seed 20261028), anchored on public facts: NSA 2023 regional populations, SATS targets, WHO ICD-10 codes, the IDSR priority-disease list, and the 2025–2026 Namibian malaria upsurge (AllAfrica, 2026; WHO Regional Office for Africa, 2026). Facility names are towns; GBHS and its hospitals are fictitious and **no real hospital's data are represented**.	No real person or health record is used (§3.1.7, §3.8). The generator logs the ground truth of every injected defect and planted effect, so DQ rules and analytics can be scored for precision, recall and recovery (§3.3, §3.5).
A2	External data: the **NSA 2023 census** regional populations are the real published values (their total, 3,022,401, reconciles to the national figure in check R09). **NASA POWER** daily weather is the real file format. The build environment blocked the API, so the generator wrote stand-in values in the exact JSON layout, including the −999 fill values.	`01_data/fetch_public_data.py` downloads the real NASA POWER files into the same paths with no code change. We dropped World Bank indicators rather than fabricate a benchmark (§3.3).
A3	Calendars: calendar year for management KPIs; ISO epidemiological weeks (Monday–Sunday) and epi years for IDSR; fiscal year April–March, as for the Namibian Government; rainy season July–June.	All are in `dim_date` (§3.2). Surveillance KPIs use epi weeks and epi years (§3.4).
A4	Data cut-off: 30 September 2026 (epi week 2026-W39). A day-2 extract dated 1 October 2026 tests incremental loading. It contains SCD2 changes, discharge milestones of open spells, late IDSR reports, inferred patient members and a Type 1 correction.	The 2026 figures are year-to-date and labelled as such everywhere.
A5	KPI definitions follow SATS: RED counts as "immediate" if the patient is seen within 5 minutes; ORANGE 10, YELLOW 60, GREEN 240 minutes (South African Triage Group, 2012). The readmission definition is in §3.1. An IDSR report is on time if received by Tuesday 12:00 after the epi week, which is a GBHS standard operating procedure we assumed.	Definitions are in §3.1 and the data dictionary (Appendix A).
A6	Facility catchment shares (12–35 % of the regional population) are planning assumptions held in master data, not measurements.	Incidence rates are "per catchment population" and labelled as estimates (§3.5).
A7	Users and user principal names (`fm.rundu@gbhs.example` etc.) are fictitious and use the reserved `.example` top-level domain (RFC 2606).	RLS and OLS tests use these accounts (§3.4).
A8	Deployment target: PostgreSQL 16 on a GBHS server, an on-premises data gateway, a Power BI Pro workspace and a Power BI app.	Licensing and governance in §3.8.
A9	The brief has two inconsistencies. The header says "Due: Week 16 (28 October 2026)" while section 4 refers to "the Week 14 deadline"; we follow the dated header and ask the lecturer to confirm. §3.7 asks for an evaluation "against the KPIs in Section 4.1", which does not exist; we read it as the KPIs of §3.1.	See §3.7 and the submission checklist (§4).

3.1 Business case and requirements

Organisation profile

Green Basket Health Services (Pty) Ltd (GBHS) is the health-services subsidiary of Green Basket cc. It is a private, mission-driven provider that works alongside the public system under a referral agreement with the Ministry of Health and Social Services (MoHSS):

- **Five hospitals.** One referral hospital in Windhoek and four district hospitals, each with a 24-hour emergency department (ED) and inpatient wards: 424 beds in 21 wards after the 1 October 2026 load.
- **Five primary-care clinics.** Katutura, Eenhana, Nkurenkuru, Otjiwarongo and Keetmanshoop.
- **Coverage.** The facilities span nine regions, from the malaria-endemic north-east (Kavango East, Kavango West, Zambezi) to the non-endemic south and coast.
- **Payers.** Patients pay through medical aid (PSEMAS and private schemes), statutory cover (SSC, MVA Fund), MoHSS referral or cash.
- **Surveillance duty.** As a health facility, every GBHS site must notify priority diseases weekly through Integrated Disease Surveillance and Response (IDSR). This duty comes from the Public and Environmental Health Act, 2015, in force since September 2020 (Republic of Namibia, 2015, 2020; WHO Regional Office for Africa, 2019).

Table 2. GBHS facility portfolio (current SCD Type 2 ward versions; attendances = ED for hospitals, outpatient for clinics, 2025)

facility_code	facility	facility_type	region	zone	malaria_stratum	wards	beds_now	has_ed	catchment_share_pct	attendances_2025	admissions_2025
GBH-WHK	Windhoek	Referral hospital	Khomas	Central	Non-endemic	6	134	true	16	22,709	9,281
GBH-RUN	Rundu	District hospital	Kavango East	North-East	Endemic - high	4	92	true	35	18,467	5,944
GBH-OSH	Oshakati	District hospital	Oshana	North-Central	Endemic - low	5	102	true	30	18,141	7,494
GBH-KAT	Katima Mulilo	District hospital	Zambezi	North-East	Endemic - high	3	48	true	30	13,072	3,330
GBH-WVB	Walvis Bay	District hospital	Erongo	West	Non-endemic	3	48	true	25	9,223	3,477
GBC-KTU	Katutura	Clinic	Khomas	Central	Non-endemic	0	0	false	12	12,902	0
GBC-NKU	Nkurenkuru	Clinic	Kavango West	North-East	Endemic - high	0	0	false	25	10,234	0
GBC-EEN	Eenhana	Clinic	Ohangwena	North-Central	Endemic - low	0	0	false	15	8,674	0
GBC-OTJ	Otjiwarongo	Clinic	Otjozondjupa	Central	Endemic - low	0	0	false	20	7,017	0
GBC-KMP	Keetmanshoop	Clinic	//Kharas	South	Non-endemic	0	0	false	30	5,756	0

The setting is realistic:

- Namibian public referral hospitals report bed occupancy above 100 % and long waits (The Namibian, n.d.; AllAfrica, 2025).
- In 2026 the country had a major malaria upsurge. Nearly 28,000 cases were recorded in the first ten weeks after unusual rains; Katima Mulilo and Nkurenkuru were among the hardest-hit areas (WHO Regional Office for Africa, 2026). By early September there were more than 122,000 confirmed cases and 122 deaths, 30.6 % of the cases in Kavango East (AllAfrica, 2026).
- The national response included extended clinic hours and redeploying staff to the worst-hit areas, the same lever that GBHS tested in 2026 (§3.5).

GBHS's synthetic data reproduce these pressures (assumption A1).

Strategy and objectives

GBHS's 2025–2028 strategy, *"Right care, right time, every region"*, has three pillars:

- timely access to emergency care;
- efficient and safe use of beds and staff;
- early detection of disease threats in the communities it serves.

The BI programme supports the five measurable objectives below. Each has a baseline taken from the warehouse and a target agreed with the executive committee (scenario assumption).

Table 3. Strategic objectives, KPIs, 2025 baseline (warehouse) and target for 2027

#	Objective (pillar)	KPI (definition in Appendix A / §3.4)	Baseline 2025	Target 2027
O1	Timely emergency care	Seen within SATS triage target (% of triaged ED patients)	67.5 %	≥ 80 %
		Left without being seen (LWBS, % of ED attendances)	6.2 %	≤ 3 %
		Median door-to-doctor time (min)	63	≤ 45
O2	Efficient bed use	Bed occupancy (patient days / available bed days)	81.8 %	≤ 85 % per hospital
		Discharged before noon (% of discharges)	26.9 %	≥ 40 %
		Average length of stay (days, by discharge date)	4.10	≤ 4.0
O3	Safe transitions of care	30-day emergency readmission rate (% of eligible discharges)	13.4 %	≤ 12 %
O4	Sustainable workforce	Nursing hours per patient day (NHPPD, inpatient wards)	5.37	≥ 5.5
		Overtime hours (% of nursing hours)	5.1 %	≤ 4 %
		ED arrivals per doctor-shift	15.2	≤ 13
O5	Early detection and response	IDSR weekly report completeness / timeliness	98.7 % / 92.9 %	≥ 80 % / ≥ 80 % (WHO AFRO)
		Malaria early warning raised before the seasonal peak (weeks)	3–6	≥ 4
		Malaria incidence in the catchment (cases per 1,000 person-years)	25.2	Reported weekly; no internal target (MoHSS sets elimination goals)

Business problem

GBHS manages its hospitals from monthly Excel packs that each facility compiles by hand from the PAS, paper ward returns and the e-roster. Consolidating them takes about a week, and the numbers disagree because each site counts differently. The profiling of the raw extracts (§3.3) shows why:

- **Duplicates and versions.** There are 2,966 duplicated attendance rows from overlapping yearly PAS exports, and 16,223 superseded inpatient-spell images from the change-data-capture export. Resubmitted ward returns (109) and IDSR reports (73) double-count unless versions are resolved.
- **Inconsistent formats and codes.**
  - 16,477 timestamps are in day-first format;
  - 15,904 ICD-10 codes are badly formatted (e.g. `A09-`, `j18.9`);
  - there are 30 spellings of the 9 IDSR conditions, 84 of the ward codes and 18 of the payer classes;
  - Nkurenkuru reports epi weeks as `W05/2024`.
- **Implausible values.**
  - 807 “seen by a doctor before arrival” timestamps;
  - 280 ED stays over 72 hours;
  - occupied-bed counts keyed ×10;
  - negative roster hours;
  - IDSR returns with more deaths than cases.
- **Missing data.** There are 291 missing ward returns, 2,875 attendances without a triage category, and 25 spells open for over 60 days that are probably missing their discharge.

The consequence is that management could not tell why ED performance fell in 2025, whether the cause was demand, beds or staffing, nor answer the MoHSS with confidence about the malaria situation in the north-east. **Problem statement:** *GBHS lacks a single, trusted and timely view that connects patient flow, bed and staff utilisation and disease surveillance. Without it, it cannot target scarce doctors, nurses and beds where demand and disease are rising, nor show regulators that its notifiable-disease reporting is complete and on time.*

Business questions and KPIs

Table 4. Business questions, KPIs and where they are answered

#	Business question	KPIs (measures)	Answered in
BQ1	**Patient flow.** Are ED patients seen within their triage target, and what drives the delays: demand, staffing, beds (access block) or time of day?	Seen Within Target %, Median / P90 Door-to-Doctor, LWBS Rate %, ED Arrivals per Doctor Shift	Page 1, Page 2; §3.5 technique 4 (wait drivers)
BQ2	**Resource utilisation.** Are beds and nursing hours matched to demand by facility, ward and season?	Bed Occupancy %, Average Daily Census, ALOS, Discharged Before Noon %, NHPPD, Overtime %, Agency %	Page 2 (drill-down Region › Facility › Ward); §3.5 drill-across
BQ3	**Disease surveillance.** Where and when are notifiable diseases rising, and is IDSR reporting complete and on time?	Notified Cases, Malaria Incidence per 1,000 PY / per 100k, Test Positivity %, IDSR Completeness % and Timeliness %, Outbreak Alerts	Page 3; §3.5 technique 3 (outbreak detection)
BQ4	**Preparedness.** How many malaria cases should the north-east expect in the next 4–13 weeks, and how reliable is that forecast?	Forecast Cases with 80 % interval, Backtest WAPE %	Page 4; §3.5 technique 1 (forecasting)
BQ5	**Safe transitions.** Which discharged patients are most likely to be readmitted within 30 days, so that follow-up can be targeted?	Readmission Rate 30d %, Top-20% Risk Capture %, calibration by decile	Page 4; §3.5 technique 2 (classification)

Key KPI definitions (the full set is in Appendix A and in the DAX of §3.4):

- **Seen within target.** The time from arrival to the first doctor assessment is at most the SATS time for the triage category: RED 5 minutes (counting “immediate”), ORANGE 10, YELLOW 60, GREEN 240 (South African Triage Group, 2012). Patients who were not triaged or not seen are excluded from the denominator.
- **30-day readmission rate.** Emergency readmissions within 30 days of discharge, divided by eligible discharges. An eligible discharge is alive, home or against medical advice, not maternity, and has 30 days of follow-up. Both are counted by the discharge date of the index spell.
- **Bed occupancy.** Midnight occupied beds summed over days, divided by available (open) beds summed over days.
- **IDSR completeness / timeliness.** Weekly reports received / on time, divided by the reports expected from each active facility for each epi week due by the cut-off. Timeliness is judged on the first version received: a later correction does not make a report late.

Stakeholder analysis

Table 5. Stakeholders, their information needs and how the solution serves them (T = technical, N = non-technical)

Stakeholder	Type	Information need	How the solution serves them
CEO and executive committee	N	Monthly view of access, utilisation and risk across all sites; trend versus targets	Page 1 (Executive Overview), Executive role, monthly app subscription
Facility managers (5 hospitals, 5 clinics)	N	Their own facility only: ED performance, beds, staffing; drill to ward	Pages 1–2 with dynamic RLS (Facility Manager role), drill-through *Facility Detail*
ED nursing managers and the clinical director	N	Waits by triage category and hour; LWBS; doctors per shift	Page 2 hour and triage visuals; §3.5 wait-driver model
Nursing services and ward managers	N	Occupancy, census, discharges before noon, NHPPD, overtime, agency	Page 2 matrix, semi-additive census measures
Surveillance officers and the MoHSS regional surveillance unit	N	Weekly notifiable cases, malaria trend and incidence, outbreak alerts, completeness and timeliness of IDSR	Page 3, outbreak alerts on page 4; Surveillance Partner role (OLS, aggregate only)
Health Information Manager and clinical coding lead	N/T	Data-quality issues (coding, duplicates, missing discharges) and their trend	DQ rule catalogue, `audit.dq_issue_log`, page 4 DQ table (§3.3, §3.8)
Finance and the revenue cycle (payers)	N	Activity by payer group for medical-aid negotiations	Patient payer attribute (SCD2, as-was) in all patient facts
BI team (data engineer, Power BI developer, analyst)	T	Reproducible pipeline, lineage, tests, performance	`run_all.py`, audit schema, reconciliation, PBIP in git (§3.3, §3.7)
Information security officer and IT	T	Least privilege, access reviews, encryption, incident response	DB roles, RLS/OLS, `sec.user_facility_access`, governance checklist (§3.8)
Patients and communities (indirect)	N	Timely, safe care and privacy of their health information	KPIs aimed at waits and readmissions; pseudonymisation, data minimisation, small-cell suppression (§3.8)

Process model

Figure 1 is a BPMN-style flowchart of the patient journey and the reporting processes. Each swimlane is an actor, and the labels S1–S9 mark where each data source captures data. Three design insights come from it:

1. **Every KPI timestamp exists only in the PAS.** Arrival, triage, seen by doctor and departure are all recorded there, so ED waits are measured per attendance. That makes the attendance the natural transaction grain.
2. **An admission accumulates milestones over days.** Admission, coding, discharge and the end of the 30-day follow-up arrive at different times, which calls for an accumulating snapshot that is updated in place.
3. **Ward census and IDSR are periodic returns** (daily and weekly). They need periodic-snapshot facts, plus a coverage fact to measure *missing* reports.

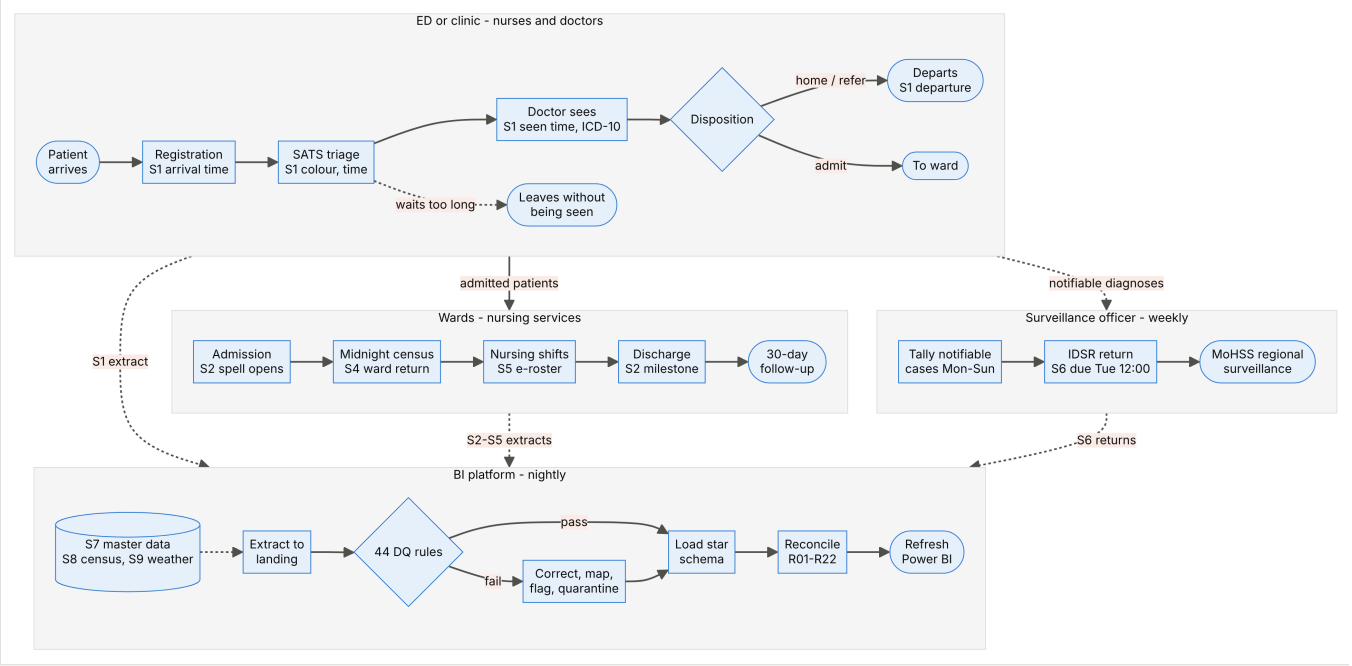


Figure 1. Patient-flow and reporting process (BPMN-style flowchart) with the data capture points S1-S9 (Mermaid source: `06_report/diagrams/bpmn.mmd`)

Data sources, ethical and legal use of data

The solution integrates **nine sources** of all three required types. The brief needs at least one transactional, one master/reference and one external source; we have six transactional, one master/reference and two external.

Table 6. Data sources (type, format, volume in the full load, provenance and terms of use)

#	Source	Type	Format and volume	Provenance and terms
S1	PAS ED and clinic attendances	Transactional	5 yearly gzip CSV extracts that overlap at year ends; 563,209 rows	Synthetic (A1)
S2	PAS inpatient spells	Transactional	Change-data-capture export, gzip CSV; 151,774 spell images	Synthetic
S3	PAS patient index and change log	Transactional / master	gzip CSV; 107,332 + 11,661 rows; pseudonymised IDs only	Synthetic
S4	Ward midnight census returns	Transactional (periodic)	CSV; 36,232 ward-days	Synthetic
S5	Nursing e-roster shift summary	Transactional (periodic)	gzip CSV; 97,502 unit-shift rows	Synthetic
S6	IDSR weekly returns	Transactional (periodic)	CSV; 22,410 facility × condition × week rows	Synthetic; follows the WHO AFRO IDSR priority diseases
S7	GBHS master data workbook	Master / reference	Excel, 10 sheets: regions, facilities, wards and their history, SATS, ICD-10, IDSR conditions, payers, code mappings	Synthetic; ICD-10 titles from WHO (2019); SATS from South African Triage Group (2012)
S8	NSA 2023 Population and Housing Census, population by region	External (public)	CSV as published, with spaces as thousands separators and title and footnote rows; 14 regions	**Real values**, Namibia Statistics Agency (2024); public statistics, cited
S9	NASA POWER daily weather (rainfall, Tmax, Tmin, humidity) at each facility	External (public)	JSON from the POWER API; 10 points × 1,856 days	NASA POWER format (NASA Langley Research Center, n.d.); values are stand-ins until `fetch_public_data.py` is run (A2). NASA data are free and open, with attribution requested

Ethical and legal-use confirmation (required by the brief). The team confirms that:

1. **No personal information about any real person is collected, stored, processed or published.** All patient records are generated by a seeded program. Names, national ID numbers, addresses, telephone numbers and exact dates of birth do not exist anywhere in the data. Patients carry a random pseudonym (`P` followed by 12 hexadecimal characters), and only the birth year is kept for age bands.
2. **The two external datasets are public aggregate statistics** (NSA census, NASA POWER). We use them within their terms of use and cite them (APA, §References). Neither contains personal data.
3. **No real health facility's performance is represented.** GBHS is fictitious, and its facilities only borrow town names (A1). Any resemblance to real hospital figures is coincidental.
4. **The design treats the data as if they were real special-category health data.** It applies data minimisation, pseudonymisation, least-privilege access, row-level and object-level security, small-cell suppression for external sharing, and an access-review process. This follows the principles of Article 13 (privacy) of the Namibian Constitution, Namibia's draft Data Protection Bill, POPIA, the GDPR and the HIPAA de-identification standard (§3.8).
5. **Notifiable-disease reporting is a legal duty.** In a real deployment, the IDSR data flows to the MoHSS would rest on the Public and Environmental Health Act, 2015. Clinical data would be used for analytics only under the provider's governance approvals, and any research use would need ethics committee approval.

3.2 Data warehouse design

Architecture

We follow Kimball's bus architecture (Kimball & Ross, 2013): a set of process-oriented fact tables that share conformed dimensions, held in PostgreSQL 16 and loaded by ELT. Python only lands the raw files; all cleansing, conforming and loading runs as set-based SQL inside the database, where it is auditable and repeatable (Figure 2). Power BI imports a thin `mart` layer of views, so that the model never depends on warehouse internals. Analytics outputs written back by Python (forecasts, alerts, model results) live in the same `mart` schema, so Power BI and the report read one source of truth.

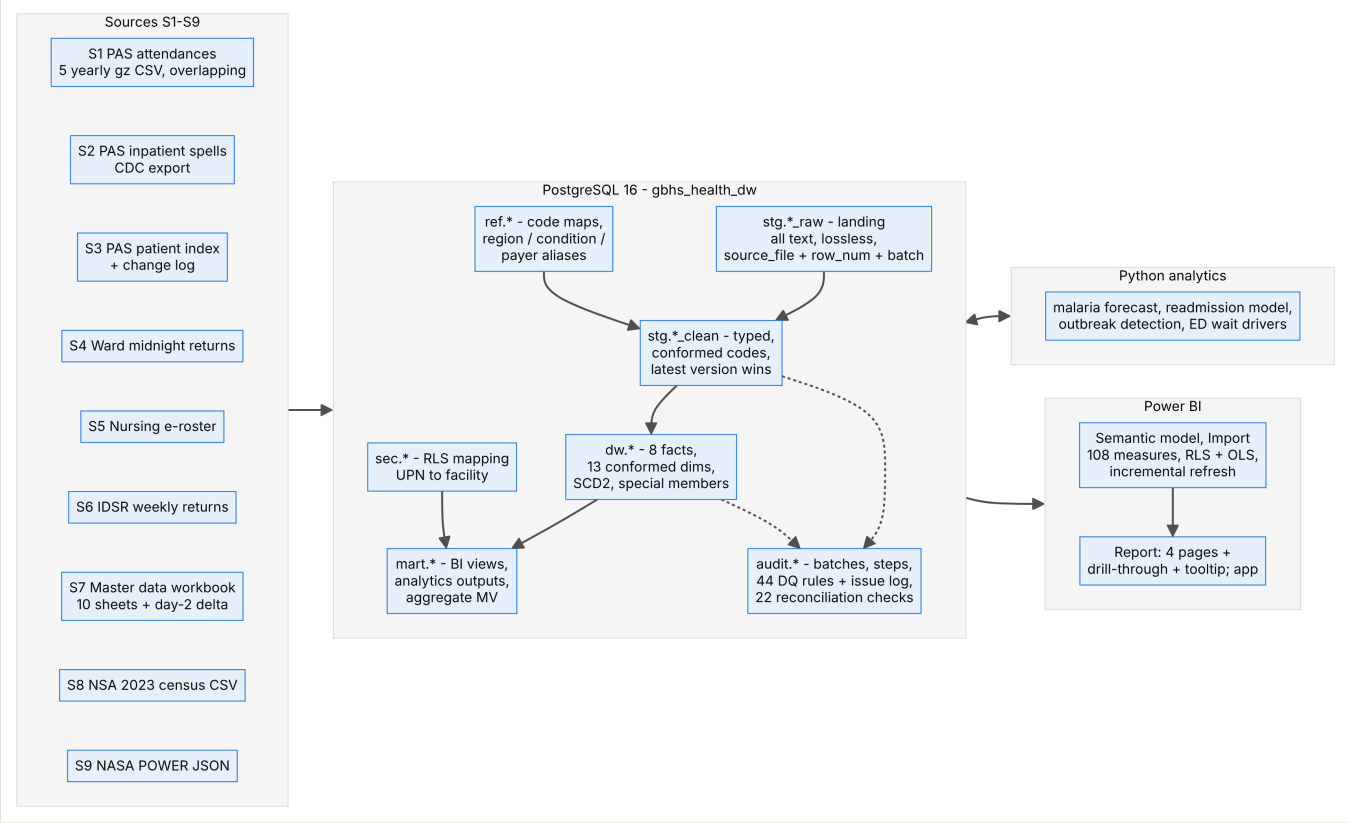


Figure 2. Solution architecture: sources, PostgreSQL layers, analytics and Power BI (Mermaid source: `06_report/diagrams/architecture.mmd`)

Table 7. Schemas and their role (least privilege per schema; grants in 02_sql/10_security_grants.sql)

Schema	Content	Who can read
`stg`	Landing tables (`*_raw`: every column as text, plus file name, row number and batch) and typed `*_clean` tables, merged by natural key across batches	ETL role only
`ref`	Mapping tables: code maps, region, condition and payer aliases	ETL role only
`dw`	The dimensional model: 13 dimensions and 8 facts (264 columns, all documented with a comment)	ETL role, BI developers
`mart`	Views and tables consumed by Power BI (no pseudonyms, no timestamps); analytics outputs; aggregate materialised view	Power BI reader role (`gbhs_pbi_reader`)
`audit`	Batch log, step log with row counts and timings, DQ rule catalogue and issue log, reconciliation results	ETL role, data stewards
`sec`	Row-level-security mapping of user principal names to facilities	ETL role; exposed to Power BI only through `mart.v_sec_user_facility`
`etl`	Functions and procedures of the pipeline	ETL role

Bus matrix

Eight business processes share thirteen conformed dimensions (bus matrix below). ● marks a direct foreign key. ○ marks a dimension reached through a snowflaked parent: Care Unit › Facility › Region, and Diagnosis › Condition. Conformed dimensions make **drill-across** possible. For example, nursing hours (e-roster) are divided by patient days (ward census) through Date and Care Unit to give NHPPD, and ED attendances are divided by doctor-shifts to give load per doctor (§3.5).

Table 8. Enterprise bus matrix: business processes x conformed dimensions

Business process (fact table)	Date	Hour	Care Unit	Facility	Region	Patient	Diagnosis	Condition	Triage	Encounter Profile	Admission Profile	Age Band	Shift
ED and clinic attendances (`fact_encounter`)	●	●	●	○	○	●	●	○	●	●		●	
Inpatient spells (`fact_admission`)	● ×2		●	○	○	●	●	○			●	●	
Ward midnight census (`fact_ward_census_daily`)	●		●	○	○								
Staffing (`fact_staff_roster`)	●		●	○	○								●
Notifiable-disease surveillance (`fact_surveillance_weekly`)	●			●	○			●					
IDSR reporting compliance (`fact_idsr_report`)	●			●	○								
Population at risk, external NSA (`fact_population`)	●				●								
Weather, external NASA POWER (`fact_weather_daily`)	●			●	○								

Grain of every fact table

Table 9. Declared grain, fact type, measures and additivity (row counts after the 1 October 2026 load)

Fact table	Type	Grain (one row per ...)	Measures and additivity	Rows
`fact_encounter`	Transaction	ED or clinic attendance (PAS attendance number, a degenerate dimension)	`encounter_count`, `is_lwbs`, `is_admitted`, `is_died`, `seen_within_target`: additive. `wait_to_triage_min`, `door_to_doctor_min`, `ed_los_min`: additive in sums, analysed as medians/percentiles. `age_years`: non-additive	560,557
`fact_admission`	Accumulating snapshot	Inpatient spell from admission to discharge. Milestones: admitted, discharged (role-playing discharge date), 30-day follow-up	`admission_count`, `los_days` (= patient days of the spell), `is_discharged`, `is_died`, readmission flags: additive. `prior_admissions_12m`: non-additive	135,629
`fact_ward_census_daily`	Periodic snapshot	Ward × day (midnight census)	`occupied_beds`, `beds_available`, `beds_complement`, `beds_closed`: **semi-additive** (sum across wards, never across days; summed over days they become patient days and bed days). `admissions`, `discharges`, `deaths`: additive flows	36,435
`fact_staff_roster`	Periodic snapshot	Care unit × day × shift	Hours (planned, nurse, RN, doctor, overtime, agency, sick leave): additive. `nurses_on_duty`, `doctors_on_duty`: semi-additive	97,365
`fact_surveillance_weekly`	Periodic snapshot	Facility × IDSR condition × epi week (key = Monday of the week)	Cases (<5, 5+, total), deaths, tests done and positive: additive	21,771
`fact_idsr_report`	Factless coverage	Facility × epi week **expected** to report (by the cut-off)	`report_expected`, `report_received`, `report_on_time`: additive counts, so completeness = received / expected. This fact records *what did not happen* (a missing report)	2,470
`fact_population`	Periodic snapshot (annual, external)	Region × year (key = 1 July, mid-year)	`population`: **semi-additive** (sum across regions, never across years)	98
`fact_weather_daily`	Periodic (external)	Facility location × day	`rainfall_mm`: additive over days, averaged across locations. `tmax_c`, `tmin_c`, `rh_pct`: non-additive	18,560

The day-2 load proves the accumulating behaviour: 87 spells that were open at the 30 September cut-off received their discharge milestone in place on 1 October. They were not inserted again (extract below).

Table 10. Accumulating snapshot: spells open at the cut-off and completed by the 1 October load (extract)

admission_id	unit_code	admitted	discharged	los_days	discharge_status_code	updated_by_batch
OSHA2605228	OSH-PAE1	2026-09-10	2026-10-01	21	HOME	2
WHKA2606771	WHK-PAE1	2026-09-15	2026-10-01	16	HOME	2
RUNA2604658	RUN-SUR1	2026-09-23	2026-10-01	8	HOME	2
RUNA2604667	RUN-MED1	2026-09-23	2026-10-01	8	HOME	2
OSHA2605533	OSH-MED1	2026-09-24	2026-10-01	7	HOME	2
OSHA2605542	OSH-SUR1	2026-09-25	2026-10-01	6	HOME	2
WHKA2607052	WHK-MAT1	2026-09-25	2026-10-01	6	HOME	2
WVBA2602619	WVB-MED1	2026-09-25	2026-10-01	6	HOME	2

Dimensional model (ERD)

The model is a **fact constellation** of three subject areas that share dimensions. Figures 3–7 show the ERDs at column level for keys and the main attributes; Appendix A lists every column.

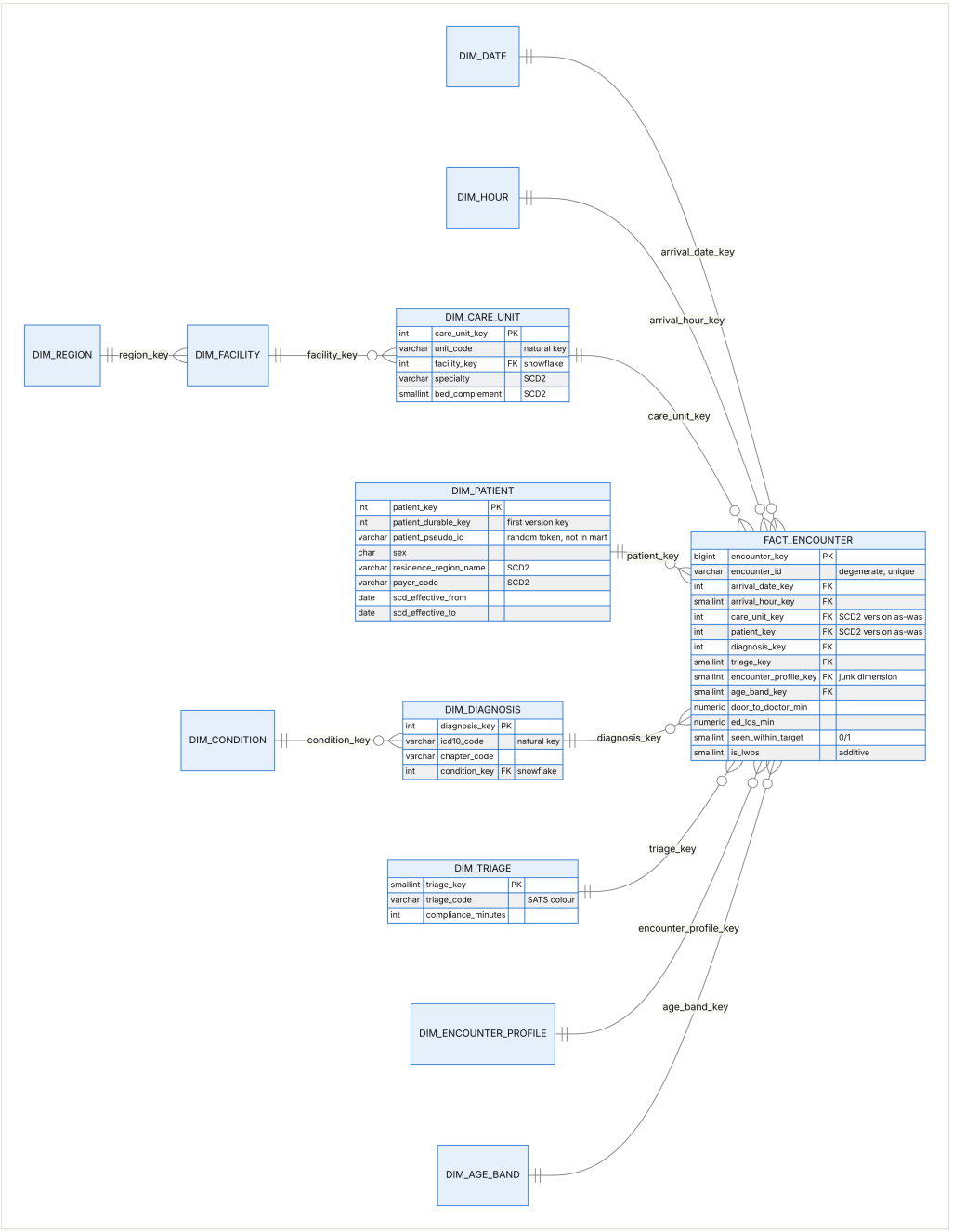


Figure 3. ERD 1 - Patient flow (a): ED and clinic attendances, a transaction fact with eight dimensions (Mermaid source: `06_report/diagrams/erd_flow.mmd`)

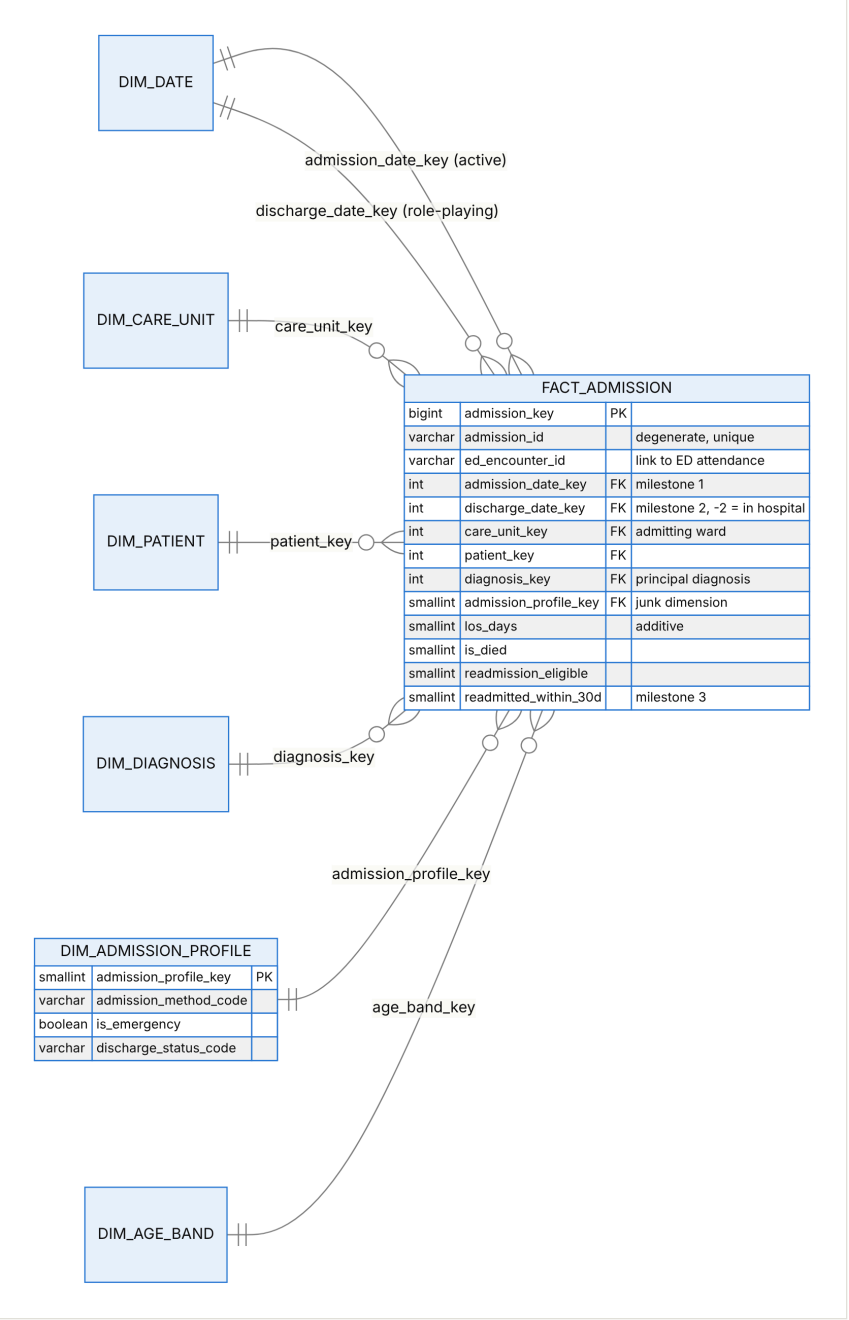


Figure 4. ERD 2 - Patient flow (b): inpatient spells, an accumulating snapshot with a role-playing date (Mermaid source: `06_report/diagrams/erd_spell.mmd`)

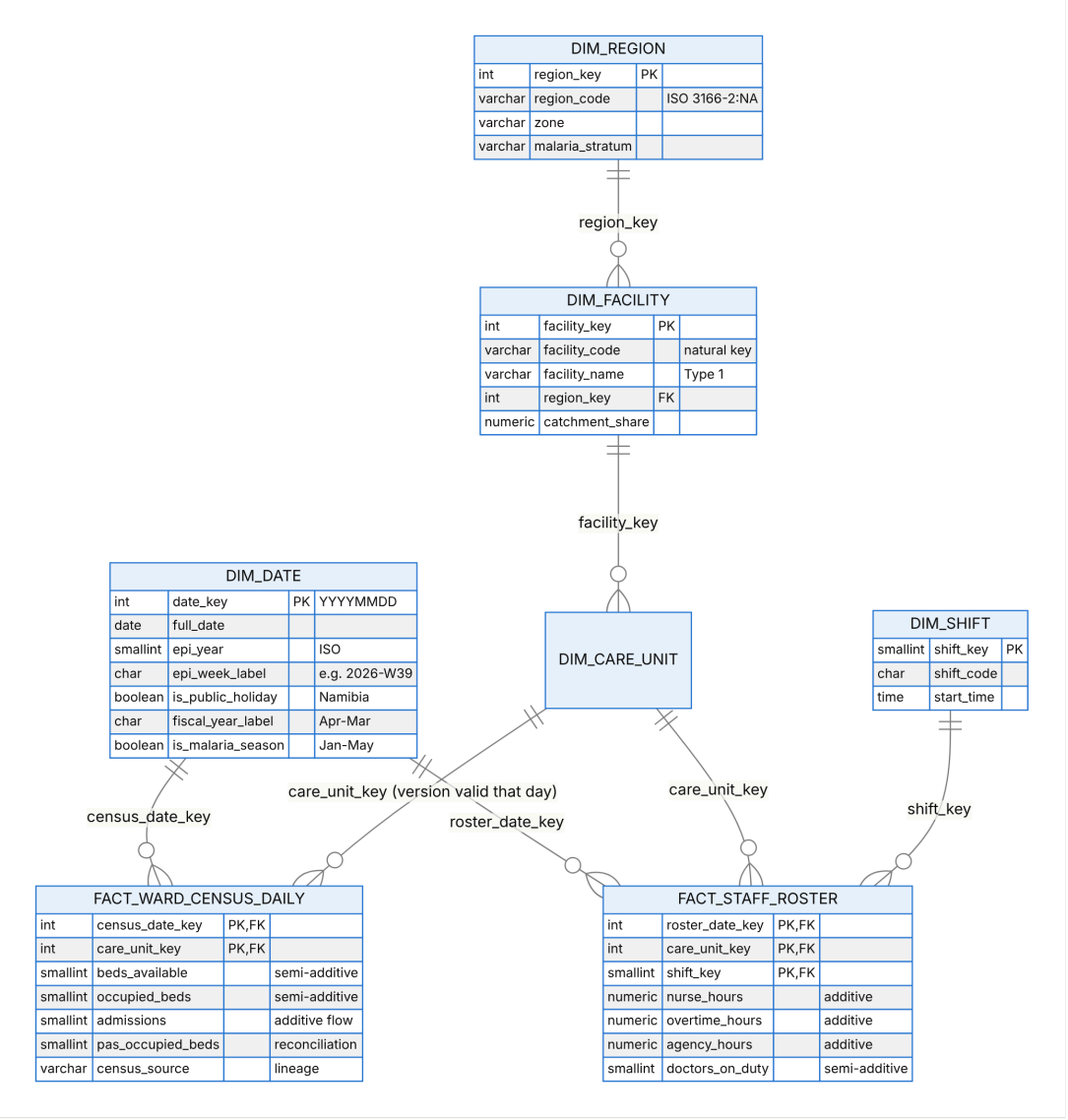


Figure 5. ERD 3 - Resource utilisation: ward census and e-roster (periodic snapshots) sharing Date and Care Unit (Mermaid source: `06_report/diagrams/erd_resource.mmd`)

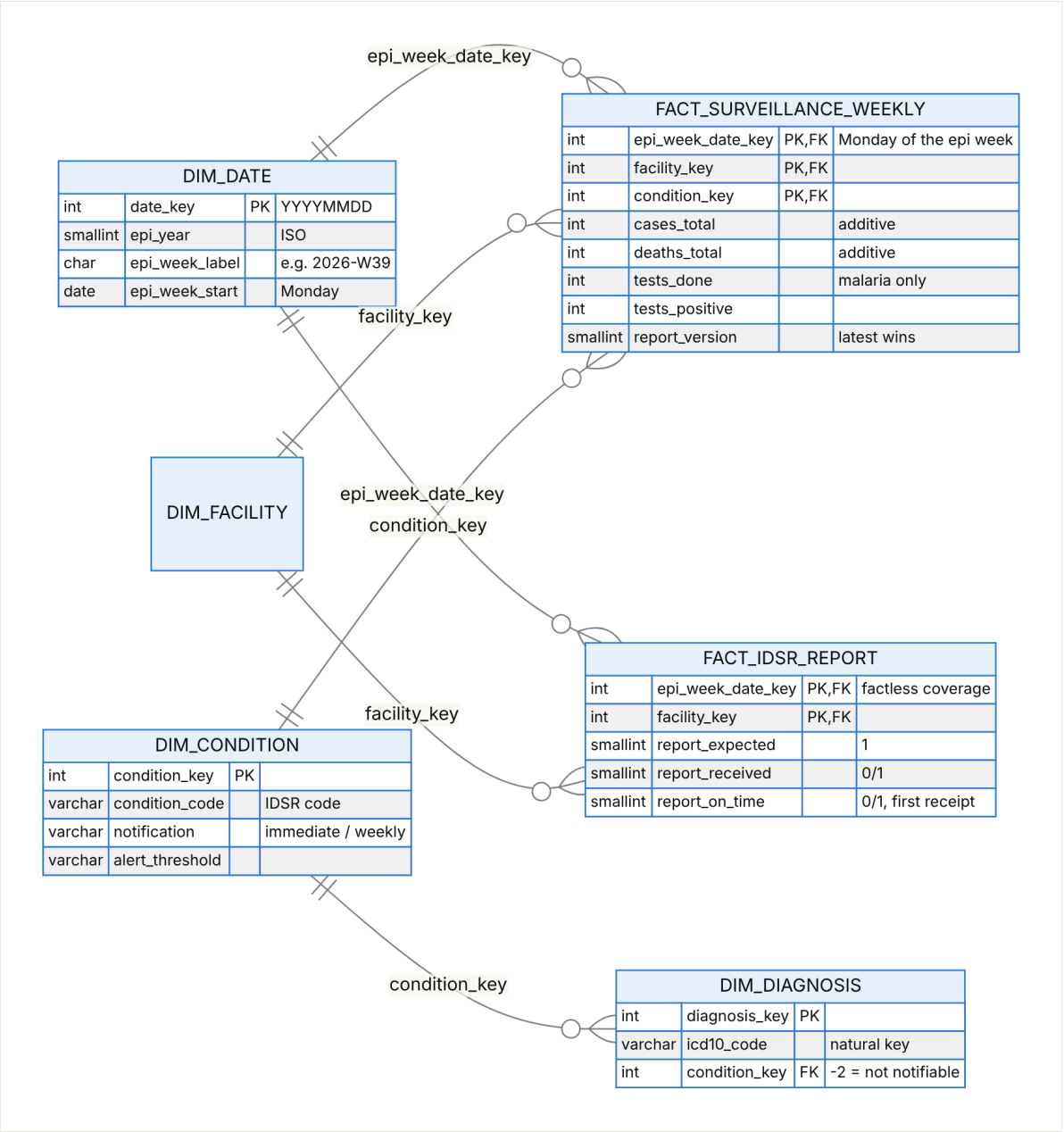


Figure 6. ERD 4 - Disease surveillance (a): weekly IDSR cases and reporting coverage, with the Diagnosis → Condition snowflake (Mermaid source: `06_report/diagrams/erd_surveillance.mmd`)

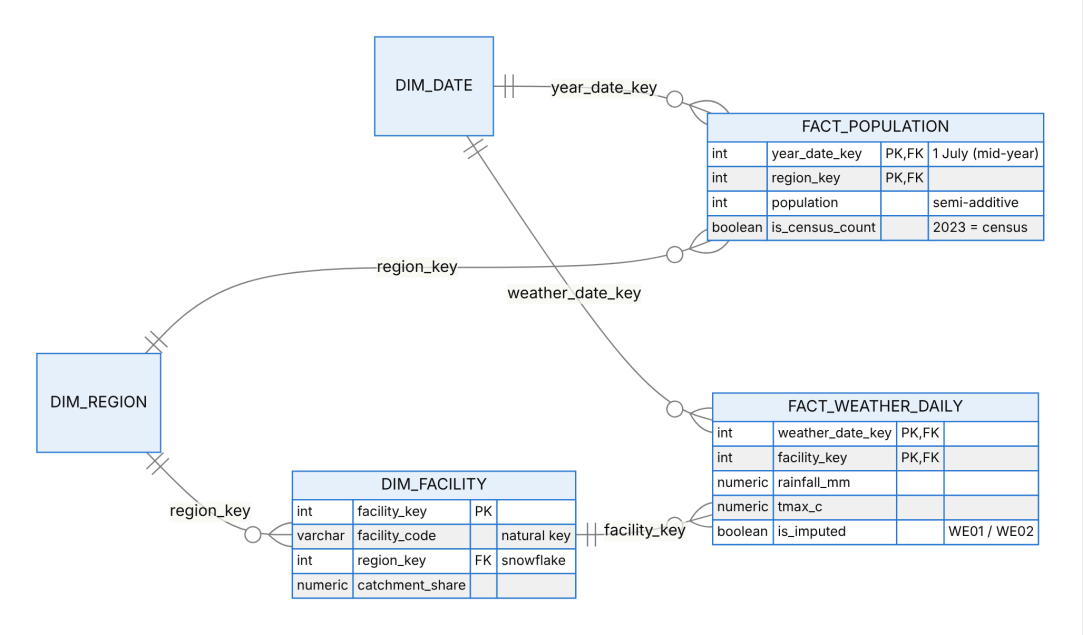


Figure 7. ERD 5 - Disease surveillance (b): external context, NSA population by region and NASA POWER weather by facility (Mermaid source: `06_report/diagrams/erd_external.mmd`)

Dimensions

Table 11. The 13 conformed dimensions

Dimension	Rows	Natural key	Main attributes	History
`dim_date`	2,558	calendar date	Day, ISO weekday, Namibian public holiday and name, working day; ISO epi year/week/label/start; month, quarter, year; fiscal year (April–March) and quarter; rainy season (July–June); malaria-season flag; after-cut-off flag	Static, generated
`dim_hour`	24	hour 0–23	Label, day-part band, nursing shift, after-hours flag	Static
`dim_region`	15	ISO 3166-2:NA code	Region, management zone, malaria-endemicity stratum	Type 1
`dim_facility`	11	facility code	Name, short name, type, town, coordinates, ED flag, catchment share, opening date	Type 1 (name corrected on 1 October)
`dim_care_unit`	41	unit code	Ward name, unit type, facility, **specialty, bed complement**	**Type 2**
`dim_patient`	118,916	pseudonymised patient ID	Sex, birth year, **residence region, payer class**, payer group, registration date, inferred flag, durable key	**Type 2** + inferred members
`dim_diagnosis`	65	ICD-10 code	WHO title, category, block, chapter, severity class, default specialty, chronic flag, IDSR condition	Type 1
`dim_condition`	11	IDSR condition code	Name, IDSR category, notification timeline, alert threshold, ICD-10 codes	Type 1
`dim_triage`	7	SATS colour	Label, priority, target and compliance minutes, colour	Static
`dim_encounter_profile`	56	department × arrival mode × disposition	Junk dimension: low-cardinality flags in one small table instead of three foreign keys	Static
`dim_admission_profile`	36	admission method × discharge status	Junk dimension, including the emergency-admission flag	Static
`dim_age_band`	7	age band	IDSR age group (under 5 / 5 and over), life stage	Static
`dim_shift`	2	shift code	Day 07:00–19:00, Night 19:00–07:00	Static

A complete date dimension.

- `dim_date` holds every day from 1 January 2021 to 31 December 2027 (2,556 days, checked by reconciliation R15), plus the special members −1 and −2.
- The keys are integers in YYYYMMDD format. They read as dates and fold into SQL predicates for incremental refresh.
- **ISO epidemiological weeks** are first-class attributes, because IDSR reports Monday–Sunday weeks that cross month and year boundaries. For example, 30 December 2024 belongs to 2025-W01.
- **Namibian public holidays** are computed, not typed in: the moveable feasts are derived from the Easter date, and a holiday that falls on a Sunday is observed on the Monday.

File: `02_sql/04_etl_functions.sql` (lines 62–82)

```
        CREATE OR REPLACE FUNCTION etl.fn_namibian_holiday(p_date date)
        RETURNS text LANGUAGE plpgsql IMMUTABLE AS $$
        DECLARE y int := extract(year FROM p_date); e date; fixed record; obs date;
        BEGIN
            e := etl.fn_easter_sunday(y);
            IF p_date = e - 2 THEN RETURN 'Good Friday'; END IF;
            IF p_date = e + 1 THEN RETURN 'Easter Monday'; END IF;
            IF p_date = e + 39 THEN RETURN 'Ascension Day'; END IF;
            FOR fixed IN
                SELECT * FROM (VALUES (1, 1, 'New Year''s Day'), (3, 21, 'Independence Day'), (5, 1, 'Workers'' Day'),
                                      (5, 4, 'Cassinga Day'), (5, 25, 'Africa Day'), (5, 28, 'Genocide Remembrance Day'),
                                      (8, 26, 'Heroes'' Day'), (12, 10, 'Human Rights Day'), (12, 25, 'Christmas Day'),
                                      (12, 26, 'Family Day')) AS t(m, d, n)
            LOOP
                IF fixed.n = 'Genocide Remembrance Day' AND y < 2025 THEN CONTINUE; END IF;
                obs := make_date(y, fixed.m, fixed.d);
                IF p_date = obs THEN RETURN fixed.n; END IF;
```

```
            IF extract(isodow FROM obs) = 7 AND p_date = obs + 1 THEN RETURN fixed.n || ' (observed)'; END IF;
        END LOOP;
        RETURN NULL;
    END $$;
```

Snowflaked dimensions and their trade-offs

The core is a star. We kept two snowflakes on purpose, because facts attach at more than one level of these hierarchies:

1. **Care Unit › Facility › Region.**
   - Attendances, spells, the census and the roster are recorded per ward (care unit).
   - IDSR returns, IDSR coverage, weather and catchment population are recorded per **facility**, and the NSA population per **region**.
   - Flattening facility into the care unit would leave the facility-grain facts without a dimension. It would also repeat facility attributes in every SCD2 ward version, and a facility rename (Type 1) would have to update every version.
2. **Diagnosis › Condition.** PAS codes diagnoses in ICD-10, but IDSR reports *conditions*. Condition is the **conformed** dimension that joins clinical activity to surveillance; check R08 reconciles IDSR malaria cases with PAS diagnoses B50–B54 through it.

Table 12. Star (flattened) versus snowflake for the Care Unit > Facility > Region hierarchy

Criterion	Flattened star (facility and region columns inside `dim_care_unit`)	Snowflake (chosen)
Storage	Facility and region attributes repeated in 41 ward versions (negligible)	Each attribute stored once (negligible either way)
Query performance	One join fewer for ward-grain facts	One extra join to tables of 11 and 15 rows; no measurable cost in PostgreSQL (§3.7) or VertiPaq (relationships on integer keys)
Facility-grain facts	Need a separate facility dimension that is not conformed with the ward one, with a risk of two "Facility" slicers	One Facility dimension filters ward-grain **and** facility-grain facts (single slicer, drill-across)
Maintainability	A facility rename (Type 1) rewrites every ward version; region changes ripple through SCD2 rows	Type 1 change in one row; SCD2 on wards stays independent of facility attributes
Power BI usability	Simpler field list	Single-direction relationships Care Unit → Facility → Region; RLS on Facility propagates down to wards and to every fact

The patient's residence region is deliberately *flattened* (no relationship to `dim_region`). A second path Patient → Region would make filter propagation ambiguous, because a facility's region and a patient's home region are different roles. The attribute is therefore a text column in `dim_patient`.

Surrogate keys, special members and natural keys

- **Surrogate keys.** Every dimension has an integer surrogate key generated by the warehouse. It is independent of source codes, which change and differ between systems: ward codes had 84 spellings. Facts carry only surrogate keys, plus degenerate identifiers (`encounter_id`, `admission_id`) for traceability.
- **Special members.**
  - −1 = *Unknown*: for example, 180 attendances without a patient ID (EN11) and 840 invalid ICD-10 codes (EN08).
  - −2 = *Not applicable*: for example, the discharge date of a spell that is still in hospital.
  - Facts therefore never contain NULL foreign keys, and totals reconcile (R12–R14).
- **Durable key.** `patient_durable_key` is the surrogate key of the patient's first version. It is constant across SCD2 versions, so *distinct patients* are counted correctly even when a patient changes payer or region.
- **Smart date key.** The YYYYMMDD integer is the only "smart" key, by Kimball convention.

Slowly changing dimension strategy

Table 13. SCD decision per dimension attribute

Dimension	Attribute(s)	SCD type	Justification
Care Unit	specialty, bed complement	**Type 2**	Occupancy must use the beds *valid that day*. When Rundu's medical ward grew from 24 to 32 beds in April 2025, occupancy before and after must not be recomputed with today's 36 beds. Specialty changes, such as a ward converted to COVID-19 isolation, change the case mix.
Care Unit	ward name, unit type, facility	Type 1	Corrections, not business events
Patient	residence region, payer class	**Type 2**	Payer mix and catchment analyses must show the payer and region *at the time of the visit* (as-was), e.g. for medical-aid negotiations
Patient	sex, birth year (corrections)	Type 1	Data corrections (MD02, MD03)
Facility, Region, Diagnosis, Condition	all	Type 1	Reference data; corrections overwrite. On 1 October the facility name "GBHS Katima-Mulilo Hospital" was corrected in place
Date, Hour, Triage, Shift, Age Band, junk dimensions	–	Type 0 (static)	Fixed by definition

How Type 2 is implemented. The master data carry a **change history**: each change has an effective date. The procedure creates the initial version and then *replays* the events in date order. It closes the current version (`scd_effective_to` = effective date − 1), inserts a new version with `scd_version + 1`, and skips events that do not change the MD5 hash of the tracked attributes. Re-running a batch therefore creates no duplicate versions.

- **Facts and as-was versions.** Facts look up the version valid on the event date (`event_date BETWEEN scd_effective_from AND scd_effective_to`). A spell admitted in February 2022 to ward WHK-MED2 is attributed to its *Isolation (COVID-19)* version.
- **Patients.** The patient dimension uses the same logic in a set-based form, because it has 107,229 patients and 11,686 later versions.
- **Integrity.** Checks R10 and R11 found no gaps, no overlaps and exactly one current version per natural key.

File: `02_sql/06_etl_load_dimensions.sql` (lines 222–242)

```
    -- 2) replay Type 2 change events in date order (one new version per unit per effective date)
    FOR ev IN
        SELECT unit_code, effective_date FROM stg.md_unit_history_clean
        WHERE attribute IN ('specialty', 'bed_complement')
        GROUP BY unit_code, effective_date ORDER BY effective_date, unit_code
    LOOP
```

```
            SELECT * INTO cur FROM dw.dim_care_unit WHERE unit_code = ev.unit_code AND scd_is_current;
            CONTINUE WHEN NOT FOUND OR ev.effective_date <= cur.scd_effective_from;
            SELECT * INTO src FROM stg.md_unit_clean WHERE unit_code = ev.unit_code;
            v_spec := etl.fn_unit_attr_asof(ev.unit_code, 'specialty', ev.effective_date, src.specialty);
            v_beds := etl.fn_unit_attr_asof(ev.unit_code, 'bed_complement', ev.effective_date, src.bed_complement::text);
            v_hash := md5(concat_ws('|', v_spec, v_beds));
            CONTINUE WHEN v_hash = cur.scd_row_hash;
            UPDATE dw.dim_care_unit SET scd_effective_to = ev.effective_date - 1, scd_is_current = false, dw_batch_id = p_batch, dw_updated_at = now()
            WHERE care_unit_key = cur.care_unit_key;
            INSERT INTO dw.dim_care_unit (care_unit_key, unit_code, unit_name, unit_type, facility_key, specialty, bed_complement,
                                          scd_effective_from, scd_is_current, scd_version, scd_row_hash, dw_batch_id)
            VALUES ((SELECT max(care_unit_key) + 1 FROM dw.dim_care_unit), cur.unit_code, cur.unit_name, cur.unit_type, cur.facility_key,
                    v_spec, v_beds::smallint, ev.effective_date, true, cur.scd_version + 1, v_hash, p_batch);
            n_versions := n_versions + 1;
        END LOOP;
```

Table 14. SCD Type 2 history of three wards (the day-2 load created RUN-MED1 v3 and KAT-PAE1 v2)

unit_code	scd_version	specialty	bed_complement	scd_effective_from	scd_effective_to	scd_is_current
KAT-PAE1	1	Paediatric	10	1900-01-01	2026-09-30	false
KAT-PAE1	2	Paediatric	14	2026-10-01	9999-12-31	true
RUN-MED1	1	Medical	24	1900-01-01	2025-04-06	false
RUN-MED1	2	Medical	32	2025-04-07	2026-09-30	false
RUN-MED1	3	Medical	36	2026-10-01	9999-12-31	true
WHK-MED2	1	Medical	24	1900-01-01	2022-01-02	false
WHK-MED2	2	Isolation (COVID-19)	24	2022-01-03	2022-03-13	false
WHK-MED2	3	Medical	24	2022-03-14	9999-12-31	true

Table 15. Business effect of SCD2: spells of ward WHK-MED2 attributed to the version valid on the admission date

unit_code	specialty	first_admission	last_admission	admissions	covid_admissions
WHK-MED2	Medical	2021-12-19	2022-01-01	20	3
WHK-MED2	Isolation (COVID-19)	2022-01-03	2022-03-13	177	8
WHK-MED2	Medical	2022-03-14	2022-06-30	329	9

Table 16. SCD2 on the patient dimension: one patient (pseudonym masked) who changed payer; attendances follow the version

patient	scd_version	payer_code	residence_region_name	scd_effective_from	scd_effective_to	scd_is_current	attendances_on_version
P****8747	1	MOHSS	Khomas	1900-01-01	2022-06-30	false	11
P****8747	2	PMA	Khomas	2022-07-01	9999-12-31	true	69

Table 17. SCD2 summary after the day-2 load

dimension	natural_keys	versions	later_versions	inferred_now
dim_patient	107,229	118,915	11,686	0
dim_care_unit	31	40	9	0

Inferred members (late-arriving dimensions). Some attendances refer to a patient who is not yet in the patient index (EN10): the PAS created the attendance before the registration was exported. The load does not reject these attendances. It creates an **inferred** patient member that carries only the pseudonym and the flag `is_inferred`. The 1 October extract delivered the missing records, and the load **completed the same members in place**, so the facts did not need to be re-pointed (table below).

Table 18. Inferred members: created in the full load, completed by the day-2 load

inferred_in_full_load	completed_by_delta	still_inferred_after_delta
40	40	0

Alternative designs considered

Table 19. Comparison with alternative designs

Design	Strengths	Weaknesses for GBHS	Decision
Kimball constellation with conformed dimensions (chosen)	Fast, understandable star joins; Power BI-native; facts at their natural grain; drill-across through conformed Date, Care Unit, Facility and Condition	Needs disciplined conformance (reference maps, SCD) and more ETL logic	**Chosen.** The questions cross processes (flow × beds × staff × disease), which is exactly what conformed dimensions solve
Inmon 3NF enterprise warehouse + data marts	Single integrated, normalised model; good for many future subject areas	Two modelling layers to build and maintain; users still need marts; slower to deliver for a five-person team in one semester	Rejected: too heavy for the scope; we keep normalised reference data in `ref`
Data Vault 2.0 (hubs, links, satellites)	Auditability, full history, parallel loading, resilient to source change	Many tables; still needs a star layer for Power BI; steep learning curve	Rejected: our `audit` schema, lossless landing and SCD2 give enough auditability
One big flat table per report	Simple for a single dashboard	Mixed grains (attendance, spell, ward-day, week) cannot share one table without double counting; no conformed slicers; semi-additive measures break	Rejected
Single "patient episode" fact merging ED and inpatient	One table for the patient journey	ED-only attendances (≈ 80 %) have no spell; spell milestones arrive days later; null-heavy and with ambiguous grain	Rejected: two facts with a degenerate link (`ed_encounter_id`) and cross-source check R06

Data dictionary

The full data dictionary is **Appendix A**: all 22 tables of the `dw` and `sec` schemas (269 columns), with data type, nullability, key, business definition and lineage. `03_analytics/build_data_dictionary.py` generates it from the PostgreSQL catalogue comments, so it cannot drift from the DDL; it is also delivered as `data_dictionary.md` and `.xlsx`. The same script maps the definitions onto the mart views, and `build_pbip.py` writes them into the semantic model, so **every table, measure and visible column in Power BI has a description** (§3.4).

3.3 ETL, data integration and data quality

Pipeline: source → staging → warehouse

The pipeline is **ELT**:

1. Python lands each file unchanged.
2. Every transformation, quality rule and load runs as a set-based PostgreSQL procedure.
3. Every step writes its row counts and timings to the `audit` schema.

`run_all.py` rebuilds everything from the raw files in about 5 minutes (full load 128 s; day-2 delta 6 s). A batch is the unit of work: it carries an **extract cut-off timestamp**, and every landed row, issue, log line and loaded fact row is stamped with its `batch_id` (lineage).

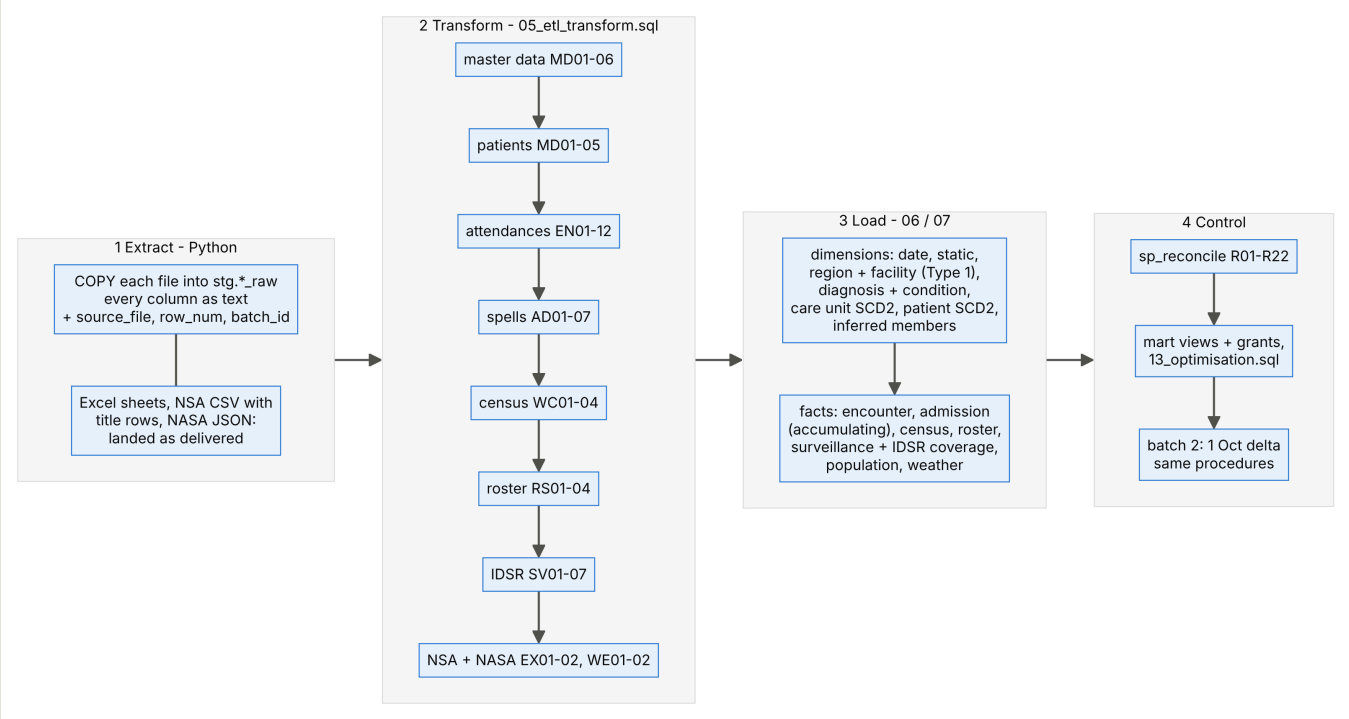


Figure 8. Pipeline steps per batch (02_sql/run_pipeline.py calls etl.sp_run_load, then etl.sp_reconcile) (Mermaid source: `06_report/diagrams/pipeline.mmd`)

Table 20. Scripts of the pipeline (all submitted in 02_sql/ and 01_data/)

Step	Script	What it does
Generate	`01_data/generate_synthetic_data.py`	Writes the synthetic source extracts, the delta files, `generator_manifest.json` and `dq_ground_truth.json` (seed 20261028)
Fetch public data	`01_data/fetch_public_data.py`	Downloads the real NASA POWER files into the same paths; checks the NSA regional totals against the national total
Database	`02_sql/00_create_database.sql`, `01_ddl_schemas_audit.sql`	Database, least-privilege roles, schemas, audit tables, catalogue of 44 DQ rules
Staging DDL	`02_sql/02_ddl_staging_ref.sql`	`stg.*_raw` landing tables, `stg.*_clean` typed tables, `ref.*` mapping tables
Warehouse DDL	`02_sql/03_ddl_warehouse.sql`	13 dimensions, 8 facts, constraints, a `COMMENT` on every column, RLS mapping table
Functions	`02_sql/04_etl_functions.sql`	Parsing (timestamps, epi weeks, numbers), ICD-10 cleaning, holidays, SCD look-ups, batch and step logging
Transform	`02_sql/05_etl_transform.sql`	One procedure per source: applies the DQ rules, logs every issue, merges into `stg.*_clean` by natural key
Load	`02_sql/06_etl_load_dimensions.sql`, `07_etl_load_facts.sql`	Dimensions (Type 1, SCD2, inferred members) and facts (surrogate-key look-ups as-was, accumulating upsert); `sp_run_load` orchestrates
Reconcile	`02_sql/08_reconciliation.sql`	22 checks: row counts, control totals, cross-source, integrity, completeness, plausibility
Publish	`02_sql/09_mart_views.sql`, `10_security_grants.sql`	Power BI views and least-privilege grants
Optimise	`02_sql/13_optimisation.sql`	B-tree and BRIN indexes, aggregate materialised view (§3.7)
Orchestrate	`02_sql/run_pipeline.py`, `run_all.py`	Extract and land, call the procedures, refresh the aggregate, run analytics, evidence and backup

Extract and landing. Every source lands in a `stg.*_raw` table with **all columns as text**, plus `source_file`, `row_num` and `batch_id`. Landing therefore never fails on a bad value, and every warehouse row can be traced back to its file and line.

- Overlapping yearly PAS files are landed in full; the overlap is resolved by EN01.
- The spell export lands every change-data-capture (CDC) image; the latest image is kept by AD01.
- The NSA CSV lands with its title, total and footnote rows; these are rejected by EX02.
- The ten NASA POWER JSON documents land as one row each and are expanded with `jsonb_each_text` in SQL.

Table 21. Files landed by the full load (batch 1)

source_file	target_table	rows_landed
GBHS_Master_Data.xlsx	stg.md_*_raw	221
pas_patient_index.csv.gz	stg.patient_raw	107,332
pas_patient_changes.csv.gz	stg.patient_change_raw	11,661
pas_ed_opd_encounters_2022.csv.gz	stg.encounter_raw	108,944
pas_ed_opd_encounters_2023.csv.gz	stg.encounter_raw	110,997
pas_ed_opd_encounters_2024.csv.gz	stg.encounter_raw	113,090
pas_ed_opd_encounters_2025.csv.gz	stg.encounter_raw	127,109
pas_ed_opd_encounters_2026.csv.gz	stg.encounter_raw	103,069
pas_inpatient_spells.csv.gz	stg.spell_raw	151,774
ward_midnight_census.csv	stg.census_raw	36,232
eroster_shift_summary.csv.gz	stg.roster_raw	97,502
idsr_weekly_returns.csv	stg.idsr_raw	22,410
nsa_census2023_population_by_region.csv	stg.nsa_census_raw	19
POWER_Point_Daily_*.json (10 files)	stg.weather_json_raw	10

Data profiling

Before writing any rule, we profiled the landed raw tables with `03_analytics/data_profiling.py`. The output drove the rule catalogue.

- **PAS attendances (stg.encounter_raw)** - rows: 563,209; distinct keys: 560,243; duplicate rows: 2,966; pct missing triage ed: 0.51; pct missing patient: 0.032; pct missing seen time: 3.86; dayfirst timestamps: 16,477; nonstandard icd codes: 16,598; distinct disposition spellings: 15
- **PAS inpatient spells (stg.spell_raw)** - rows: 151,774; distinct keys: 135,551; duplicate rows: 16,223; pct missing discharge: 10.92; pct missing diagnosis: 0.81; distinct method spellings: 10; distinct ward spellings: 84
- **Patient index (stg.patient_raw)** - rows: 107,332; distinct keys: 107,172; duplicate rows: 160; distinct sex spellings: 4; distinct payer spellings: 18; distinct region spellings: 37; implausible birth years: 53
- **Ward census (stg.census_raw)** - rows: 36,232; distinct keys: 36,123; duplicate rows: 109; blank or negative: 14; max occupied value: 340
- **E-roster (stg.roster_raw)** - rows: 97,502; distinct keys: 97,308; duplicate rows: 194; distinct shift spellings: 4; negative hours: 29
- **IDSR returns (stg.idsr_raw)** - rows: 22,410; distinct condition spellings: 30; weeks in w slash format: 2,187; negative counts: 3; positives gt tests: 4

Door-to-doctor minutes on raw timestamps: median 73, P90 205, P99 424, P99.9 675; minimum -21,038,394 (negative = seen before arrival, EN05) and maximum 1,440 (> 24 h, EN06).

Two counts differ slightly from the rule hits, by design:

- **Day-first timestamps.** Profiling counts *rows* with a day-first timestamp (16,477), which includes duplicate copies. Rule EN02 counts the attendances after duplicates are removed (16,388).
- **Nkurenkuru's `W05/2024` epi weeks.** Profiling counts 2,187 rows, one per condition. Rule SV01 counts the 234 weekly reports they belong to.

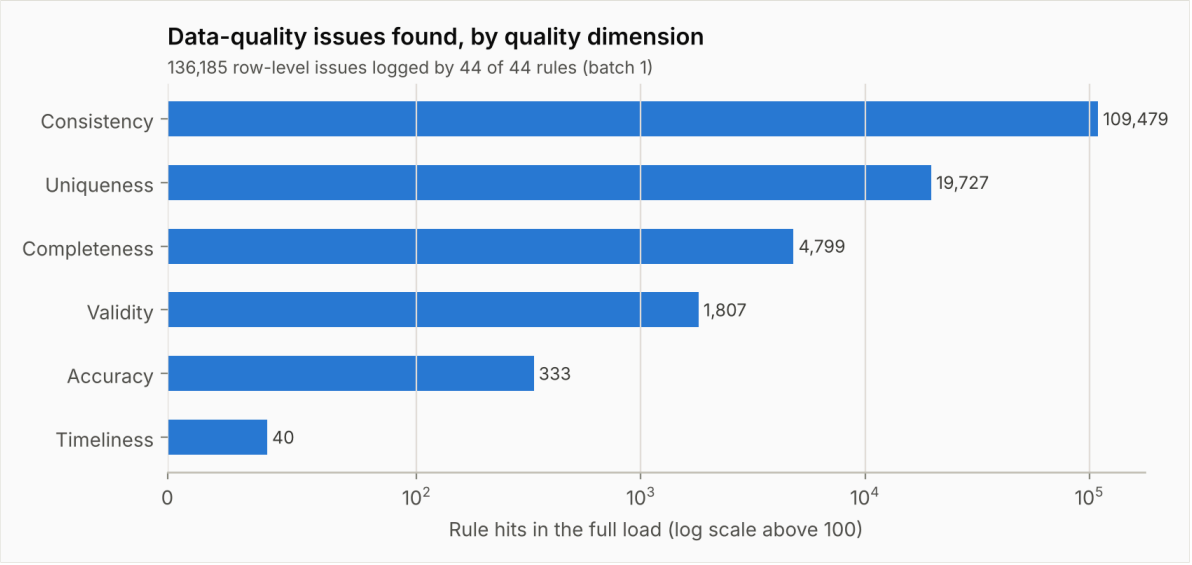


Figure 9. Data-quality issues logged in the full load, by rule and DQ dimension (log scale)

Data-quality rules and how each problem was handled

The catalogue `audit.dq_rule` holds **44 rules**. Each rule has:

- a DQ dimension (completeness, uniqueness, validity, accuracy, consistency, timeliness);
- an **action**: REJECT, CORRECT, MAP, IMPUTE, FLAG, KEEP LATEST or INFER;
- an **owner role** (data steward).

Every hit is written to `audit.dq_issue_log` with the source key, the raw value and the action taken. A problem is therefore never fixed silently. The brief's four problem classes are handled as follows.

Table 22. How the four problem classes required by the brief were handled

Problem class	Examples found (full load)	Handling	Rules
Missing values	2,875 attendances without a triage category; 180 without a patient ID; 1,084 spells not yet coded; 291 ward returns missing; 344 NASA fill values (−999)	**Flag** and load to the *Unknown* member (−1), so totals stay complete but are visible as "Not recorded". **Impute** census from PAS spells (the PAS midnight census is an independent count). Interpolate temperature and humidity linearly between neighbouring observed days; set missing dry-season rainfall (May–September) to 0 and leave rainy-season gaps NULL	EN04, EN11, AD05, WC01, WC04, WE01, WE02
Duplicates and versions	2,966 duplicate attendance rows (overlapping files); 16,223 superseded CDC spell images; 160 superseded patient-index rows; 194 duplicate roster rows; 109 resubmitted ward returns; 73 corrected IDSR reports	**Reject** exact duplicates. **Keep latest** by `last_updated` or report version; the latest image of a spell wins, including across batches (accumulating snapshot)	EN01, AD01, MD04, RS03, WC03, SV02
Outliers and implausible values	Seen-by-doctor before arrival (807); ED stay over 72 h (280); discharge before admission (40); year typos such as 2035 (27); occupied beds keyed ×10 (18); IDSR counts keyed ×10 (8); negative counts or hours; deaths > cases	**Reject** the impossible timestamp but keep the attendance (the KPI is excluded, not the patient). **Correct** keying errors when a second source proves the true value: ×10 is detected only if the value is far from the PAS count *and* one tenth of it is close. **Flag** clinically possible but suspicious values for steward review	EN05, EN06, AD02, AD03, WC02, SV06, SV07, RS02, RS04, SV04, SV05, MD03
Inconsistent formats and codes	16,388 day-first timestamps; 15,904 badly formatted ICD-10 codes; 11,204 facility-code variants; 38,706 code variants (triage, disposition, arrival mode); 30 condition spellings; 84 ward-code spellings; 18 payer spellings; `W05/2024` epi weeks; numbers with space thousands separators	**Correct** formats with strict parsers: a regular-expression timestamp parser accepts only valid calendar dates in two layouts. **Map** spellings through `ref.*` alias tables to conformed codes. Unmapped values go to *Unknown* and are logged	EN02, EN03, EN07, EN09, SV01, SV03, AD04, AD07, MD01, MD02, MD05, RS01, EX01

Table 23. The complete rule catalogue with the hits of the latest batch

rule_code	rule_name	dq_dimension	action	owner_role	hits
AD01	Superseded spell image (CDC)	Uniqueness	KEEP LATEST	Health Information Manager	16,223
AD02	Discharge before admission	Validity	CORRECT	Health Information Manager	40
AD03	Implausible length of stay	Accuracy	CORRECT	Health Information Manager	27
AD04	Ward code variant	Consistency	CORRECT	Health Information Manager	2,711
AD05	Principal diagnosis not coded	Completeness	FLAG	Clinical Coding Lead	1,084
AD06	Open spell older than 60 days	Completeness	FLAG	Nursing Services Manager	25
AD07	Admission method variant	Consistency	MAP	Health Information Manager	2,711
EN01	Duplicate attendance record	Uniqueness	REJECT	Health Information Manager	2,966
EN02	Timestamp in day-first format	Consistency	CORRECT	Health Information Manager	16,388
EN03	Facility code variant	Consistency	CORRECT	Health Information Manager	11,204
EN04	Triage category missing	Completeness	FLAG	ED Nursing Manager	2,875
EN05	Doctor-seen time before arrival	Validity	REJECT	ED Nursing Manager	807
EN06	Implausible length of stay	Accuracy	REJECT	ED Nursing Manager	280
EN07	ICD-10 code formatting	Consistency	CORRECT	Clinical Coding Lead	15,877
EN08	ICD-10 code invalid or missing	Validity	FLAG	Clinical Coding Lead	840
EN09	Code variant (triage, disposition, arrival mode)	Consistency	MAP	Health Information Manager	38,705
EN10	Patient not in patient index (late-arriving)	Timeliness	INFER	Health Information Manager	40
EN11	Patient identifier missing	Completeness	FLAG	Health Information Manager	180
EN12	Arrival date outside the extract window	Validity	REJECT	Health Information Manager	6
EX01	Census number formatting	Validity	CORRECT	External Data Steward	15
EX02	Census aggregate or note row	Consistency	REJECT	External Data Steward	4
MD01	Region name variant	Consistency	MAP	Master Data Steward	2,146
MD02	Sex code variant	Consistency	MAP	Health Information Manager	2,143
MD03	Birth year implausible	Validity	CORRECT	Health Information Manager	53
MD04	Superseded patient-index record	Uniqueness	KEEP LATEST	Health Information Manager	160
MD05	Payer code variant	Consistency	MAP	Revenue Cycle Steward	2,143
MD06	Duplicate or untrimmed ICD-10 code	Uniqueness	CORRECT	Clinical Coding Lead	2
RS01	Shift code variant	Consistency	MAP	Workforce Planning Officer	13,872
RS02	Negative hours	Validity	CORRECT	Workforce Planning Officer	29
RS03	Duplicate roster row	Uniqueness	REJECT	Workforce Planning Officer	194
RS04	Overtime exceeds hours worked	Consistency	FLAG	Workforce Planning Officer	29
SV01	Epi-week format variant	Consistency	CORRECT	Surveillance Officer	234
SV02	Superseded IDSR report version	Uniqueness	KEEP LATEST	Surveillance Officer	73
SV03	Condition name variant	Consistency	MAP	Surveillance Officer	1,305
SV04	Deaths exceed cases	Consistency	FLAG	Surveillance Officer	4
SV05	Positive tests exceed tests done	Consistency	FLAG	Surveillance Officer	4
SV06	Case count keyed x10	Accuracy	CORRECT	Surveillance Officer	8
SV07	Negative case count	Validity	CORRECT	Surveillance Officer	3
WC01	Ward return missing	Completeness	IMPUTE	Nursing Services Manager	291
WC02	Occupied beds keyed x10	Accuracy	CORRECT	Nursing Services Manager	18
WC03	Resubmitted ward return	Uniqueness	KEEP LATEST	Nursing Services Manager	109
WC04	Occupied beds negative or blank	Validity	IMPUTE	Nursing Services Manager	14
WE01	Weather fill value (temperature, humidity)	Completeness	IMPUTE	External Data Steward	265
WE02	Weather fill value (rainfall)	Completeness	IMPUTE	External Data Steward	79

Table 24. Examples from the issue log (one per rule type)

rule_code	source_object	column_name	raw_value	action_taken
AD02	PAS inpatient spells	discharge_datetime	2021-06-07 15:07	Corrected to 2022-06-07 15:07
AD03	PAS inpatient spells	discharge_datetime	2035-02-11 18:32	Corrected to 2025-02-11 18:32
EN02	pas_ed_opd_encounters_2023.csv.gz	arrival_datetime	10/02/2023 20:55	Parsed as DD/MM/YYYY HH:MI
EN03	pas_ed_opd_encounters_2026.csv.gz	facility_code	gbh-kat	Corrected to GBH-KAT
EN05	pas_ed_opd_encounters_2022.csv.gz	seen_datetime	2022-03-08 14:34	Seen time before arrival: timestamp rejected
EN07	pas_ed_opd_encounters_2025.csv.gz	primary_icd10	A09-	Corrected to A09
EN09	pas_ed_opd_encounters_2023.csv.gz	disposition	LWOBS	Mapped to LWBS
MD02	PAS patient index	sex	Female	Mapped to F
MD05	PAS patient index	payer_class	MoHSS	Mapped to MOHSS
SV01	IDSR weekly returns	epi_week	W03/2022	Parsed as 2022-W03
SV03	IDSR weekly returns	condition	Measles (suspected)	Mapped to MEA
SV06	IDSR weekly returns	cases_5plus	110	Divided by 10 (clinical count 11)
WC02	Ward midnight census	occupied_midnight	150	Corrected return (value 15)

The timestamp parser shows the strict style of the rules. It is a pure SQL expression, so PostgreSQL inlines it into the set-based transform (§3.7). It accepts ISO and day-first layouts only when the date is a real calendar date, including leap years. Anything else becomes NULL and is logged, never guessed:

File: `02_sql/04_etl_functions.sql` (lines 96–110)

```
        CREATE OR REPLACE FUNCTION etl.fn_parse_ts(p_txt text)
        RETURNS timestamp LANGUAGE sql IMMUTABLE PARALLEL SAFE AS $$
            SELECT CASE
                WHEN p_txt ~ '^[0-9]{4}-(0[1-9]|1[0-2])-(0[1-9]|1[0-9]|2[0-8])[ T]([01][0-9]|2[0-3]):[0-5][0-9](:[0-5][0-9])?$'
                  OR p_txt ~ '^[0-9]{4}-(0[13-9]|1[0-2])-(29|30)[ T]([01][0-9]|2[0-3]):[0-5][0-9](:[0-5][0-9])?$'
                  OR p_txt ~ '^[0-9]{4}-(0[13578]|1[02])-31[ T]([01][0-9]|2[0-3]):[0-5][0-9](:[0-5][0-9])?$'
                  OR p_txt ~ '^20(0[048]|[2468][048]|[13579][26])-02-29[ T]([01][0-9]|2[0-3]):[0-5][0-9](:[0-5][0-9])?$'
                    THEN p_txt::timestamp
                WHEN p_txt ~ '^(0[1-9]|1[0-9]|2[0-8])/(0[1-9]|1[0-2])/[0-9]{4} ([01][0-9]|2[0-3]):[0-5][0-9]$'
                  OR p_txt ~ '^(29|30)/(0[13-9]|1[0-2])/[0-9]{4} ([01][0-9]|2[0-3]):[0-5][0-9]$'
                  OR p_txt ~ '^31/(0[13578]|1[02])/[0-9]{4} ([01][0-9]|2[0-3]):[0-5][0-9]$'
                  OR p_txt ~ '^29/02/20(0[048]|[2468][048]|[13579][26]) ([01][0-9]|2[0-3]):[0-5][0-9]$'
                    THEN to_timestamp(p_txt, 'DD/MM/YYYY HH24:MI')::timestamp
            END
        $$;
```

The ×10 keying rule for ward returns uses the PAS midnight census as an independent witness, so that a genuinely full ward is not “corrected”:

File: `02_sql/05_etl_transform.sql` (lines 422–432)

```
            -- WC02: a x10 keying error is recognised when the value is far from the PAS census but one tenth of it is close
            -- (genuine overcrowding is close to PAS, so it is never "corrected"); values above 3 x complement are imputed
            ALTER TABLE tmp_cen ADD COLUMN x10 boolean;
            UPDATE tmp_cen SET x10 = occ >= 10 AND abs(occ - pas) > greatest(5, 0.5 * pas)
                                   AND abs(occ / 10.0 - pas) <= greatest(2, 0.2 * pas) AND occ / 10.0 <= 1.6 * complement
            WHERE rn = 1;
            UPDATE tmp_cen SET
                rule = CASE WHEN occ IS NULL OR occ < 0 THEN 'WC04' WHEN x10 OR occ > 3 * complement THEN 'WC02' END,
                occ_final = CASE WHEN occ IS NULL OR occ < 0 THEN pas WHEN x10 THEN round(occ / 10.0)
                                 WHEN occ > 3 * complement THEN pas ELSE occ END,
                src = CASE WHEN occ IS NULL OR occ < 0 THEN 'Imputed from PAS' WHEN x10 THEN 'Corrected return'
```

Validating the rules against ground truth

The generator damaged its own output on purpose and wrote down **which records it damaged** (`01_data/raw/dq_ground_truth.json`). `03_analytics/validate_dq_rules.py` compares that list with the keys the rules logged. This turns data quality from a claim into a measured result:

40 rules with a record-level truth list: 134,964 of 134,992 injected records found (recall 0.9998), 0 false positives (precision 1.0000); 38 rules match their truth list exactly. Count-level checks for the external sources are listed at the end.

rule	defect	injected	flagged	tp	fp	fn	precision	recall
AD01	Superseded spell image (CDC)	16223	16223	16,223	0	0	1.000	1.000
AD02	Discharge before admission	40	40	40	0	0	1.000	1.000
AD03	Implausible length of stay	27	27	27	0	0	1.000	1.000
AD04	Ward code variant	2711	2711	2,711	0	0	1.000	1.000
AD05	Principal diagnosis not coded	1084	1084	1,084	0	0	1.000	1.000
AD06	Open spell older than 60 days	25	25	25	0	0	1.000	1.000
AD07	Admission method variant	2711	2711	2,711	0	0	1.000	1.000
EN01	Duplicate attendance record	2966	2966	2,966	0	0	1.000	1.000
EN02	Timestamp in day-first format	16388	16388	16,388	0	0	1.000	1.000
EN03	Facility code variant	11204	11204	11,204	0	0	1.000	1.000
EN04	Triage category missing	2875	2875	2,875	0	0	1.000	1.000
EN05	Doctor-seen time before arrival	807	807	807	0	0	1.000	1.000
EN06	Implausible length of stay	280	280	280	0	0	1.000	1.000
EN07	ICD-10 code formatting	15904	15877	15,877	0	27	1.000	0.998
EN08	ICD-10 code invalid or missing	840	840	840	0	0	1.000	1.000
EN09	Code variant (triage, disposition, arrival mode)	37848	37847	37,847	0	1	1.000	1.000

rule	defect	injected	flagged	tp	fp	fn	precision	recall
EN10	Patient not in patient index (late-arriving)	40	40	40	0	0	1.000	1.000
EN11	Patient identifier missing	180	180	180	0	0	1.000	1.000
EN12	Arrival date outside the extract window	6	6	6	0	0	1.000	1.000
EX01	Census numbers with thousands separators	15	15					
MD01	Region name variant	2145	2145	2,145	0	0	1.000	1.000
MD02	Sex code variant	2143	2143	2,143	0	0	1.000	1.000
MD03	Birth year implausible	53	53	53	0	0	1.000	1.000
MD04	Superseded patient-index record	160	160	160	0	0	1.000	1.000
MD05	Payer code variant	2143	2143	2,143	0	0	1.000	1.000
MD06	Duplicate or untrimmed ICD-10 code	2	2	2	0	0	1.000	1.000
RS01	Shift code variant	13872	13872	13,872	0	0	1.000	1.000
RS02	Negative hours	29	29	29	0	0	1.000	1.000
RS03	Duplicate roster row	194	194	194	0	0	1.000	1.000
RS04	Overtime exceeds hours worked	29	29	29	0	0	1.000	1.000
SV01	Epi-week format variant	234	234	234	0	0	1.000	1.000
SV02	Superseded IDSR report version	73	73	73	0	0	1.000	1.000
SV03	Condition name variant	1305	1305	1,305	0	0	1.000	1.000
SV04	Deaths exceed cases	4	4	4	0	0	1.000	1.000
SV05	Positive tests exceed tests done	4	4	4	0	0	1.000	1.000
SV06	Case count keyed x10	8	8	8	0	0	1.000	1.000
SV07	Negative case count	3	3	3	0	0	1.000	1.000
WC01	Ward return missing	291	291	291	0	0	1.000	1.000
WC02	Occupied beds keyed x10	18	18	18	0	0	1.000	1.000
WC03	Resubmitted ward return	109	109	109	0	0	1.000	1.000
WC04	Occupied beds negative or blank	14	14	14	0	0	1.000	1.000
WE01+WE02	NASA POWER -999 fill values (all parameters)	344	344					

All 28 “missed” records were in fact caught, by another rule:

- **27 EN07 records.** The generator first mis-formatted these ICD-10 codes and then overwrote them with an invalid code (`XYZ`, `000`, `R50.99`), which rule EN08 caught.
- **1 EN09 record.** It also had an arrival date in 2062, so EN12 rejected the whole row before the code mapping ran.

Every damaged record was therefore flagged, and no clean record was flagged. Two rules were rewritten after this validation exposed weaknesses in the first version:

- **WC02.** A plain “more than 3 × the bed complement” test missed ×10 errors on small counts (2 → 20 in a 24-bed ward). We added the PAS cross-check.
- **SV04.** “Deaths > cases” fired on legitimate deaths reported in a later week than the case. We now allow a margin of 3, and the injected defects use +5.

Conformed keys and integration

- **Conformed codes.** Each source’s own spelling of the same thing is mapped through `ref.code_map`, `ref.region_alias`, `ref.condition_alias` and `ref.payer_alias` to the master-data code. Examples: `gbh-kat` → `GBH-KAT`; `LWOBS` → `LWBS`; `Measles (suspected)` → `MEA`; `MoHSS` → `MOHSS`; `Caprivi` → `NA-CA` (Zambezi). The NSA census names its regions differently from the master data; the same alias table maps them to ISO codes.
- **Surrogate-key look-ups as-was.** Facts take the surrogate key of the dimension version **valid on the event date** (`BETWEEN scd_effective_from AND scd_effective_to`). Keys that cannot be resolved go to the special members −1 and −2, never NULL.
- **Cross-source conformance.** Condition links ICD-10 diagnoses in PAS to IDSR returns, and Care Unit links attendances, spells, the census and the roster. Facility links IDSR, weather and catchment population, and Region links the NSA census.
- **Durable patient key.** It survives SCD2 versions and inferred members (§3.2).

Reconciliation

`etl.sp_reconcile` runs **22 checks** after every batch and stores them in `audit.recon_result`. Each check gives PASS, WARN (explained difference within tolerance) or FAIL (a difference that stops publication). The checks cover:

- **Row counts:** landed = accepted + rejected; staging = fact.
- **Control totals:** door-to-doctor minutes, length-of-stay days, nursing hours, the census population.
- **Cross-source checks:** ED admissions against spells; ward-return bed-days against PAS spells; IDSR malaria against PAS diagnoses B50–B54.
- **SCD2 and referential integrity**, **completeness** and **plausibility**.

Table 25. Reconciliation of the latest batch (day-2 load, 1 October 2026): 20 PASS, 2 WARN, 0 FAIL

check_code	check_name	check_type	source_value	target_value	status	explanation
R01	Attendance rows landed = accepted + rejected (this batch)	ROW_COUNT	320	320	PASS	320 accepted, 0 rejected (EN01 duplicates, EN12 out-of-window, unknown facility)
R02	Attendances: clean staging = fact_encounter	ROW_COUNT	560,557	560,557	PASS	
R03	Control total: door-to-doctor minutes, staging vs fact	CONTROL_TOTAL	51,932,017	51,932,017	PASS	
R04	Inpatient spells: distinct spells in staging = fact_admission	ROW_COUNT	135,629	135,629	PASS	16223 superseded CDC images rejected over all batches (AD01)

check_code	check_name	check_type	source_value	target_value	status	explanation
R05	Control total: length-of-stay days of discharged spells, staging vs fact	CONTROL_TOTAL	551,247	551,247	PASS	
R06	Cross-source: ED attendances admitted = spells linked to them	CROSS_SOURCE	72,144	72,144	PASS	Differences = admissions recorded in the ED but missing (or not yet exported) in the inpatient spell extract
R07	Cross-source: occupied bed-days, ward returns vs PAS spells	CROSS_SOURCE	552,033	569,275	WARN	PAS exceeds the ward returns by 17242 bed-days; 25 spells are still open > 60 days in PAS (AD06, probable missing discharges)
R08	Cross-source: malaria cases, IDSR returns vs PAS attendances (B50-B54)	CROSS_SOURCE	43,695	43,643	PASS	IDSR / PAS = 99.9%: transcription differences on the weekly forms (rows corrected under SV06 are already fixed)
R09	Control total: 14 regions = NSA national total (2023 census)	CONTROL_TOTAL	3,022,401	3,022,401	PASS	
R10	SCD2 integrity dim_patient: gaps/overlaps + keys without exactly one current version	INTEGRITY	0	0	PASS	118915 versions for 107229 patients
R11	SCD2 integrity dim_care_unit: gaps/overlaps + keys without exactly one current version	INTEGRITY	0	0	PASS	40 versions for 31 units
R12	Referential: attendances on patient -1 = unidentified patients logged (EN11)	REFERENTIAL	180	180	PASS	
R13	Referential: attendances on diagnosis -1 = invalid/missing codes logged (EN08)	REFERENTIAL	840	840	PASS	
R14	Referential: facts without a care unit or (spells) without a patient	REFERENTIAL	0	0	PASS	
R15	Completeness: dim_date has every day 2021-01-01..2027-12-31	COMPLETENESS	2,556	2,556	PASS	
R16	Completeness: one census row per ward per day	COMPLETENESS	36,435	36,435	PASS	305 ward-days imputed from PAS (WC01, WC02, WC04)
R17	Completeness: weather one row per facility per day, no temperature gaps	COMPLETENESS	18,560	18,560	PASS	253 facility-days imputed (WE01/WE02)
R18	Completeness: IDSR coverage fact = facilities x due epi weeks	COMPLETENESS	2,470	2,470	PASS	completeness 97.9%, timeliness 87.2%
R19	Integrity: spells discharged after admission, milestone keys consistent	INTEGRITY	0	0	PASS	344 spells still in hospital at the cut-off
R20	Control total: nursing hours worked, staging vs fact	CONTROL_TOTAL	4,026,374	4,026,374	PASS	
R21	Mart view v_fact_encounter = dw.fact_encounter	ROW_COUNT	560,557	560,557	PASS	
R22	Plausibility: ward-days above 200% occupancy (WARN up to 0.5% of ward-days)	PLAUSIBILITY	0	3	WARN	6002 ward-days above 100% (overflow beds) are plausible and kept

Both WARN results are real, explained business findings, not pipeline errors:

- **R07.** The PAS shows 3 % more bed-days than the ward returns. The 25 spells still “open” for more than 60 days account for 17,562 PAS bed-days, which is more than the whole 17,242-day gap. They are probable missing discharges (AD06, flagged to the Nursing Services Manager).
- **R22.** Three ward-days exceed 200 % occupancy. 6,002 ward-days above 100 % are plausible overflow, using trolleys and beds in corridors.

Load evidence

Table 26. Batches (audit.etl_batch)

batch_id	batch_type	description	extract_cutoff	seconds	status
1	FULL	Full rebuild from raw extracts (cut-off 2026-09-30)	2026-09-30 23:59	128.4	SUCCEEDED
2	INCREMENTAL	Day-2 delta 2026-10-01 (attendances, spells, census, roster, IDSR, master data)	2026-10-01 23:59	5.5	SUCCEEDED

Table 27. Step log of the full load (audit.etl_step_log): rows in, rows out, rejected and time per step

step_name	rows_in	rows_out	rows_rejected	ms	note
transform master data	120	119	0	9	
load dim_date		2,558	0	107	
load static dimensions		92	0	5	
load dim_region / dim_facility (Type 1)	10	11	0	2	0 facility rows updated in place (Type 1)
load dim_condition / dim_diagnosis (snowflake)	64	65	0	3	
load dim_care_unit (SCD2)	31	39	0	6	31 initial rows, 7 new versions, 0 Type 1 updates
transform patient index (MD01-MD05)	107,332	107,172	160	2,160	11661 change-log rows
load dim_patient (SCD2)	107,172	118,834	0	8,246	118833 initial rows, 0 new versions, 0 inferred members completed
transform ED/clinic attendances (EN01-EN12)	563,209	560,237	2,972	29,103	
transform inpatient spells (AD01-AD07)	151,774	135,551	16,223	5,859	135551 distinct spells in batch
inferred patient members (EN10)	40	40	0	207	
transform ward census (WC01-WC04)	36,232	36,414	109	2,273	291 ward-days imputed from PAS
transform e-roster (RS01-RS04)	97,502	97,308	194	2,682	
transform IDSR weekly returns (SV01-SV07)	22,410	21,753	657	2,026	
transform external sources (EX01-EX02, WE01-WE02)	19	18,574	0	289	
load fact_encounter	560,237	560,237	0	30,499	
load fact_admission (accumulating snapshot)	135,551	135,551	0	13,891	135551 new spells, 0 existing spells updated (milestones)
load fact_ward_census_daily (periodic snapshot)	36,414	36,414	0	648	
load fact_staff_roster	97,308	97,308	0	1,780	

step_name	rows_in	rows_out	rows_rejected	ms	note
load fact_surveillance_weekly + fact_idsr_report (coverage)	21,753	21,753	0	314	2470 expected reports in coverage fact
load fact_population (external, NSA)	14	98	0	2	national intercensal growth 3.027% per year
load fact_weather_daily (external, NASA POWER)	18,560	18,560	0	211	

Table 28. Step log of the day-2 incremental load (1 October 2026): SCD2 versions, milestones, late reports

step_name	rows_in	rows_out	note
transform master data	3	119	
load static dimensions		92	
load dim_region / dim_facility (Type 1)	10	11	1 facility rows updated in place (Type 1)
load dim_condition / dim_diagnosis (snowflake)	64	65	
load dim_care_unit (SCD2)	31	41	0 initial rows, 2 new versions, 0 Type 1 updates
transform patient index (MD01-MD05)	82	82	25 change-log rows
load dim_patient (SCD2)	82	118,916	17 initial rows, 25 new versions, 40 inferred members completed
transform ED/clinic attendances (EN01-EN12)	320	320	
transform inpatient spells (AD01-AD07)	165	165	165 distinct spells in batch
transform ward census (WC01-WC04)	21	21	0 ward-days imputed from PAS
transform e-roster (RS01-RS04)	57	57	
transform IDSR weekly returns (SV01-SV07)	18	18	
load fact_encounter	320	320	
load fact_admission (accumulating snapshot)	165	165	78 new spells, 87 existing spells updated (milestones)
load fact_ward_census_daily (periodic snapshot)	21	21	
load fact_staff_roster	57	57	
load fact_surveillance_weekly + fact_idsr_report (coverage)	18	18	2470 expected reports in coverage fact
load fact_population (external, NSA)	14	98	national intercensal growth 3.027% per year

The day-2 load exercised every incremental path with the **same procedures** as the full load:

- 320 new attendances;
- 165 spell images, of which 78 were new spells and 87 were milestone updates of open spells;
- 2 new care-unit versions: RUN-MED1 grew from 32 to 36 beds and KAT-PAE1 from 10 to 14;
- 25 new patient versions, and 40 inferred patient members completed in place;
- 1 Type 1 facility-name correction;
- 18 late IDSR returns, which moved two reports from “missing” to “received late”.

Re-running a batch is idempotent: it uses `ON CONFLICT` keys, hash-compared SCD2 and “latest version wins” merges.

Table 29. Warehouse tables after both loads (row counts and size)

table_name	row_count	total_size
dw.fact_encounter	560,557	118 MB
dw.fact_admission	135,629	54 MB
dw.fact_staff_roster	97,365	14 MB
dw.fact_ward_census_daily	36,435	4368 kB
dw.fact_surveillance_weekly	21,771	3232 kB
dw.fact_weather_daily	18,560	1872 kB
dw.fact_idsr_report	2,470	568 kB
dw.fact_population	98	64 kB
dw.dim_patient	118,916	65 MB
dw.dim_date	2,558	1216 kB
dw.dim_diagnosis	65	88 kB
dw.dim_encounter_profile	56	40 kB
dw.dim_care_unit	41	56 kB
dw.dim_admission_profile	36	40 kB
dw.dim_hour	24	24 kB
dw.dim_region	15	40 kB
dw.dim_condition	11	40 kB
dw.dim_facility	11	40 kB
dw.dim_age_band	7	40 kB
dw.dim_triage	7	40 kB
dw.dim_shift	2	40 kB

Table 30. Sample rows of fact_encounter (Rundu ED, 15 September 2026) with their dimension labels

arrival_date_key	hour_label	unit_code	triage_code	icd10_code	disposition_code	age_band	door_to_doctor_min	ed_los_min	seen_within_target
20260915	00:00	RUN-ED	GREEN	B24	LWBS	65+		144	
20260915	01:00	RUN-ED	YELLOW	T14.9	DIS	5-14	195	409	0
20260915	03:00	RUN-ED	GREEN	S01.9	DIS	65+	54	102	1
20260915	06:00	RUN-ED	GREEN	J45.9	DIS	15-24	211	275	1
20260915	09:00	RUN-ED	YELLOW	J06.9	DIS	0-4	21	119	1
20260915	09:00	RUN-ED	YELLOW	J18.9	ADM	25-44	30	259	1
20260915	10:00	RUN-ED	YELLOW	A09	DIS	45-64	19	123	1
20260915	10:00	RUN-ED	GREEN	T30.0	DIS	0-4	182	210	1

3.4 Semantic model and DAX (Power BI)

The semantic model is delivered as a Power BI Project (`04_powerbi/GBHS_Health.pbip`), with TMDL for the model and PBIR for the report (Microsoft, 2024d, 2024e). It is generated from a declarative specification (`model_spec.py`) by `build_pbip.py`, which validates it before release:

- Microsoft's Tabular Object Model (TOM) deserialises the model in Power BI mode and writes it back to TMDL. This resolves every table, column, relationship, sort-by column, hierarchy level, role and the refresh policy.
- Every report file is validated against Microsoft's PBIR JSON schemas.
- Every visual field reference is checked against the model.
- A static check runs on all 113 DAX expressions.

`04_powerbi/BUILD_GUIDE.md` takes the team from the PBIP to the `.pbix` and `.pbit` in Power BI Desktop. It also gives the values each card must show, computed independently in SQL (`expected_values.py`).

Storage mode

Table 31. Storage-mode decision

Criterion	Import (chosen)	DirectQuery	Composite (Import + DirectQuery)
Data volume	About 0.9 M fact rows compress to roughly 12.5 MB (§3.7): far below Pro limits	No size limit	Same as Import for imported tables
Visual response	In-memory VertiPaq: milliseconds	Every visual sends SQL through the gateway to PostgreSQL: seconds, and load on the operational database	Mixed
DAX	Full: time intelligence, iterators, semi-additive patterns	Restricted functions and performance traps (iterators, time intelligence)	Full on Import tables
Freshness needed	Daily for management; IDSR weekly (Tuesday)	Real time (not needed)	Real time for the detail only
Security	RLS and OLS in the model	RLS in the model plus database permissions	Both
Cost and operations	Scheduled refresh; **incremental refresh** keeps it to the last 10 days	Gateway capacity, database tuning	Most complex

Decision: Import for all tables, with an incremental-refresh policy on `Attendance` (Microsoft, 2024c). The policy archives 5 years and refreshes the last 10 days. Its `RangeStart` and `RangeEnd` filter on the integer `arrival_date_key`, which folds into the SQL `WHERE` clause. A composite model, with a DirectQuery detail table under Import aggregations, is the scale-out path if volumes grow tenfold (§3.7).

Model structure and relationships

The model has **32 tables**:

- 13 dimensions;
- 9 fact tables: the 8 warehouse facts, plus the derived `Catchment Population` (NSA population × catchment share);
- 5 analytics outputs: malaria forecast, outbreak alerts, readmission deciles and drivers, ED wait drivers;
- 2 governance tables: DQ summary and reconciliation;
- the hidden RLS mapping table;
- `_Measures`;
- the `Metric Selector` field parameter.

The relationship rules follow Kimball's star pattern in Power BI:

- **Many-to-one, single direction.** The dimension (one side) filters the fact (many side). There are no bidirectional or many-to-many relationships, so filter paths are unambiguous and fast.
- **One active path** between any dimension and any fact. The snowflakes Care Unit → Facility → Region and Diagnosis → Condition give a single path from a facility or condition to every fact.
- **Role-playing date.** `Admission` has an active relationship on the admission date and an **inactive** one on the discharge date. Discharge-based measures (ALOS, mortality, readmission) activate it with `USERELATIONSHIP`.
- **Disconnected tables** (model outputs, DQ, reconciliation, security mapping) have no relationships, so they cannot filter facts by accident.

Table 32. The 38 relationships of the semantic model (all many-to-one, single cross-filter direction)

#	From (many side)	To (one side)	Cardinality	Cross-filter	Active
1	Attendance[Arrival Date Key]	Date[Date Key]	many-to-one	single	yes
2	Attendance[Arrival Hour Key]	Hour[Hour Key]	many-to-one	single	yes
3	Attendance[Care Unit Key]	Care Unit[Care Unit Key]	many-to-one	single	yes
4	Attendance[Patient Key]	Patient[Patient Key]	many-to-one	single	yes
5	Attendance[Diagnosis Key]	Diagnosis[Diagnosis Key]	many-to-one	single	yes
6	Attendance[Triage Key]	Triage[Triage Key]	many-to-one	single	yes
7	Attendance[Encounter Profile Key]	Encounter Profile[Encounter Profile Key]	many-to-one	single	yes
8	Attendance[Age Band Key]	Age Band[Age Band Key]	many-to-one	single	yes
9	Admission[Admission Date Key]	Date[Date Key]	many-to-one	single	yes
10	Admission[Discharge Date Key]	Date[Date Key]	many-to-one	single	no (USERELATIONSHIP)
11	Admission[Care Unit Key]	Care Unit[Care Unit Key]	many-to-one	single	yes
12	Admission[Patient Key]	Patient[Patient Key]	many-to-one	single	yes
13	Admission[Diagnosis Key]	Diagnosis[Diagnosis Key]	many-to-one	single	yes
14	Admission[Admission Profile Key]	Admission Profile[Admission Profile Key]	many-to-one	single	yes
15	Admission[Age Band Key]	Age Band[Age Band Key]	many-to-one	single	yes

#	From (many side)	To (one side)	Cardinality	Cross-filter	Active
16	Ward Census[Census Date Key]	Date[Date Key]	many-to-one	single	yes
17	Ward Census[Care Unit Key]	Care Unit[Care Unit Key]	many-to-one	single	yes
18	Staff Roster[Roster Date Key]	Date[Date Key]	many-to-one	single	yes
19	Staff Roster[Care Unit Key]	Care Unit[Care Unit Key]	many-to-one	single	yes
20	Staff Roster[Shift Key]	Shift[Shift Key]	many-to-one	single	yes
21	Surveillance[Epi Week Date Key]	Date[Date Key]	many-to-one	single	yes
22	Surveillance[Facility Key]	Facility[Facility Key]	many-to-one	single	yes
23	Surveillance[Condition Key]	Condition[Condition Key]	many-to-one	single	yes
24	IDSR Report[Epi Week Date Key]	Date[Date Key]	many-to-one	single	yes
25	IDSR Report[Facility Key]	Facility[Facility Key]	many-to-one	single	yes
26	Regional Population[Year Date Key]	Date[Date Key]	many-to-one	single	yes
27	Regional Population[Region Key]	Region[Region Key]	many-to-one	single	yes
28	Catchment Population[Year Date Key]	Date[Date Key]	many-to-one	single	yes
29	Catchment Population[Facility Key]	Facility[Facility Key]	many-to-one	single	yes
30	Weather[Weather Date Key]	Date[Date Key]	many-to-one	single	yes
31	Weather[Facility Key]	Facility[Facility Key]	many-to-one	single	yes
32	Malaria Forecast[Week Date Key]	Date[Date Key]	many-to-one	single	yes
33	Outbreak Alerts[Week Date Key]	Date[Date Key]	many-to-one	single	yes
34	Outbreak Alerts[Facility Key]	Facility[Facility Key]	many-to-one	single	yes
35	Outbreak Alerts[Condition Key]	Condition[Condition Key]	many-to-one	single	yes
36	Care Unit[Facility Key]	Facility[Facility Key]	many-to-one	single	yes
37	Facility[Region Key]	Region[Region Key]	many-to-one	single	yes
38	Diagnosis[Condition Key]	Condition[Condition Key]	many-to-one	single	yes

Usability: hidden columns, folders, sort order, hierarchies and descriptions

- **Hidden technical columns.** 142 of the 258 columns are hidden. These are all surrogate and foreign keys, sort keys and the raw numeric fact columns. `discourageImplicitMeasures` is on, so users aggregate only through the 108 curated measures and never through implicit sums of semi-additive columns.
- **Display folders.** Measures sit in eight folders: *0 Targets & Labels*, *1 Patient Flow*, *2 Inpatient Care*, *3 Resource Utilisation*, *4 Disease Surveillance*, *5 Time Intelligence*, *6 Advanced Analytics*, *7 Data Quality & Security*. Date attributes are foldered as *Calendar*, *Epidemiological week (IDSR)*, *Fiscal (Apr–Mar)* and *Seasons*. SCD history columns sit in *SCD history*.
- **Sort-by-column on 17 labels.** For example: Day Name → Day of Week; Year Month → Year Month Key; Epi Week Label → Epi Week Key; Fiscal Year → Fiscal Year Start; Triage Category → Priority (RED first, not alphabetical); Age Band → Age Band Key; ICD-10 Chapter → Chapter Number (I, II, … XXII, not alphabetical); Hour Band → Hour Band Sort; Metric → Metric Order.
- **Hierarchies.**
  - Date: *Calendar* (Year › Quarter › Month › Date), *Epidemiological* (Epi Year › Epi Week › Date) and *Fiscal*;
  - Region: *Geography* (Zone › Region);
  - Care Unit: *Unit* (Unit Type › Specialty › Ward);
  - Diagnosis: *ICD-10* (Chapter › Block › Diagnosis).
- **Metadata.** `Date` is marked as the date table (key column `Date`) and auto date/time is off.
  - Geographic columns carry data categories (City, StateOrProvince, Latitude, Longitude).
  - Every table, every visible column and every measure has a **description**. Column descriptions come from the warehouse data dictionary (§3.2), so a hover in the field list shows the business definition and source.
- **Field parameter.** `Metric Selector` lets a user switch one chart between nine KPIs: ED attendances, seen within target, median door-to-doctor, LWBS, bed occupancy, ALOS, readmissions, NHPPD and malaria cases.

DAX measures

The model has **108 measures** (`04_powerbi/measures.dax`; all are listed in Appendix B). The brief asks for at least 10, including time intelligence and `CALCULATE` with filter modification. The following measures demonstrate the required patterns; each is used on a report page.

Table 33. Brief requirements and the measures that meet them

Requirement / pattern	Measures (examples)	Page
≥ 10 measures	108 measures in 8 folders	All
`CALCULATE` with filter modification: `KEEPFILTERS`, `REMOVEFILTERS`, `USERELATIONSHIP`, explicit filter arguments	ED Attendances, LWBS Rate %, NHPPD, Discharges, ALOS (days), Readmission Rate 30d %, Occupied Beds (Latest Day), Catchment Person-Years, Census Population 2023, Malaria Cases	1–4
Time intelligence: `TOTALYTD`, `SAMEPERIODLASTYEAR`, `DATESINPERIOD`, `DATEADD`, `EDATE`	ED Attendances YTD, ED Attendances YTD PY (like-for-like), ED Attendances YTD YoY %, Seen Within Target % PY, Bed Occupancy % Rolling 12M, Malaria Cases Same Week Last Year, Malaria Cases 4-Week Average, Rainfall 4-8 Weeks Earlier (mm)	1, 3
Custom calendar (ISO epi year)	Malaria Cases Epi-YTD, Malaria Cases Epi-YTD PY, Malaria Cases Epi-YTD YoY %	3
Semi-additive (snapshot)	Average Daily Census, Occupied Beds (Latest Day), Bed Occupancy % (Latest Day), Catchment Population	2, 3
Drill-across two facts	NHPPD (roster ÷ census), ED Arrivals per Doctor Shift (attendances ÷ roster)	2
Statistical	Median Door-to-Doctor (min), P90 Door-to-Doctor (min) (`MEDIAN`, `PERCENTILE.INC`)	1, 2
Privacy	Notified Cases (Suppressed), Notified Cases Label (small-cell suppression < 5)	3
Security test	Current User, Visible Facilities	4

Flow KPIs: `CALCULATE` with `KEEPFILTERS`. `KEEPFILTERS` intersects the department filter with whatever the user has selected. A plain filter argument would override the selection, so that selecting *Clinic outpatient* would still show ED values.

File: `04_powerbi/measures.dax` (generated from `model_spec.py`)

```
-- Display folder: 1 Patient Flow | Format: #,0
-- All ED and clinic attendances (one row per attendance).
Attendances =
    SUM ( 'Attendance'[Encounter Count] )

-- Display folder: 1 Patient Flow | Format: #,0
-- Emergency-department attendances (CALCULATE with filter modification on the department).
ED Attendances =
    CALCULATE ( [Attendances], KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" ) )

-- Display folder: 1 Patient Flow | Format: 0.0%
-- Share of triaged ED patients seen by a doctor within the SATS compliance time of their category (RED 5, ORANGE 10, YELLOW 60, GREEN 240 minutes).
Seen Within Target % =
    DIVIDE ( SUM ( 'Attendance'[Seen Within Target] ), COUNT ( 'Attendance'[Seen Within Target] ) )

-- Display folder: 1 Patient Flow | Format: #,0
-- Median minutes from ED arrival to first doctor assessment (patients seen).
Median Door-to-Doctor (min) =
    CALCULATE ( MEDIAN ( 'Attendance'[Door to Doctor (min)] ), KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" ) )

-- Display folder: 1 Patient Flow | Format: 0.0%
-- Share of ED attendances that left without being seen.
LWBS Rate % =
    CALCULATE ( DIVIDE ( SUM ( 'Attendance'[Is LWBS] ), SUM ( 'Attendance'[Encounter Count] ) ), KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED"
) )
```

Role-playing date: discharge measures. These are counted by **discharge** date by activating the inactive relationship:

File: `04_powerbi/measures.dax` (generated from `model_spec.py`)

```
-- Display folder: 2 Inpatient Care | Format: #,0
-- Spells discharged, by DISCHARGE date (activates the inactive role-playing relationship).
Discharges =
    CALCULATE ( SUM ( 'Admission'[Is Discharged] ), USERELATIONSHIP ( 'Admission'[Discharge Date Key], 'Date'[Date Key] ) )

-- Display folder: 2 Inpatient Care | Format: #,0.00
-- Average length of stay (midnights) of spells discharged in the period.
ALOS (days) =
    CALCULATE ( AVERAGE ( 'Admission'[LOS (days)] ), USERELATIONSHIP ( 'Admission'[Discharge Date Key], 'Date'[Date Key] ), KEEPFILTERS ( 'Admission'[Is
Discharged] = 1 ) )

-- Display folder: 2 Inpatient Care | Format: 0.0%
-- Eligible spells (alive home/AMA, not maternity, 30 days follow-up) followed by an emergency readmission within 30 days, by discharge date of the index
spell.
Readmission Rate 30d % =
    CALCULATE ( DIVIDE ( SUM ( 'Admission'[Readmitted Within 30d] ), SUM ( 'Admission'[Readmission Eligible] ) ), USERELATIONSHIP ( 'Admission'[Discharge
Date Key], 'Date'[Date Key] ) )
```

Semi-additive census. - *Average Daily Census* sums occupied beds across wards and **averages across days**. - *Occupied Beds (Latest Day)* takes the last census date in the period. `REMOVEFILTERS` on the ward hierarchy makes every ward use the same "latest" date. - *NHPPD* is a drill-across of two facts through the conformed Date and Care Unit dimensions.

File: `04_powerbi/measures.dax` (generated from `model_spec.py`)

```
-- Display folder: 3 Resource Utilisation | Format: #,0.0
-- Average number of occupied beds per day (semi-additive: summed over wards, averaged over days).
Average Daily Census =
    AVERAGEX ( VALUES ( 'Date'[Date] ), CALCULATE ( SUM ( 'Ward Census'[Occupied Beds] ) ) )

-- Display folder: 3 Resource Utilisation | Format: #,0
-- Occupied beds on the last census date in the period, across the selected wards (semi-additive closing balance).
Occupied Beds (Latest Day) =
    VAR _d =
        CALCULATE (
            MAX ( 'Ward Census'[Census Date Key] ),
            REMOVEFILTERS ( 'Care Unit' ), REMOVEFILTERS ( 'Facility' ), REMOVEFILTERS ( 'Region' )
        )
    RETURN
        CALCULATE ( SUM ( 'Ward Census'[Occupied Beds] ), 'Ward Census'[Census Date Key] = _d )

-- Display folder: 3 Resource Utilisation | Format: #,0.00
-- Nursing hours per patient day on inpatient wards (drill-across: e-roster x ward census through Date and Care Unit).
NHPPD =
    DIVIDE ( CALCULATE ( [Nurse Hours], KEEPFILTERS ( 'Care Unit'[Unit Type] = "Inpatient ward" ) ), [Patient Days] )
```

Time intelligence. *Year-to-date* stops at the last data date, and the prior-year window is cut at the same day (`EDATE(…, -12)`, like-for-like). Otherwise 2026 year-to-date (data until 1 October) would be compared with the whole of 2025. Both *rolling 12 months* and *same week last year* use the standard date functions on the marked date table.

File: `04_powerbi/measures.dax` (generated from `model_spec.py`)

```
-- Display folder: 5 Time Intelligence | Format: d mmm yyyy
-- Latest attendance date in the model (ignores all filters except RLS).
Last Data Date =
    VAR _k =
        CALCULATE ( MAX ( 'Attendance'[Arrival Date Key] ), REMOVEFILTERS () )
    RETURN
        IF ( NOT ISBLANK ( _k ), DATE ( INT ( _k / 10000 ), MOD ( INT ( _k / 100 ), 100 ), MOD ( _k, 100 ) ) )

-- Display folder: 5 Time Intelligence | Format: #,0
-- Year-to-date ED attendances; blank for periods after the last data date.
ED Attendances YTD =
    IF ( MIN ( 'Date'[Date] ) <= [Last Data Date], [ED Attendances YTD Raw] )

-- Display folder: 5 Time Intelligence | Format: #,0
-- Same year-to-date window one year earlier, cut at the same day (like-for-like).
```

```
ED Attendances YTD PY =
    VAR _cutoffPY =
        EDATE ( [Last Data Date], -12 )
    RETURN
        CALCULATE (
            [ED Attendances YTD Raw],
            SAMEPERIODLASTYEAR ( 'Date'[Date] ),
            KEEPFILTERS ( 'Date'[Date] <= _cutoffPY )
        )

-- Display folder: 5 Time Intelligence | Format: +0.0%;-0.0%;0.0%
-- Year-to-date growth versus the like-for-like prior-year window.
ED Attendances YTD YoY % =
    DIVIDE ( [ED Attendances YTD] - [ED Attendances YTD PY], [ED Attendances YTD PY] )

-- Display folder: 5 Time Intelligence | Format: 0.0%
-- Occupancy over the trailing 12 months ending at the context end (capped at the last data date).
Bed Occupancy % Rolling 12M =
    VAR _last = [Last Data Date]
    VAR _end =
        MIN ( MAX ( 'Date'[Date] ), _last )
    RETURN
        IF (
            MIN ( 'Date'[Date] ) <= _last,
            CALCULATE ( [Bed Occupancy %], DATESINPERIOD ( 'Date'[Date], _end, -12, MONTH ) )
        )

-- Display folder: 5 Time Intelligence | Format: #,0
-- Malaria cases 52 weeks (364 days) earlier: the same epi week of the previous year.
Malaria Cases Same Week Last Year =
    IF (
        MIN ( 'Date'[Date] ) <= [Last Epi Week Start] + 6,
        CALCULATE ( [Malaria Cases], DATEADD ( 'Date'[Date], -364, DAY ) )
    )
```

Custom time intelligence for the epidemiological year. IDSR years run on ISO weeks: 2025-W01 started on 30 December 2024. `TOTALYTD` would therefore cut weeks in half, so the epi-year-to-date measure finds the epi year of the context end and filters the `Epi Year` attribute instead:

File: `04_powerbi/measures.dax` (generated from `model_spec.py`)

```
-- Display folder: 5 Time Intelligence | Format: #,0
-- Malaria cases from epi week 1 of the epi year to the end of the context (capped at the last reported week).
Malaria Cases Epi-YTD =
    VAR _end =
        MIN ( MAX ( 'Date'[Date] ), [Last Epi Week Start] + 6 )
    VAR _y =
        CALCULATE ( MAX ( 'Date'[Epi Year] ), REMOVEFILTERS ( 'Date' ), 'Date'[Date] = _end )
    RETURN
        IF (
            MIN ( 'Date'[Date] ) <= _end,
            CALCULATE ( [Malaria Cases], REMOVEFILTERS ( 'Date' ), 'Date'[Epi Year] = _y, 'Date'[Date] <= _end )
        )

-- Display folder: 5 Time Intelligence | Format: #,0
-- Same number of epi weeks one epi year earlier (like-for-like).
Malaria Cases Epi-YTD PY =
    VAR _end =
        MIN ( MAX ( 'Date'[Date] ), [Last Epi Week Start] + 6 )
    VAR _y =
        CALCULATE ( MAX ( 'Date'[Epi Year] ), REMOVEFILTERS ( 'Date' ), 'Date'[Date] = _end )
    VAR _endPY = _end - 364
    RETURN
        IF (
            MIN ( 'Date'[Date] ) <= _end,
            CALCULATE ( [Malaria Cases], REMOVEFILTERS ( 'Date' ), 'Date'[Epi Year] = _y - 1, 'Date'[Date] <= _endPY )
        )
```

Incidence with person-time. An incidence rate needs a denominator in *person-years at risk*. The measure walks the epi years in the context and multiplies each year's catchment population (NSA population × catchment share) by the number of days of that year in the context. This uses `REMOVEFILTERS` plus an explicit filter, because the population fact is keyed on 1 July. One measure is therefore correct for a week, a quarter, a year or five years. For a single epi year it equals the annual population, so the matrix on page 3 reproduces the SQL pivot of §3.5 exactly.

File: `04_powerbi/measures.dax` (generated from `model_spec.py`)

```
-- Display folder: 4 Disease Surveillance | Format: #,0
-- Person-time at risk in the period: catchment population of each epi year x days of that year in context / 365, within the surveillance window.
Catchment Person-Years =
    VAR _first = [First Surveillance Date]
    VAR _last = [Last Epi Week Start] + 6
    RETURN
        SUMX (
            VALUES ( 'Date'[Epi Year] ),
            VAR _y = 'Date'[Epi Year]
            VAR _days =
                CALCULATE ( COUNTROWS ( 'Date' ), KEEPFILTERS ( 'Date'[Date] >= _first && 'Date'[Date] <= _last ) )
            VAR _pop =
                CALCULATE (
                    SUM ( 'Catchment Population'[Catchment Population] ),
                    REMOVEFILTERS ( 'Date' ),
                    'Date'[Calendar Year] = _y
                )
            RETURN
                DIVIDE ( _pop * _days, 365 )
        )

-- Display folder: 4 Disease Surveillance | Format: #,0
-- Average catchment population over the period (NSA regional population x facility catchment share; time-weighted across epi years).
Catchment Population =
    VAR _first = [First Surveillance Date]
    VAR _last = [Last Epi Week Start] + 6
    VAR _days =
        CALCULATE ( COUNTROWS ( 'Date' ), KEEPFILTERS ( 'Date'[Date] >= _first && 'Date'[Date] <= _last ) )
    RETURN
```

```
        DIVIDE ( [Catchment Person-Years] * 365, _days )

-- Display folder: 4 Disease Surveillance | Format: #,0.0
-- Annualised malaria incidence: cases per 1 000 person-years at risk (comparable across periods of different length).
Malaria Incidence per 1,000 PY =
    DIVIDE ( [Malaria Cases] * 1000, [Catchment Person-Years] )
```

Small-cell suppression. Counts of 1–4 are blanked, or shown as "<5", before any external sharing of surveillance data (§3.8):

File: `04_powerbi/measures.dax` (generated from `model_spec.py`)

```
-- Display folder: 4 Disease Surveillance | Format: #,0
-- Small-cell suppression for external sharing: counts of 1-4 are blanked to prevent re-identification.
Notified Cases (Suppressed) =
    VAR _c = [Notified Cases]
    RETURN
        IF ( _c >= 1 && _c <= 4, BLANK (), _c )

-- Display folder: 4 Disease Surveillance | Format: text
-- Suppressed count as text: "<5" for 1-4 cases.
Notified Cases Label =
    VAR _c = [Notified Cases]
    RETURN
        IF ( NOT ISBLANK ( _c ), IF ( _c >= 1 && _c <= 4, "<5", FORMAT ( _c, "#,0" ) ) )
```

Row-level and object-level security

The model defines four roles. A user is assigned to **one** role in the Power BI service (§3.8).

Table 34. Security roles: RLS filters (DAX) and object-level security

Role	Row-level filters (RLS, DAX)	Object-level security (OLS)	Purpose
Executive (all facilities)	(none - unrestricted rows)	-	Unrestricted read for the executive committee, the CEO and the BI team.
Facility Manager (dynamic)	**Facility**: `'Facility'[Facility Key] IN CALCULATETABLE ( VALUES ( 'Security User Facility'[Facility Key] ), 'Security User Facility'[User Principal Name] = LOWER ( USERPRINCIPALNAME () ) )` **Security User Facility**: `'Security User Facility'[User Principal Name] = LOWER ( USERPRINCIPALNAME () )`	-	Dynamic RLS: users see the facilities mapped to their UPN in sec.user_facility_access (and every ward, attendance, spell, roster, census and surveillance row of those facilities). They can read only their own mapping rows.
North-East Zone (static)	**Region**: `'Region'[Zone] = "North-East"`	-	Static RLS: the North-East malaria task team sees Zambezi and the two Kavango regions only.
Surveillance Partner (aggregate only)	(none - unrestricted rows)	Attendance, Admission, Patient, Readmission Deciles, Readmission Drivers, Security User Facility hidden (metadataPermission = none)	Object-level security for the MoHSS surveillance unit: patient-level tables (attendances, spells, patients, model outputs on patients, the access list) are invisible - not just empty. Aggregated IDSR, population and weather data only.

- **Dynamic RLS (Facility Manager; Microsoft, 2024a).**
  - The filter on `Facility` keeps the facilities that `sec.user_facility_access` maps to the signed-in user's UPN. This is compared in lower case, because Entra ID UPNs are case-insensitive.
  - Because the relationships run Facility → Care Unit → every ward-grain fact, and Facility → every facility-grain fact, one filter secures attendances, spells, census, roster, IDSR, weather and catchment data.
  - A user with no mapping sees nothing (fail closed).
  - A second filter lets managers read only their own rows of the mapping table.
  - The mapping is maintained in the warehouse by the information security officer and refreshed with the model, so access changes need no new version of the report.
- **Static RLS (North-East Zone).** One fixed filter, `Region[Zone] = "North-East"`, gives the malaria task team Zambezi and the two Kavango regions.
- **Object-level security (Surveillance Partner).** The MoHSS surveillance unit sees aggregated IDSR, population and weather data, but the patient-level tables are **invisible** (`metadataPermission = none`; Microsoft, 2024b): `Attendance`, `Admission`, `Patient` and the readmission model outputs. Their names do not appear in the field list or in *Analyze in Excel*, which is stronger than RLS returning empty rows.

Test matrix. The team runs each of the tests below with *Modeling › View as* in Desktop and with *Test as role* in the service. The expected values were computed in SQL from the same mapping table. The facility managers' totals add up exactly to the executive totals, because every attendance and IDSR return belongs to one facility.

Table 35. RLS / OLS test matrix with expected values (from 04_powerbi/expected_values.py)

Other user (UPN)	Role	Visible Facilities	ED Attendances 2025	Malaria cases, epi year 2025
(none)	Executive (all facilities)	10	81,612	14,571
fm.coast.south@gbhs.example	Facility Manager (dynamic)	2	9,223	84
fm.katima@gbhs.example	Facility Manager (dynamic)	1	13,072	4,899
fm.north@gbhs.example	Facility Manager (dynamic)	3	18,141	1,249
fm.rundu@gbhs.example	Facility Manager (dynamic)	2	18,467	7,875
fm.windhoek@gbhs.example	Facility Manager (dynamic)	2	22,709	464
intruder@gbhs.example (not mapped)	Facility Manager (dynamic)	(blank)	(blank)	(blank)
(none)	North-East Zone (static)	3	31,539	12,774
(none)	Surveillance Partner (aggregate only)	10	error: table Attendance is not visible (OLS)	14,571

Validation status and expected values

What was verified automatically at build time:

- the TOM round trip;
- schema validity of all 83 report files;
- every field reference;
- the static DAX check (113 expressions);
- the RLS test-matrix values (SQL).

DAX **execution** needs the Analysis Services engine in Power BI Desktop, so the build guide lists the expected card values for the team to confirm after the first refresh. The model also holds the day-2 load of 1 October 2026 (163 ED attendances), so its 2026 values differ slightly from the 30 September figures in §3.1 and §3.7. For example, 66.8 % of patients were seen within target against 66.7 %.

Table 36. Expected Executive Overview cards (Calendar Year slicer) - computed in SQL with the DAX logic

Card (measure)	2025	2026 (to Last Data Date)	All years
ED Attendances YTD	81,612	67,035	67,035 (latest year)
ED Attendances YTD YoY %	14.1 % (PY 71,507)	7.3 % (PY to 20251001: 62,487)	7.3 %
Seen Within Target %	67.5 %	66.8 %	73.6 %
Median Door-to-Doctor (min)	63	65	58
LWBS Rate %	6.2 %	5.9 %	4.5 %
Bed Occupancy %	81.8 %	84.8 %	80.1 %
Readmission Rate 30d % (by discharge date)	13.4 %	13.6 %	13.3 %
ALOS (days) (by discharge date)	4.10	4.13	4.07
ED Attendances (no slicer)			360,377

Table 37. Expected Disease Surveillance cards (Epi Year slicer)

Card (measure)	2025	2026 (to last epi week)
Malaria Cases Epi-YTD	14,571	17,514
Malaria Cases Epi-YTD YoY %	374.5 % (PY 3,071)	33.5 % (PY same weeks 13,122)
Malaria Incidence per 1,000 PY	25.2 (364 days x 579,079)	39.2 (273 days x 596,611)
Malaria Test Positivity %	62.9 %	66.2 %
IDSR Completeness %	98.7 %	98.7 %
IDSR Timeliness %	92.9 %	92.1 %

3.5 Advanced analytics

OLAP operations

Each OLAP operation is shown twice:

- as SQL on the warehouse (`02_sql/11_olap_queries.sql`), whose captured output follows the query;
- as the Power BI interaction that performs it, as designed in §3.6. The page mock-ups are in §3.6; the screenshots from Power BI Desktop replace them under the same file names (`BUILD_GUIDE.md` step 9).

Roll-up

Roll-up aggregates up a hierarchy: here quarter → year → total. In SQL this is `GROUP BY ROLLUP`, and `GROUPING()` marks each level. In Power BI, the user collapses the *Calendar* hierarchy on the visual (drill up) or removes the quarter level from the axis.

File: `02_sql/11_olap_queries.sql` (lines 8–20)

```sql
-- name: olap_rollup
-- ROLL-UP along the time hierarchy: quarter -> year -> total of both years (ED attendances and % seen within target)
SELECT coalesce(d.calendar_year::text, '2024-2025')    AS year,
       CASE WHEN GROUPING(d.calendar_year) = 1 THEN '(grand total)' ELSE coalesce(d.year_quarter, '(year total)') END AS quarter,
       count(*)                                         AS ed_attendances,
       round(100.0 * sum(e.seen_within_target) / nullif(count(e.seen_within_target), 0), 1) AS pct_seen_within_target,
       GROUPING(d.calendar_year, d.year_quarter)        AS grouping_level
FROM dw.fact_encounter e
JOIN dw.dim_date d ON d.date_key = e.arrival_date_key
JOIN dw.dim_encounter_profile ep ON ep.encounter_profile_key = e.encounter_profile_key AND ep.department_code = 'ED'
WHERE d.calendar_year BETWEEN 2024 AND 2025
GROUP BY ROLLUP (d.calendar_year, d.year_quarter)
ORDER BY d.calendar_year NULLS LAST, d.year_quarter NULLS LAST;
```

Table 38. Roll-up output: ED attendances and share seen within target, quarter → year → total

year	quarter	ed_attendances	pct_seen_within_target	grouping_level
2024	2024-Q1	17,976	77.4	0
2024	2024-Q2	17,423	78.8	0
2024	2024-Q3	17,757	77.6	0
2024	2024-Q4	18,351	75.6	0
2024	(year total)	71,507	77.3	1
2025	2025-Q1	24,058	53.8	0
2025	2025-Q2	19,928	69.6	0
2025	2025-Q3	18,304	76.5	0
2025	2025-Q4	19,322	72.8	0
2025	(year total)	81,612	67.5	1
2024-2025	(grand total)	153,119	72.2	3

The roll-up shows where the 2025 decline sits. Quarter 1 of 2025 (the malaria season) fell to 53.8 % of patients seen within target, while the rest of 2025 recovered towards 2024 levels. The annual figure of 67.5 % hides this seasonality.

Drill-down

Drill-down moves from a summary to its detail along the snowflaked geography: region → facility → ward. On page 2 the matrix has the rows Region › Facility › Ward Code, and the user expands each level with the drill buttons.

File: `02_sql/11_olap_queries.sql` (lines 22–37)

```sql
-- name: olap_drilldown
-- DRILL-DOWN along the geography snowflake: region -> facility -> care unit (inpatient bed occupancy, 2025)
SELECT coalesce(r.region_name, 'ALL REGIONS')        AS region,
       CASE WHEN GROUPING(r.region_name) = 1 THEN '' ELSE coalesce(f.facility_short_name, '(region total)') END AS facility,
       CASE WHEN GROUPING(f.facility_short_name) = 1 THEN '' ELSE coalesce(cu.unit_code, '(facility total)') END AS ward,
       sum(c.occupied_beds)                           AS patient_days,
       sum(c.beds_available)                          AS available_bed_days,
       round(100.0 * sum(c.occupied_beds) / nullif(sum(c.beds_available), 0), 1) AS occupancy_pct
FROM dw.fact_ward_census_daily c
JOIN dw.dim_care_unit cu ON cu.care_unit_key = c.care_unit_key
JOIN dw.dim_facility f ON f.facility_key = cu.facility_key
JOIN dw.dim_region r ON r.region_key = f.region_key
JOIN dw.dim_date d ON d.date_key = c.census_date_key
WHERE d.calendar_year = 2025 AND r.region_code IN ('NA-KE', 'NA-CA')
GROUP BY ROLLUP (r.region_name, f.facility_short_name, cu.unit_code)
ORDER BY r.region_name NULLS LAST, f.facility_short_name NULLS LAST, cu.unit_code NULLS LAST;
```

Table 39. Drill-down output (2025, the two endemic north-east regions): region → facility → ward

region	facility	ward	patient_days	available_bed_days	occupancy_pct
Kavango East	Rundu	RUN-MAT1	5,175	6,533	79.2
Kavango East	Rundu	RUN-MED1	9,734	10,831	89.9
Kavango East	Rundu	RUN-PAE1	4,781	7,319	65.3
Kavango East	Rundu	RUN-SUR1	4,323	5,774	74.9
Kavango East	Rundu	(facility total)	24,013	30,457	78.8
Kavango East	(region total)		24,013	30,457	78.8
Zambezi	Katima Mulilo	KAT-MAT1	3,164	4,305	73.5
Zambezi	Katima Mulilo	KAT-MED1	7,074	6,520	108.5
Zambezi	Katima Mulilo	KAT-PAE1	2,948	3,614	81.6
Zambezi	Katima Mulilo	(facility total)	13,186	14,439	91.3
Zambezi	(region total)		13,186	14,439	91.3
ALL REGIONS			37,199	44,896	82.9

Katima Mulilo's hospital total of 91.3 % hides a medical ward at **108.5 %**: patients on trolleys and in overflow beds. Its 18-bed complement grew to 22 in March 2026, and SCD2 keeps the 2025 occupancy measured against 18 beds.

Slice

A **slice** fixes one member of one dimension (Condition = Malaria) and analyses the remaining sub-cube. In Power BI this is the *IDSR condition* slicer on page 3, or a page filter.

File: `02_sql/11_olap_queries.sql` (lines 39–50)

```sql
-- name: olap_slice
-- SLICE: fix one member (condition = Malaria) and compare facilities over all years (IDSR weekly returns)
SELECT f.facility_short_name AS facility, r.malaria_stratum,
       sum(s.cases_total) AS malaria_cases, sum(s.deaths_total) AS deaths,
       round(100.0 * sum(s.tests_positive) / nullif(sum(s.tests_done), 0), 1) AS test_positivity_pct,
       round(100 * sum(s.cases_total) / sum(sum(s.cases_total)) OVER (), 1) AS share_pct
FROM dw.fact_surveillance_weekly s
JOIN dw.dim_condition c ON c.condition_key = s.condition_key AND c.condition_code = 'MAL'
JOIN dw.dim_facility f ON f.facility_key = s.facility_key
JOIN dw.dim_region r ON r.region_key = f.region_key
GROUP BY f.facility_short_name, r.malaria_stratum
ORDER BY malaria_cases DESC;
```

Table 40. Slice output: malaria cases by facility, all epi years

facility	malaria_stratum	malaria_cases	deaths	test_positivity_pct	share_pct
Rundu	Endemic - high	15,056	25	64.1	34.5
Katima Mulilo	Endemic - high	13,228	16	63.5	30.3
Nkurenkuru	Endemic - high	9,671	0	63.1	22.2
Oshakati	Endemic - low	1,793	2	52.9	4.1
Eenhana	Endemic - low	1,687	0	55	3.9
Windhoek	Non-endemic	887	2	42.2	2
Katutura	Non-endemic	561	0	36.2	1.3
Otjiwarongo	Endemic - low	499	0	34.1	1.1
Walvis Bay	Non-endemic	178	0	15.2	0.4
Keetmanshoop	Non-endemic	83	0	9.3	0.2

Dice

A **dice** selects a sub-cube on several dimensions at once: the malaria season (January–May 2025) × triage YELLOW and GREEN × the North-East zone. In Power BI this is the slicers *Calendar year*, *Triage category* and *Facility* used together on page 2, with a month filter.

File: `02_sql/11_olap_queries.sql` (lines 52–68)

```sql
-- name: olap_dice
-- DICE: sub-cube defined on three dimensions - ED attendances in the 2025 malaria season (Jan-May),
-- triage YELLOW/GREEN, facilities in the North-East zone - with LWBS and waiting time
SELECT f.facility_short_name AS facility, t.triage_code AS triage, d.month_short AS month,
       count(*) AS attendances,
       round(percentile_cont(0.5) WITHIN GROUP (ORDER BY e.door_to_doctor_min)::numeric, 0) AS median_door_to_doctor_min,
       round(100.0 * avg(e.is_lwbs), 1) AS lwbs_pct
FROM dw.fact_encounter e
JOIN dw.dim_date d ON d.date_key = e.arrival_date_key
JOIN dw.dim_triage t ON t.triage_key = e.triage_key
JOIN dw.dim_care_unit cu ON cu.care_unit_key = e.care_unit_key
JOIN dw.dim_facility f ON f.facility_key = cu.facility_key
JOIN dw.dim_region r ON r.region_key = f.region_key
WHERE d.calendar_year = 2025 AND d.month_num BETWEEN 1 AND 5 AND t.triage_code IN ('YELLOW', 'GREEN')
  AND r.zone = 'North-East' AND cu.unit_type = 'Emergency department'
GROUP BY f.facility_short_name, t.triage_code, d.month_num, d.month_short
ORDER BY f.facility_short_name, t.triage_code, d.month_num;
```

Table 41. Dice output: ED waits and LWBS for YELLOW/GREEN patients in the north-east, January-May 2025

facility	triage	month	attendances	median_door_to_doctor_min	lwbs_pct
Katima Mulilo	GREEN	Jan	566	221	40.8
Katima Mulilo	GREEN	Feb	687	276	52.7
Katima Mulilo	GREEN	Mar	663	277	48.4

facility	triage	month	attendances	median_door_to_doctor_min	lwbs_pct
Katima Mulilo	GREEN	Apr	558	240	41.9
Katima Mulilo	GREEN	May	417	164	28.1
Katima Mulilo	YELLOW	Jan	693	119	3.9
Katima Mulilo	YELLOW	Feb	819	169	6
Katima Mulilo	YELLOW	Mar	889	145	5.5
Katima Mulilo	YELLOW	Apr	658	117	4.1
Katima Mulilo	YELLOW	May	457	71	1.3
Rundu	GREEN	Jan	922	286	39.2
Rundu	GREEN	Feb	822	260	37.3
Rundu	GREEN	Mar	793	233	27.1
Rundu	GREEN	Apr	641	171	22.5
Rundu	GREEN	May	554	133	10.1
Rundu	YELLOW	Jan	1,121	131	3.5
Rundu	YELLOW	Feb	940	110	3
Rundu	YELLOW	Mar	1,020	91	1.6
Rundu	YELLOW	Apr	788	67	1.6
Rundu	YELLOW	May	671	50	0

The dice explains the LWBS crisis. At Katima Mulilo in February 2025, routine (GREEN) patients waited a median of 276 minutes, above the 4-hour target, and **52.7 % left without being seen**. Urgent (YELLOW) patients waited 169 minutes against a 60-minute target. Rundu shows the same pattern a month earlier.

Pivot

A **pivot** rotates the cube so that a dimension becomes the columns: facility × epi year, with malaria incidence per 100,000 catchment population. In Power BI this is the matrix on page 3, with Facility on the rows and Epi Year on the columns. Swapping the two fields rotates it.

File: `02_sql/11_olap_queries.sql` (lines 70–86)

```
-- name: olap_pivot
-- PIVOT: rows = facility, columns = calendar year (malaria cases per 100 000 catchment population)
SELECT f.facility_short_name AS facility,
       round(100000.0 * sum(s.cases_total) FILTER (WHERE d.epi_year = 2022) / max(cp.pop) FILTER (WHERE cp.y = 2022), 0) AS "2022",
       round(100000.0 * sum(s.cases_total) FILTER (WHERE d.epi_year = 2023) / max(cp.pop) FILTER (WHERE cp.y = 2023), 0) AS "2023",
       round(100000.0 * sum(s.cases_total) FILTER (WHERE d.epi_year = 2024) / max(cp.pop) FILTER (WHERE cp.y = 2024), 0) AS "2024",
       round(100000.0 * sum(s.cases_total) FILTER (WHERE d.epi_year = 2025) / max(cp.pop) FILTER (WHERE cp.y = 2025), 0) AS "2025",
       round(100000.0 * sum(s.cases_total) FILTER (WHERE d.epi_year = 2026) / max(cp.pop) FILTER (WHERE cp.y = 2026), 0) AS "2026 (to W39)"
FROM dw.fact_surveillance_weekly s
JOIN dw.dim_condition c ON c.condition_key = s.condition_key AND c.condition_code = 'MAL'
JOIN dw.dim_date d ON d.date_key = s.epi_week_date_key
JOIN dw.dim_facility f ON f.facility_key = s.facility_key
JOIN (SELECT f2.facility_key, p.year_date_key / 10000 AS y, round(p.population * f2.catchment_share) AS pop
      FROM dw.fact_population p JOIN dw.dim_facility f2 ON f2.region_key = p.region_key) cp
  ON cp.facility_key = s.facility_key AND cp.y = d.epi_year
GROUP BY f.facility_short_name
ORDER BY "2025" DESC NULLS LAST;
```

Table 42. Pivot output: malaria incidence per 100 000 catchment population, facility x epi year

facility	2022	2023	2024	2025	2026 (to W39)
Katima Mulilo	2,692	2,205	1,977	10,806	11,563
Nkurenkuru	3,520	3,479	2,661	10,547	9,647
Rundu	1,954	2,079	1,172	5,453	7,977
Oshakati	269	251	191	853	891
Eenhana	533	326	234	816	1,262
Otjiwarongo	142	75	119	390	348
Windhoek	150	138	70	320	389
Katutura	85	110	67	310	325
Walvis Bay	27	28	16	99	110
Keetmanshoop	25	30	38	60	86

Drill-across and semi-additive aggregation

Two further operations show why the dimensions are conformed and why measures must respect their additivity.

- **Drill-across** combines two facts through conformed dimensions: e-roster nursing hours ÷ ward-census patient days, joined on Date and Care Unit. The result is NHPPD per ward type, the *NHPPD* measure in Power BI.
- **Semi-additive aggregation**: occupied beds may be summed across wards but **not across days**. The last column shows the wrong answer that an implicit `SUM` would give, which is why implicit measures are disabled (§3.4).

Table 43. Drill-across output: nursing hours per patient day (NHPPD) and occupancy by facility and specialty, 2025

facility	specialty	patient_days	nursing_hours	nhppd	occupancy_pct
Katima Mulilo	Medical	7,074	28,787	4.07	108.5
Oshakati	Medical	10,541	45,786	4.34	98
Rundu	Medical	9,734	43,831	4.50	89.9
Walvis Bay	Medical	8,432	38,359	4.55	96.6
Windhoek	Paediatric	6,471	31,202	4.82	89.5

facility	specialty	patient_days	nursing_hours	nhppd	occupancy_pct
Rundu	Maternity	5,175	26,455	5.11	79.2
Katima Mulilo	Paediatric	2,948	15,133	5.13	81.6
Oshakati	Paediatric	5,542	29,451	5.31	96
Windhoek	Maternity	7,339	39,249	5.35	85.4
Windhoek	Medical	14,787	81,773	5.53	78.6
Oshakati	Maternity	6,396	36,030	5.63	80
Windhoek	Surgical	8,219	46,833	5.70	75.9
Walvis Bay	Maternity	2,853	16,384	5.74	78.8
Katima Mulilo	Maternity	3,164	18,442	5.83	73.5
Walvis Bay	Surgical	3,314	19,893	6	65.6
Rundu	Surgical	4,323	26,286	6.08	74.9
Oshakati	Surgical	6,032	39,047	6.47	64.7
Rundu	Paediatric	4,781	31,530	6.59	65.3
Oshakati	Intensive care	1,965	16,058	8.17	67.8
Windhoek	Intensive care	1,516	17,015	11.22	52.6

Table 44. Semi-additive output: average and closing occupied beds versus the (wrong) sum over days

facility	year_month	avg_daily_occupied_beds	occupied_beds_last_day	wrong_sum_over_days_equals_patient_days
Katima Mulilo	2026-03	51.9	46	1,608
Katima Mulilo	2026-08	37.4	31	1,158
Oshakati	2026-03	79.7	71	2,471
Oshakati	2026-08	84.6	68	2,623
Rundu	2026-03	65	72	2,014
Rundu	2026-08	67.4	68	2,089
Walvis Bay	2026-03	41.3	46	1,281
Walvis Bay	2026-08	44.1	43	1,366
Windhoek	2026-03	116	126	3,597
Windhoek	2026-08	123.3	128	3,822

Technique 1 – Forecasting weekly malaria cases (BQ4)

Question. How many malaria cases will the three north-east facilities (Rundu, Katima Mulilo, Nkurenkuru) see in the next 1–4 weeks, so that rapid diagnostic tests, drugs, beds and doctors can be planned?

Method (`03_analytics/forecasting.py`, statsmodels; Seabold & Perktold, 2010). The series is the weekly confirmed malaria attendances (ICD-10 B50–B54) from the PAS, 247 weeks from 2022-W01. The PAS is used rather than IDSR because it has no missing weeks, and the two agree within 0.1 % (check R08). Three models are compared:

Model	Specification
M1 Seasonal naive	The same week last year (benchmark)
M2 Holt-Winters exponential smoothing	On log(1 + cases), additive trend, 52-week seasonality
M3 Dynamic regression SARIMAX(1,0,1)	On log cases, with two Fourier pairs for the annual cycle and rainfall summed over weeks *t−8* … *t−4* (NASA POWER) as an exogenous lead indicator, known four weeks ahead

Evaluation.

- The evaluation is **rolling-origin**: over the last 52 weeks (2025-W40 to 2026-W39), each model is refitted every 4 weeks and forecasts 1–4 weeks ahead, giving 52 forecasts per model. This is the time-series equivalent of cross-validation and avoids judging a model on one lucky split (Hyndman & Athanasopoulos, 2021).
- Metrics: MAE, RMSE, WAPE and MASE (scaled by the in-sample seasonal-naive error), and the coverage of the 80 % prediction interval.

Table 45. Rolling-origin back-test, 52 weeks x horizons 1-4

model	n_forecasts	MAE	RMSE	WAPE	MASE	coverage_80
Seasonal naive	52	95.5	159.2	29.6 %	0.97	82.7 %
Holt-Winters (ETS)	52	76.7	129.7	23.8 %	0.78	65.4 %
SARIMAX + rainfall	52	87.3	142.3	27.1 %	0.89	75.0 %

Table 46. WAPE by forecast horizon

model	h = 1 wk	h = 2 wk	h = 3 wk	h = 4 wk
Holt-Winters (ETS)	19.4 %	24.1 %	26.4 %	25.4 %
SARIMAX + rainfall	20.1 %	25.3 %	31.3 %	31.6 %
Seasonal naive	31.4 %	29.5 %	29.0 %	28.7 %

Table 47. SARIMAX coefficients (log scale)

term	estimate	p_value
log(1 + rainfall 4-8 weeks earlier)	0.405	< 0.001
AR(1)	0.956	< 0.001
MA(1)	-0.469	< 0.001

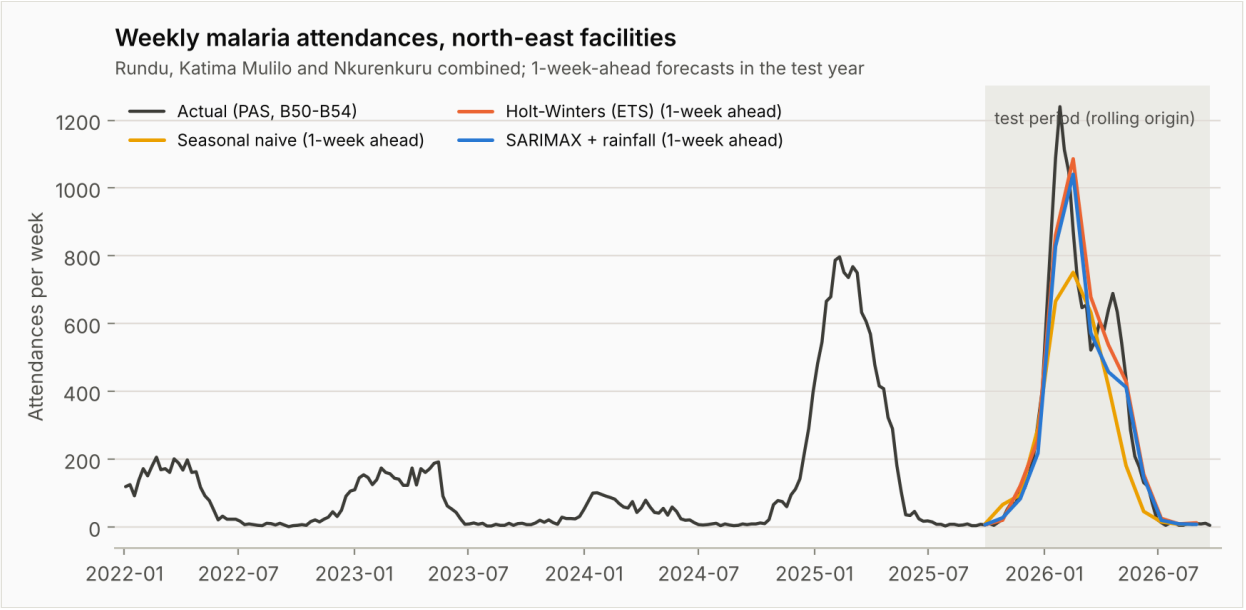


Figure 10. Rolling-origin back-test: actual weekly malaria cases (north-east) against the forecasts of the three models

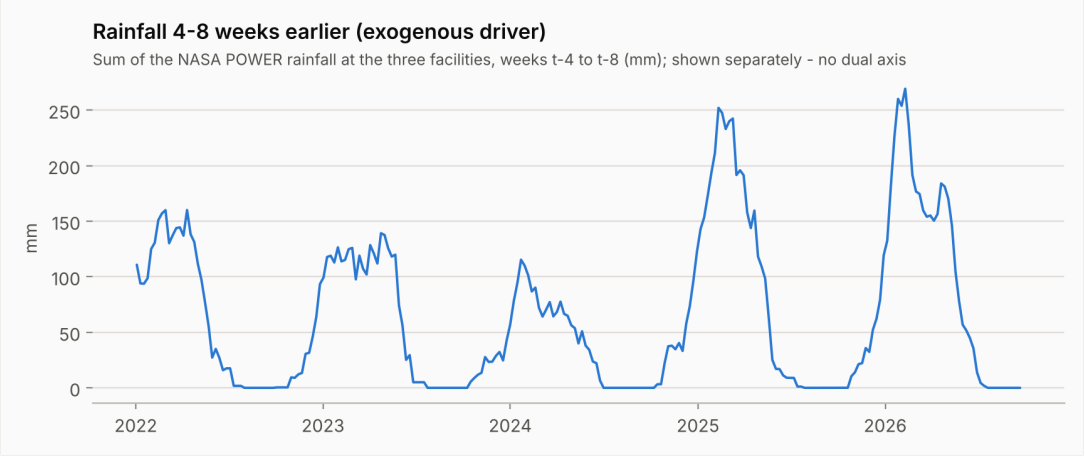


Figure 11. Rainfall 4-8 weeks earlier against weekly malaria cases: the lead indicator used by SARIMAX

Results.

- **Holt-Winters is the best model.** It has a WAPE of 23.8 % and MASE 0.78, so its error is 22 % lower than the naive benchmark. One week ahead its WAPE is **19.4 %**, which the *Forecast error* card on page 4 shows.
- **Rainfall is a significant driver.** Its coefficient is 0.405 on the log–log scale ($p < 0.001$), so 10 % more rain 4–8 weeks earlier is associated with about 4 % more cases. SARIMAX also has better-calibrated intervals (75 % coverage, against 65 % for Holt-Winters).
- **Operational choice.**
  - Publish the Holt-Winters point forecast.
  - Widen its interval, or use the SARIMAX interval, for stock planning, because the Holt-Winters 80 % interval covers only 65 % of outcomes.
  - Use rainfall on page 3 as the *lead* indicator for the season's start.

Table 48. Forward forecast (Holt-Winters), first 8 weeks after the cut-off

epi week	forecast	lower 80 %	upper 80 %
2026-W40	6	4	10
2026-W41	9	6	13
2026-W42	7	5	11
2026-W43	17	12	26
2026-W44	25	17	37
2026-W45	33	22	48
2026-W46	35	24	51
2026-W47	30	21	45

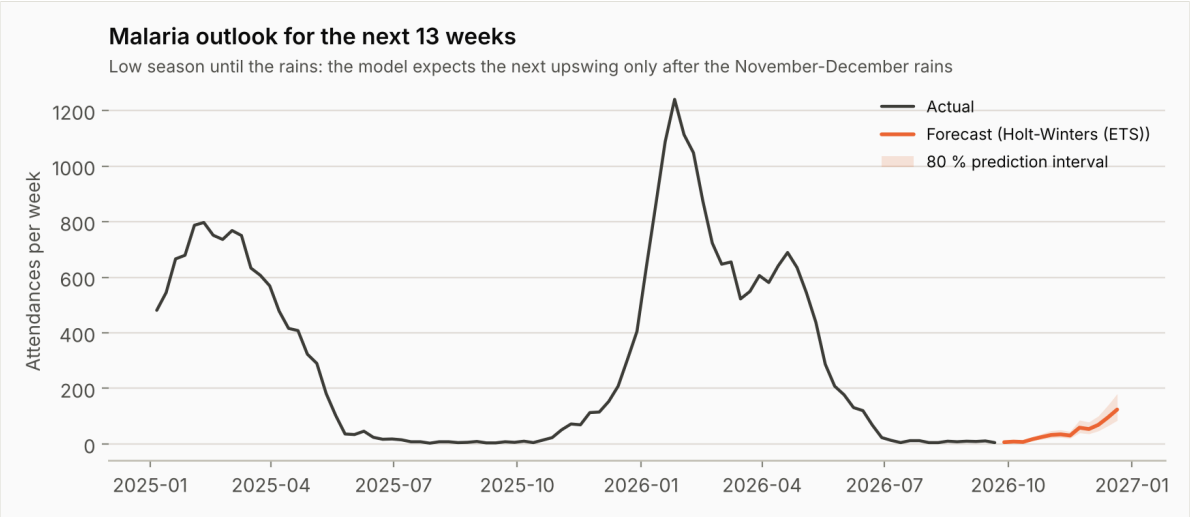


Figure 12. 13-week forward forecast with 80 % prediction interval (Holt-Winters, north-east)

Assumptions and limitations.

- Stable reporting and testing practice.
- The seasonality of the last four seasons repeats. The 2025 and 2026 seasons were exceptional, so one more “normal” season would shift the trend.
- The rainfall stand-ins must be replaced by the real NASA POWER data (`fetch_public_data.py`) before operational use.
- The forecast is for attendances at GBHS facilities, not for community incidence.

Technique 2 – Classifying 30-day readmission risk (BQ5)

Question. Which patients discharged today are most likely to come back as an emergency within 30 days, so that discharge planners can target follow-up calls and medicine reconciliation?

Method (`03_analytics/readmission_model.py`, scikit-learn; Pedregosa et al., 2011).

- **Population.** 79,457 eligible spells, i.e. discharged alive (home or against advice), not maternity, with 30 days of follow-up. The target is `readmitted_within_30d`, milestone 3 of the accumulating snapshot.
- **Features.** Only information available **at discharge**: age band, sex, payer group, specialty, ICD-10 chapter, chronic flag, severity, length of stay, prior admissions in 12 months, weekend discharge, against medical advice, emergency admission and ICU stay.
- **Models.** M1 logistic regression, with reference coding for interpretable odds ratios; M2 histogram gradient boosting.
- **Validation.** A **temporal split** prevents leakage from the future: train on discharges to 30 June 2025 (58,012), test on 1 July 2025 – 31 August 2026 (21,388).

Eligible spells: 79,457 (train 58,012 discharged before 2025-07-01, test 21,388 up to 2026-08-31). Test prevalence 13.6 %. Best model by ROC AUC: **Gradient boosting**. Operating threshold = top 20 % of predicted risk (p >= 0.192).

Table 49. Test-set performance (prevalence 13.6 %)

model	ROC AUC	PR AUC	Brier score	recall at top 20 %	precision at top 20 %
Logistic regression	0.657	0.228	0.1131	36.0 %	24.5 %
Gradient boosting	0.660	0.233	0.1128	37.0 %	25.2 %

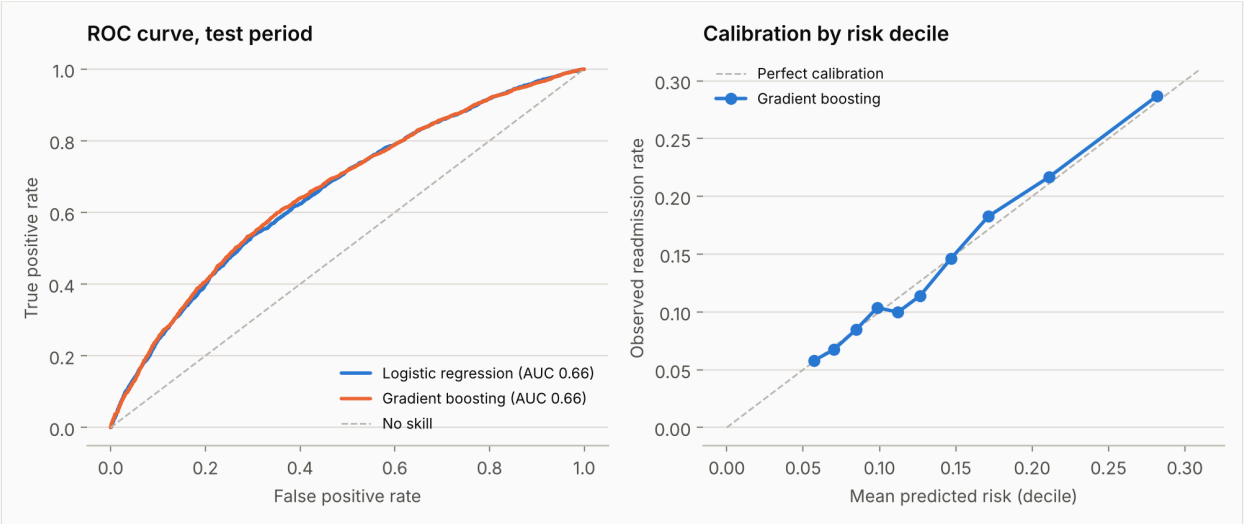


Figure 13. ROC curves and calibration by decile of predicted risk (test set)

Table 50. Largest odds ratios of the logistic model (categories versus their reference level: age 0-4, female, government payer, intensive care)

feature	odds_ratio
age band 25-44	0.47
ama	2.09
age band 15-24	0.55
age band 5-14	0.55
is chronic	1.69
specialty Medical	0.72
specialty Surgical	0.75
specialty Paediatric	0.76
specialty Isolation (COVID-19)	0.76
age band 65+	1.29
weekend discharge	1.28
chapter XI	0.79

Table 51. Fairness check at the operating threshold (top 20 % of risk)

attribute	group	spells	observed_rate	mean_predicted	tpr	fpr	flagged_share
sex	F	11958	13.7 %	13.6 %	37.8 %	17.9 %	20.6 %
sex	M	9430	13.5 %	13.6 %	36.0 %	16.6 %	19.2 %
age_band	0-4	4683	14.5 %	14.7 %	31.2 %	16.3 %	18.5 %
age_band	15-24	1563	8.1 %	9.9 %	7.9 %	3.3 %	3.6 %
age_band	25-44	4430	8.6 %	8.6 %	4.7 %	1.9 %	2.1 %
age_band	45-64	4592	13.0 %	12.4 %	20.2 %	9.2 %	10.6 %
age_band	5-14	1496	8.6 %	8.2 %	5.5 %	1.5 %	1.9 %
age_band	65+	4616	21.7 %	21.5 %	71.1 %	56.2 %	59.5 %

Results.

- **Discrimination is modest.** The AUC is 0.66 for both models, which is typical of published readmission models (0.6–0.7; Kansagara et al., 2011).
- **Calibration is good.** Predicted and observed rates agree in every decile, so the scores can be read as probabilities.
- **Targeting.** Calling the top 20 % by predicted risk reaches **37 % of all readmissions**, with a 25 % hit rate. That is about 1.9 times better than calling at random.
- **Drivers.**
  - Discharge against medical advice: odds ratio 2.09.
  - Chronic conditions: 1.69.
  - Age 65+: 1.29.
  - Weekend discharge: 1.28. This is actionable: weekend discharges need a follow-up call on Monday.
- **Fairness by sex.** Sexes are treated alike (true-positive rate 37.8 % for women, 36.0 % for men; calibration equal).
- **Fairness by age.** Older patients are flagged far more often (59.5 % of patients aged 65+). This reflects their higher observed risk (21.7 %), not miscalibration, but it should be reviewed by the clinical governance committee.

Assumptions and limitations.

- Readmissions to other providers are invisible, so the rate is a lower bound.
- There are no medication, laboratory or social features.
- An AUC of 0.66 supports **prioritising** a call list, not decisions about individual care.
- The model must be monitored for drift and re-validated yearly.

Technique 3 – Outbreak detection in IDSR surveillance (BQ3)

Question. Can GBHS detect unusual rises of priority diseases within 1–2 weeks of their start, without drowning surveillance officers in false alarms?

Method (`03_analytics/outbreak_detection.py`). Each method runs on every facility × condition weekly series:

Method	Rule
M1 **EARS C2** (CDC Early Aberration Reporting System; Hutwagner et al., 2003)	Alert when cases exceed the mean + 3 SD of weeks *t−8 … t−2*, with a one-week guard band
M2 **CUSUM** on the standardised C2 residuals (Page, 1954)	k = 1, h = 2.5; reset after an alert
M3 **Historical limits** for malaria (Stroup et al., 1989)	Alert when cases exceed the mean + 2 SD of the same epi week ± 2 weeks in previous years. This is the WHO-style malaria epidemic threshold

All methods also require at least 3 cases in the week. **Ground truth:** the generator injected 12 outbreaks of hepatitis E, measles, acute watery diarrhoea, malaria, typhoid and meningitis, with known facility, start week and size.

Table 52. Detection performance against the 12 injected outbreaks

method	outbreaks_detected	sensitivity	median_delay_weeks	false_alerts_per_100_weeks	ppv
EARS C2	9 / 12	75 %	1.0	0.84	7 %
CUSUM	11 / 12	92 %	1.0	0.97	7 %

Table 53. Outbreak-level results (delay = weeks from onset to first alert)

outbreak	facility	condition	start	weeks	extra_cases	EARS C2 detected	EARS C2 delay (wk)	CUSUM detected	CUSUM delay (wk)
OB01	GBC-KTU	HEV	2022-W28	12	72	yes	2	yes	2
OB02	GBH-OSH	MEA	2022-W40	5	24	yes	1	yes	1
OB03	GBC-KTU	AWD	2023-W06	5	140	yes	1	yes	1
OB04	GBH-WHK	MAL	2023-W16	4	30	yes	0	yes	0
OB05	GBH-OSH	MAL	2023-W32	4	28	yes	0	yes	0
OB06	GBC-OTJ	TYP	2023-W45	5	16	yes	2	yes	2
OB07	GBH-RUN	MEA	2024-W10	8	46	yes	1	yes	1
OB08	GBC-NKU	MEA	2024-W12	6	22	no		no	
OB09	GBH-WVB	HEV	2024-W33	6	26	yes	0	yes	0
OB10	GBC-EEN	AWD	2025-W03	4	60	yes	0	yes	0
OB11	GBH-KAT	MEN	2025-W33	4	9	no		yes	1
OB12	GBC-KMP	AWD	2026-W10	3	30	no		yes	2

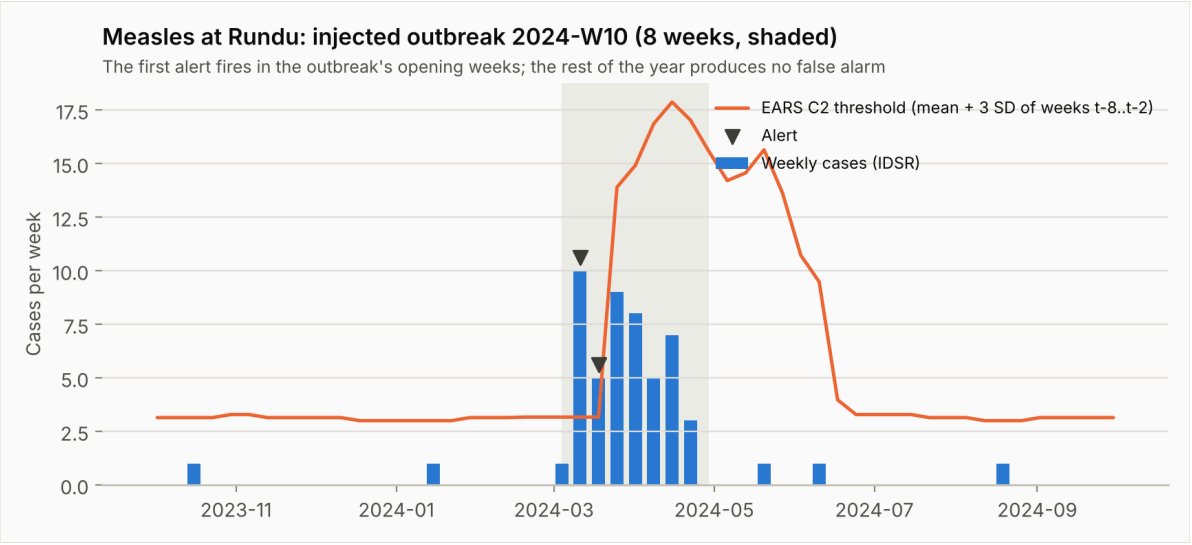


Figure 14. Measles at Rundu, 2024 (outbreak OB07): weekly cases with EARS C2 threshold and CUSUM alerts

Table 54. First malaria historical-limits alert per season, north-east facilities

facility	first alert 2024	first alert 2025	first alert 2026
Katima Mulilo	2024-W33	2025-W02	2026-W03
Nkurenkuru	2024-W39	2025-W04	2026-W03
Rundu	2024-W49	2025-W01	2026-W02

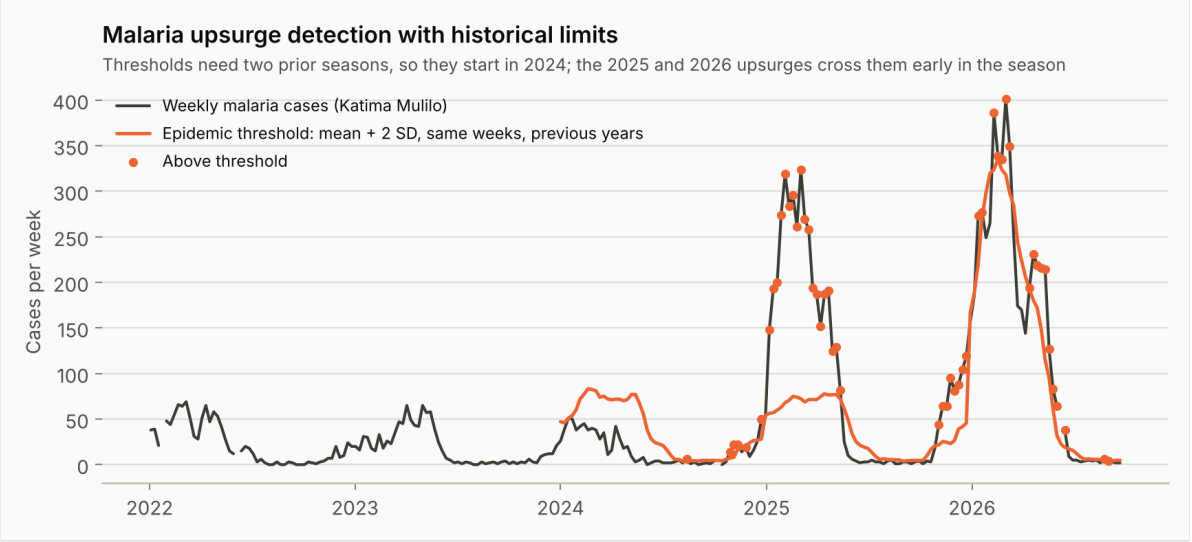


Figure 15. Malaria epidemic threshold (historical limits) at Katima Mulilo: the 2025 and 2026 upsurges cross it in epi weeks 2 and 3

Results.

- **CUSUM detected 11 of 12 outbreaks** and EARS C2 9 of 12. The median delay for both is 1 week, with fewer than 1 false alert per 100 facility-condition-weeks.
- CUSUM caught the small, slow outbreaks that C2 missed (meningitis at Katima, diarrhoea at Keetmanshoop). It accumulates small excesses that never cross a 3-SD line on their own.

- Both missed OB08 (measles at Nkurenkuru, 22 extra cases over 6 weeks). Its weekly counts stayed within the noise of a busy clinic.
- **Positive predictive value is only 7 %.** Most alerts are not injected outbreaks, so every alert needs a short triage by a surveillance officer. That is acceptable at fewer than 1 alert per 100 weeks.
- For malaria, the historical-limits alert fired in **2025-W01 to W04 and 2026-W02 to W03**, 2–6 weeks before the seasonal peaks (2025-W07, 2026-W05). Combined with the forecast, this is the trigger for the seasonal staffing uplift (§3.7).

Assumptions and limitations.

- The injected outbreaks are cleaner than real ones.
- A few weekly reporting delays and missing reports (2 % of weeks) can mask or fake a signal, which is why the completeness measure sits next to the alerts.
- Historical limits need at least two prior seasons, so a new facility has no threshold.

Technique 4 – What drives ED waiting time (BQ1)

Question. What lengthens door-to-doctor time: patient mix, load per doctor, bed occupancy (access block), night shift or the malaria season? And which levers can management pull?

Method (`03_analytics/ed_wait_drivers.py`).

- **Data.** ORANGE, YELLOW and GREEN ED attendances with a valid door-to-doctor time, **drilled across** to:
  - the e-roster: doctors on duty in the arrival shift, giving arrivals per doctor-shift;
  - the ward census: hospital occupancy on the arrival day.
- **M1, ordinary least squares on log waiting time.**
  - It adjusts for triage category and facility (fixed effects).
  - Coefficients read as percentage effects.
  - Standard errors are HC3 heteroskedasticity-robust.
  - It is fitted on a 150,000-attendance sample.
- **M2, gradient-boosted trees.** Trained on 2022–2024 and tested on 2025–2026, with **permutation importance**, the same idea as the Power BI *Key influencers* visual.

OLS: n = 150,000 ED attendances (ORANGE/YELLOW/GREEN), adjusted R2 = 0.757; gradient boosting trained on 2022-2024 (120,000 rows) explains R2 = 0.769 of log waiting time in 2025-2026 (40,000 rows). Median arrivals per doctor per shift: 15.0; median hospital occupancy: 80 %.

Table 55. OLS effects on door-to-doctor time (95 % confidence intervals, HC3)

driver	effect_pct	95 % CI	p_value
+1 arrival per doctor in the shift	+6.3 %	+6.2 to +6.4 %	< 0.001
+10 points hospital bed occupancy	+3.6 %	+3.4 to +3.9 %	< 0.001
Night shift (19:00-07:00)	+7.2 %	+6.5 to +7.9 %	< 0.001
Weekend	-0.3 %	-0.9 to +0.4 %	0.381
Malaria season (Jan-May)	+0.9 %	+0.4 to +1.5 %	0.001
Triage ORANGE vs GREEN	-93.2 %	-93.3 to -93.2 %	< 0.001
Triage YELLOW vs GREEN	-64.6 %	-64.8 to -64.4 %	< 0.001

Table 56. Gradient boosting: permutation importance on the 2025-2026 test set

driver	importance	share
Triage category	1.249	78.8 %
Arrivals per doctor (shift)	0.326	20.6 %
Hospital bed occupancy	0.007	0.5 %
Night shift	0.002	0.2 %
Malaria season	0.000	0.0 %
Facility	0.000	0.0 %
Weekend	-0.000	-0.0 %

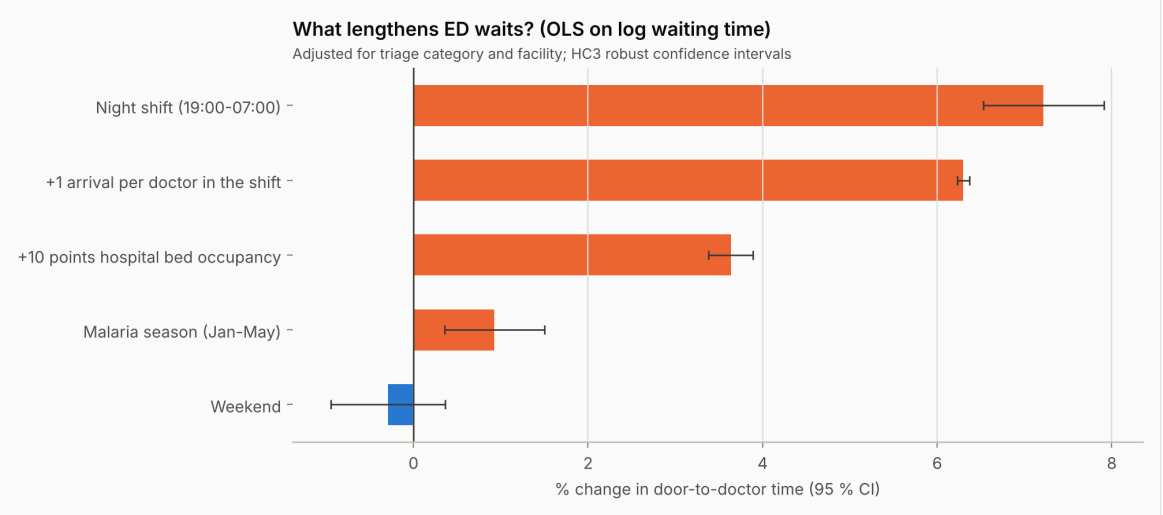


Figure 16. Operational drivers of ED waiting time (OLS, % change with 95 % CI)

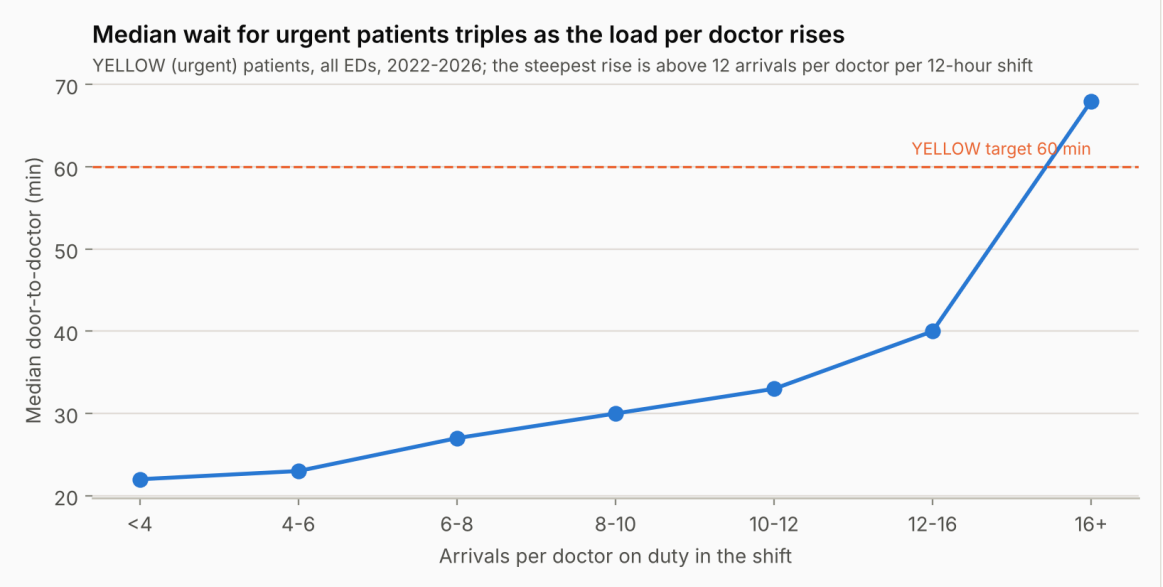


Figure 17. Median wait of YELLOW (urgent) patients by arrivals per doctor-shift

Results.

- **Case mix explains most of the variation.** Triage category accounts for 79 % of the importance, because SATS works as designed: ORANGE patients wait 93 % less than GREEN.
- **Load per doctor is the largest lever.**
  - Each extra arrival per doctor-shift adds **6.3 %** (CI 6.2–6.4 %) to the wait. It accounts for 21 % of the importance and is the only lever with real predictive power out of sample.
  - The median YELLOW wait rises from about 22 minutes below 4 arrivals per doctor to over 60 minutes above 16 (figure above).
- **Access block.** Each 10 points of hospital occupancy adds 3.6 %.
- **Night shift adds 7.2 %.** The malaria season adds only 0.9 % once load is controlled, so the season acts *through* the load.
- The weekend has no effect (p = 0.38).
- The model explains about three-quarters of the variance of log waits, both in-sample (adjusted R^2 0.757) and out of sample (R^2 0.769).

Natural experiment: the 2026 staffing intervention. In January–June 2026, Rundu and Katima Mulilo had one extra day-shift doctor; Oshakati is the comparison hospital, which had no change.

Table 57. Staffing intervention, January-June 2026 versus January-June 2025 (Oshakati = comparison hospital)

facility	calendar_year	ed_attendances	pct_seen_within_target	lwbs_pct	avg_day_shift_doctors
Katima Mulilo	2025	8,194	36.1	17.9	1
Katima Mulilo	2026	9,520	48.8	10.8	1.85
Oshakati	2025	9,200	65.3	5.5	1.93
Oshakati	2026	9,148	66.9	5.2	1.96
Rundu	2025	11,003	43.9	11.2	1.91
Rundu	2026	13,502	53.5	9	2.91

- At Katima Mulilo, day-shift doctors went from 1.0 to 1.85. The share of patients seen within target rose by **12.7 points**, and LWBS fell from 17.9 % to 10.8 %, despite 16 % more attendances.
- At Rundu, doctors went from 1.9 to 2.9. Patients seen within target rose 9.6 points, despite 23 % more attendances.

- Oshakati moved only 1.6 points.

This difference-in-differences pattern matches the regression. *Caveat:* the data are synthetic and the generator planted this intervention. The point is that the warehouse and the model **recover** it, so the same analysis will evaluate the real roster change.

Assumptions and limitations.

- This is observational: doctors are rostered *because* busy periods are expected, so the load effect may be underestimated (reverse causality).
- Unmeasured factors are not in the model: lab turnaround, senior decision-maker availability, ambulance offload.
- Effects are associations, not causal estimates. The intervention comparison is the closest thing to a causal test.

3.6 Dashboards and visual analytics

Pages

The report `GBHS_Health.Report` has **four report pages**, one drill-through page and one report-page tooltip. The brief asks for at least three: an executive, an analytical/diagnostic and an advanced/operational page. The pages contain 71 visuals; every one of the 67 data visuals has alt text, a title and a keyboard tab order.

Table 58. Report pages, audiences and the business questions they answer

Page	Type (brief)	Audience	Answers	Main visuals
1 Executive Overview	Executive	CEO, executive committee, board	BQ1–BQ3 at a glance; KPIs against targets	6 KPI cards; line chart of patients seen within target against the previous year and the 80 % target; occupancy by hospital; malaria by month; facility scorecard
2 Patient Flow & Resources	Analytical / diagnostic	Facility, ED and nursing managers	BQ1, BQ2: *why* are waits long, and where are beds and staff stretched?	Load and staffing cards; Region › Facility › Ward matrix (drill-down); median wait by arrival hour; access-block scatter of occupancy against patients seen on time; patients seen within target by triage; arrivals per doctor-shift
3 Disease Surveillance	Operational	Surveillance officers, MoHSS partner, task team	BQ3: where and when are diseases rising? Is reporting complete?	Epi-YTD, incidence, positivity, completeness and timeliness cards; weekly malaria against the same week last year; cases by condition; rainfall lead indicator; incidence pivot facility × epi year; reporting compliance by facility
4 Advanced Analytics & DQ	Advanced	Analysts, BI team, data stewards, the security officer	BQ4, BQ5, model quality, data quality, RLS check	Forecast with back-test and 80 % interval; readmission calibration by decile; ED wait drivers; outbreak alerts table; DQ issues by rule; facilities visible under RLS
Facility Detail (drill-through)	Diagnostic	Managers	One facility on one page	6 KPI cards; patients seen within target against the target; occupancy by ward; weekly malaria; staffing by year
KPI Tooltip (report-page tooltip)	–	Everyone	Context on hover	Seen within target with ▲/▼ status, median wait, LWBS, occupancy, readmissions

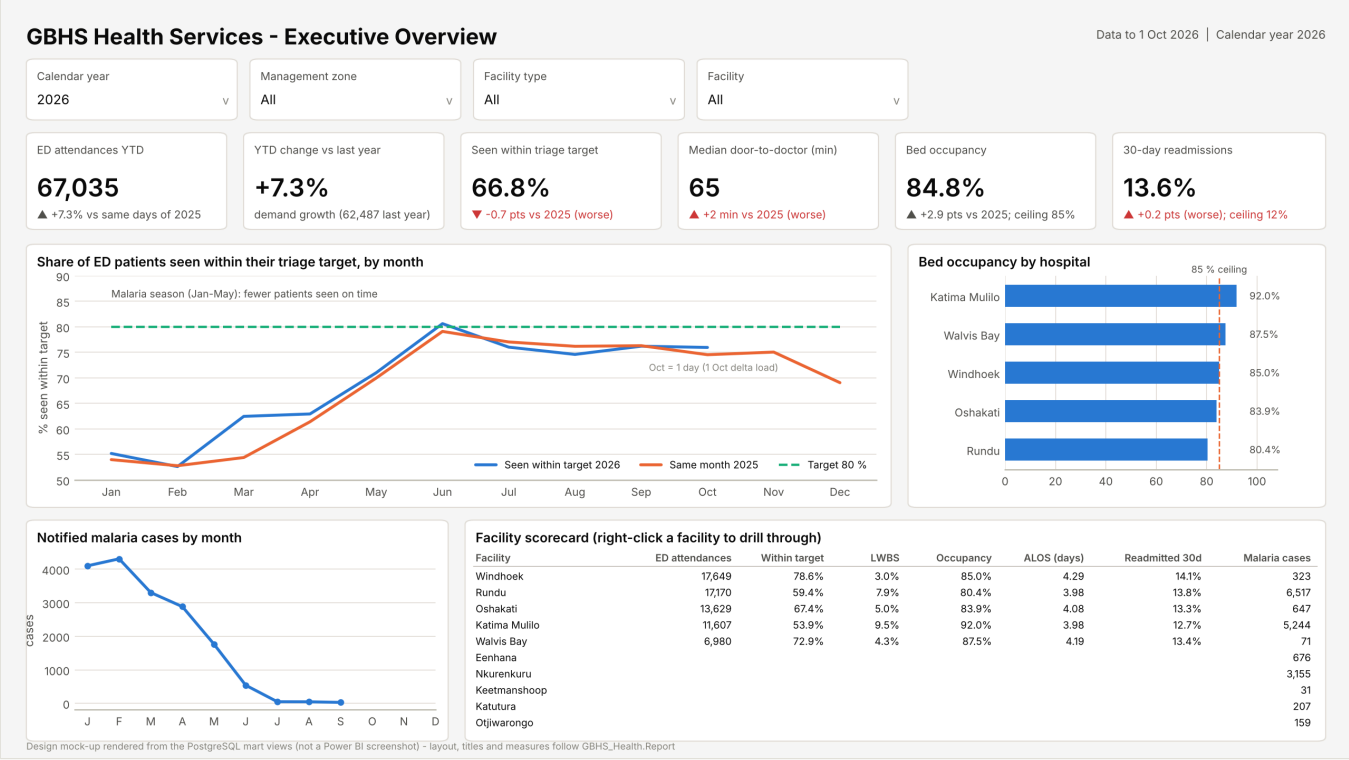


Figure 18. Page 1 - Executive Overview (calendar year 2026, design mock-up rendered from the warehouse)

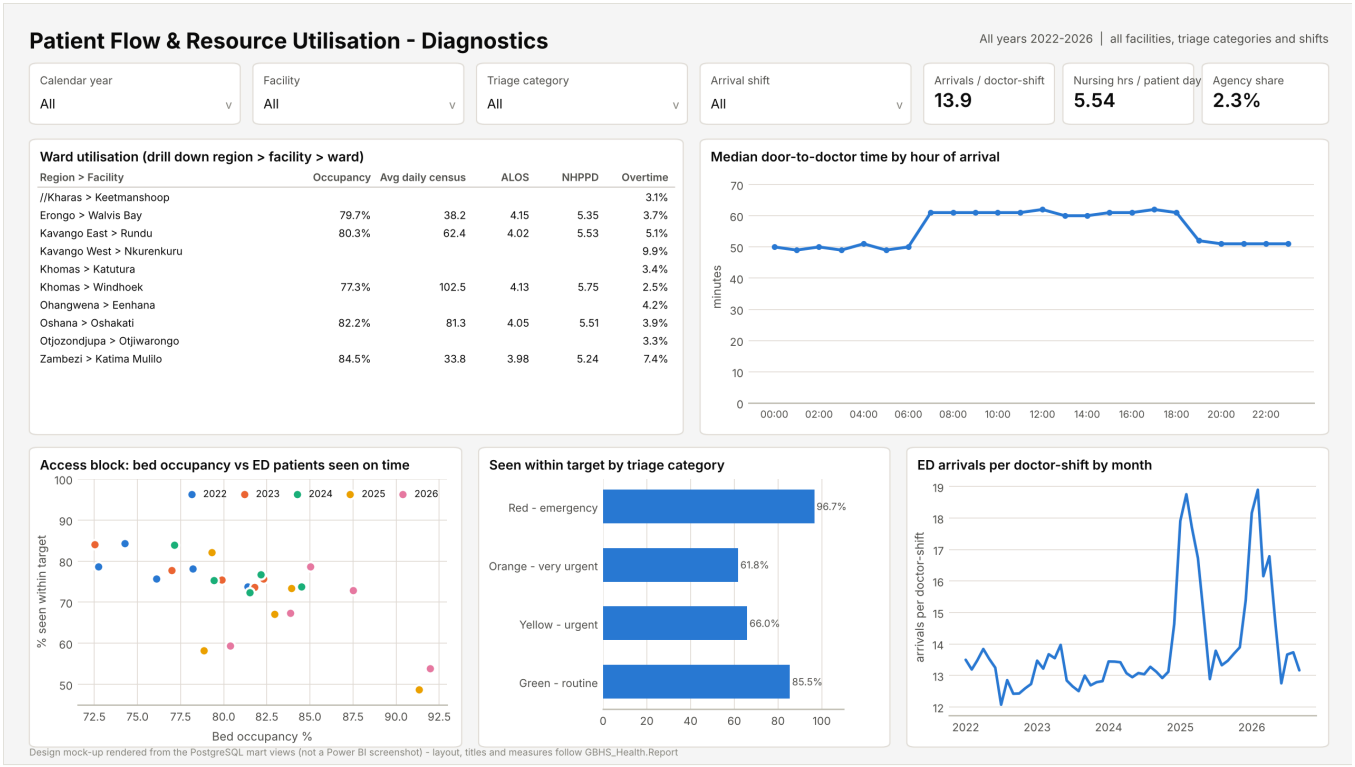


Figure 19. Page 2 - Patient Flow & Resources: drill-down matrix, wait by hour, access-block scatter, triage and load (all years)

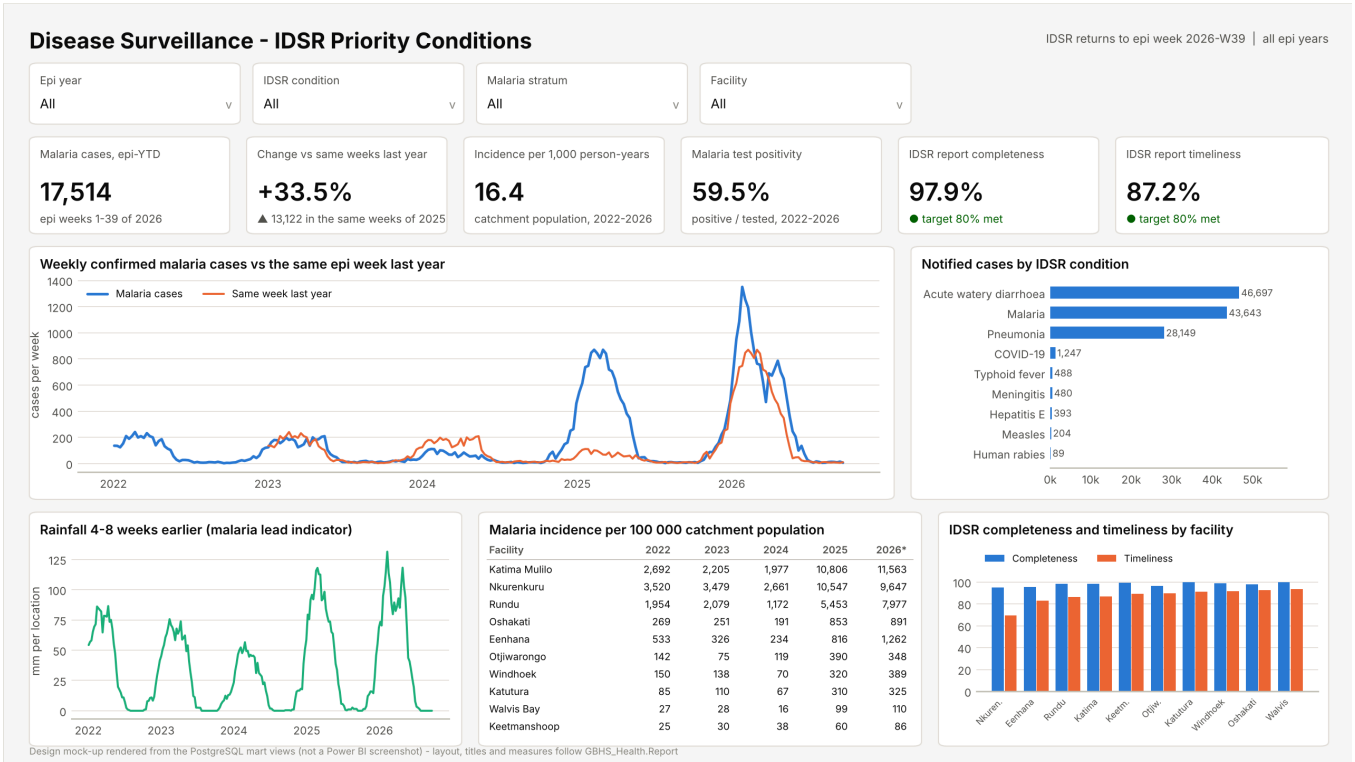


Figure 20. Page 3 - Disease Surveillance: epi-week trend, lead indicator, incidence pivot and reporting compliance

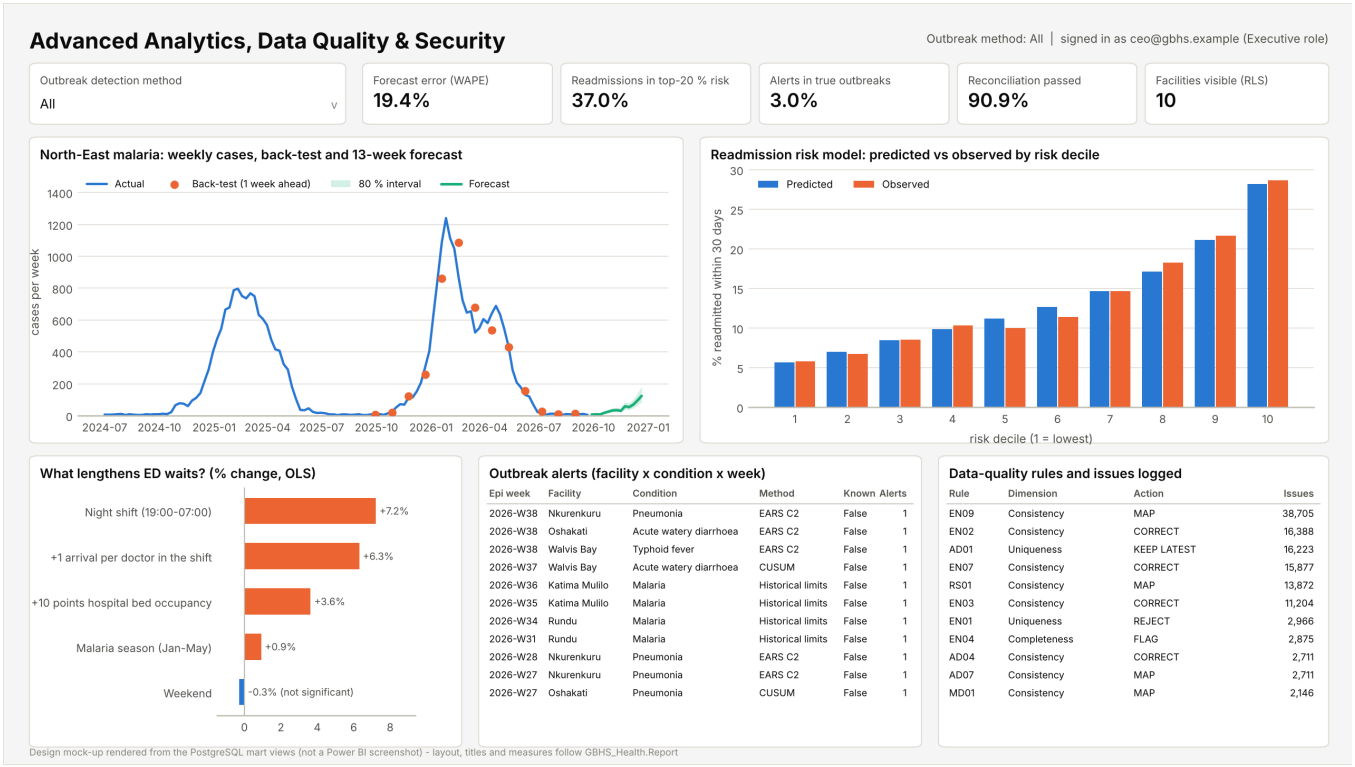


Figure 21. Page 4 - Advanced Analytics & DQ: forecast, readmission calibration, wait drivers, outbreak alerts, DQ rules

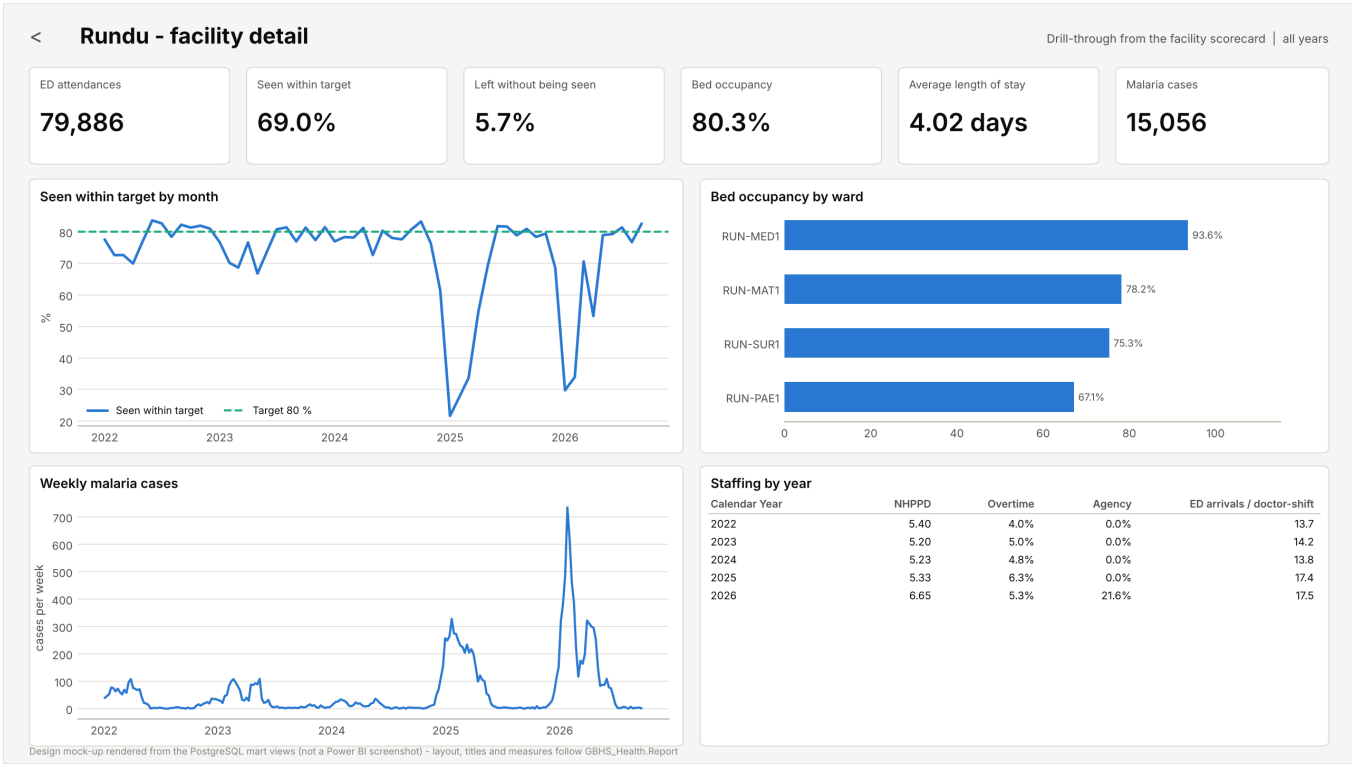


Figure 22. Drill-through page Facility Detail (from the scorecard, Rundu)

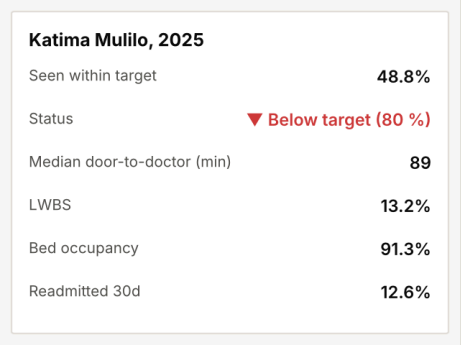


Figure 23. Report-page tooltip shown when hovering a hospital (Katima Mulilo, 2025)

The images above are **design mock-ups**. They have the same layout, titles and measures as the generated PBIR pages and are rendered from the warehouse with SQL that mirrors the DAX (`06_report/build_mockups.py`). The team replaces them with Power BI screenshots after the Desktop build (`BUILD_GUIDE.md` step 9).

Design good practice

Chart choice follows the question (Few, 2013; Knaflic, 2015):

Question	Chart
A trend over time	Line chart: monthly *seen within target*, weekly malaria
A ranking	Sorted bar chart: occupancy by hospital, cases by condition, wait drivers
A comparison of a few series	Clustered column: reporting completeness and timeliness; predicted against observed risk
A relationship	Scatter: occupancy against patients seen on time, one point per hospital-year
Detail and drill-down	Matrix and table: ward utilisation, incidence pivot, scorecard
A headline number	Card: KPIs

- **No pie charts, no gauges, no 3-D.** A line rather than a column shows the 80 % target, so the comparison is immediate.
- **No dual axes.** When two measures have different units (malaria cases and rainfall) they get two separate charts with the same weekly time axis instead of one chart with two y-axes. The two scales would otherwise invite a false visual correlation.

Visual hierarchy. Every page follows the same reading order:

1. Title and data-freshness label (top left / top right).
2. Slicers in one row.
3. KPI cards.
4. The main chart, largest and top left.
5. Supporting and detail visuals below.

The keyboard tab order is generated in the same order. At most six KPI cards and six analytical visuals appear on a page, on a 1280 × 720 canvas with a 24 px grid and white visual containers on a light grey page background.

Colour and typography.

- **One validated palette.** The theme `GBHS_Accessible.json` uses an 8-colour categorical palette that passed the colour-vision-deficiency (CVD) validator: lightness band, chroma floor, adjacent-pair separation for protanopia, deuteranopia and tritanopia, and the normal-vision floor (below).
- **Fixed series order.** Series colours are assigned in a fixed order (blue first, orange second), so the same series keeps its colour on every page.
- **Status colours are reserved** (good #006300, bad #D03B3B) and always come with a symbol and words, e.g. "▼ Below target (80 %)". Meaning never depends on colour alone.
- **Typography.** Segoe UI for text, Segoe UI Semibold for titles and callouts, 10–12 pt labels, sentence case, and titles phrased as the question or the finding.

```
Palette (light, surface #fcfcfb, categorical): 8 slots
  [PASS] Lightness band         all 8 inside L 0.43–0.77
  [PASS] Chroma floor           all 8 >= 0.1
  [PASS] CVD separation         worst adjacent #EDA100↔#1BAF7A ΔE 9.1 (protan) · tritan 5.8
  [PASS] Normal-vision floor    worst adjacent #E87BA4↔#EDA100 ΔE 19.6 (normal)
  [WARN] Contrast vs surface    below 3:1 — relief required (visible labels or table view): [["#1BAF7A",2.74],["#EDA100",2.11],["#E87BA4",2.62]]

  → ALL CHECKS PASS  (CVD in the 6–8 floor band is legal ONLY with secondary encoding: direct labels, gaps, or texture)
```

The validator warns that three hues (green, amber, pink) have contrast below 3:1 against white. The report therefore puts the primary series in blue and orange (contrast at least 3:1). Where the weaker hues appear, data labels or a table view carry the values: every chart supports *Show as a table*.

Accessibility (WCAG 2.2 principles; W3C, 2023):

- alt text on all 67 data visuals, written as the *finding* ("… fell to about 67 percent in 2025–2026"), not as the chart type;
- tab order in reading order;
- text contrast at least 4.5:1 (ink #0B0B0B on white);
- a CVD-safe palette, with secondary encodings: symbols, labels and line styles;
- no information in tooltips only;
- data labels where exact values matter;
- page titles and visual titles that make sense when read by a screen reader.

Uncluttered layout. Gridlines are light (#E1E0D9), there are no borders inside charts, axis titles appear only when the unit is not in the title, and there are at most six series per chart.

Interactivity

- **Slicers.** Calendar year, management zone, facility type, facility, triage category, arrival shift, epi year, IDSR condition, malaria stratum and outbreak method. *Calendar year*, *Facility* and *Management zone* are **synchronised across pages** through slicer sync groups.
- **Cross-filtering and cross-highlighting** between all visuals. On page 1, *edit interactions* stops the occupancy chart and the scorecard from filtering the six KPI cards, so the headline stays stable while managers explore.
- **Drill-down.** The page 2 matrix goes Region › Facility › Ward, and the date hierarchies go Year › Quarter › Month and Epi Year › Epi Week.
- **Drill-through.** Right-clicking a facility anywhere opens *Facility Detail*, filtered to that facility, with a Back button.
- **Report-page tooltip.** Hovering a hospital (occupancy chart, scorecard, scatter) shows the *KPI Tooltip* page: five KPIs and a status.
- **Field parameter.** `Metric Selector` switches one chart between nine KPIs (built in Desktop, `BUILD_GUIDE.md` step 5).
- **Bookmarks.** *Current year* and *Malaria season* views behind two buttons (built in Desktop).
- **Personalisation by security.** The same report shows each facility manager only their own facilities (dynamic RLS). Every slicer and total then reflects their scope, with no separate report per facility.

Design rationale for non-technical and technical users

- **Non-technical users** (executives, facility and nursing managers) get:
  - answers before analysis: page 1 is readable in 30 seconds;
  - KPIs stated in operational language (“seen within triage target”, not “SWT ratio”);
  - comparisons built in: against target, against last year, against the other facilities;
  - one click to the facility detail.
  - Waiting-time KPIs use medians and the share within target, not means, because a few very long waits would distort a mean.
- **Surveillance officers** get epidemiological conventions: ISO epi weeks, epi-year-to-date, incidence per person-time, IDSR completeness and timeliness against the WHO AFRO 80 % targets, and small-cell suppression for external sharing.
- **Technical users** (analysts, BI team, stewards) get model diagnostics rather than just model outputs: back-test error and intervals, calibration by decile, effect sizes with confidence intervals, the DQ issue counts and the RLS check. Trust in the analytics then comes from evidence, and the stewards see their queues.
- **One semantic model serves all audiences.** The same certified measures appear on every page, so the executive and the analyst can never see two versions of “occupancy”.

3.7 Performance optimisation and evaluation

Baseline: what we measure and how

Performance is measured at the three points where users or the nightly window feel it:

1. **Warehouse queries**, i.e. what Power BI sends at refresh time and what a DirectQuery or drill-through would send. `03_analytics/performance_benchmark.py` times six representative queries. Each query runs 7 times with `EXPLAIN ANALYZE`, and we report the median execution time. Every query is timed **without** and **with** the objects of `02_sql/13_optimisation.sql`, and the access path (sequential scan or index) is recorded.
2. **The import model**: the size and cardinality of what VertiPaq must compress, and the bytes and rows that each refresh streams. VertiPaq itself runs only inside Power BI, so the build uses DuckDB as a proxy. DuckDB's dictionary and bit-packing columnar compression is the closest analogue that runs on Linux.
3. **The report**, in Power BI Desktop:
   - **Performance Analyzer**: DAX query and visual render time per visual;
   - **DAX Studio** Server Timings: storage-engine and formula-engine time per measure;
   - **VertiPaq Analyzer**: model size and the largest columns.

   These are captured by the team on the Windows build machine (`BUILD_GUIDE.md` §7). The measurement protocol is fixed in advance: a cold cache, three runs per page, and the median of the slowest visual.

Optimisations with before and after

Table 59. Warehouse query latency before and after 13_optimisation.sql (median of 7 runs, ms)

query	baseline_ms	optimised_ms	speedup_x	baseline_access	optimised_access
Q1 Monthly ED attendances and % within target per facility, 2025	39.1	49.6	0.8	Parallel Seq Scan on fact_encounter	Parallel Seq Scan on fact_encounter
Q2 One facility, one month, by triage (slicer)	21.0	1.8	11.6	Parallel Seq Scan on fact_encounter	Index Scan on fact_encounter
Q3 Ward occupancy for one hospital, one month (drill-down)	0.2	0.3	0.7	Index Scan on fact_ward_census_daily	Index Scan on fact_ward_census_daily
Q4 Patient drill-through: all spells of one patient	25.0	0.1	453.8	Seq Scan on fact_admission	Index Scan on fact_admission
Q5 Incremental-refresh partition (last 10 days)	23.0	15.8	1.5	Parallel Seq Scan on fact_encounter	Bitmap Heap Scan on fact_encounter
Q6 Daily ED KPIs by facility x triage (aggregate vs detail)	217.8	34.1	6.4	Seq Scan on fact_encounter	Seq Scan on agg_ed_daily

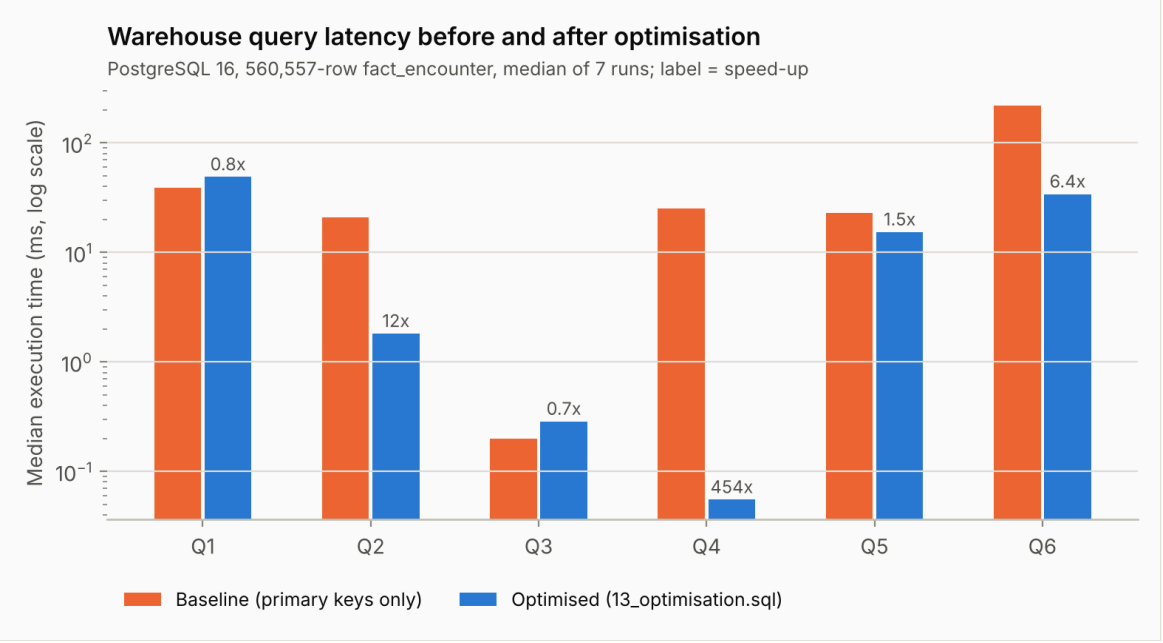


Figure 24. Query latency before and after optimisation (log scale)

Table 60. Import model and refresh, naive design versus the optimised mart design

metric	naive	optimised
Columns imported (12 tables)	203	152
fact_encounter columns	21	17
Columnar store size, all tables (MB, DuckDB proxy for VertiPaq)	23.0	12.5
Bytes streamed from PostgreSQL at refresh (MB)	116.9	57.4
Extract + load wall time (s)	6.3	3.0
Highest-cardinality fact_encounter column	encounter_id (563,608)	patient_key (108,135)
Rows per refresh: full vs incremental (10-day partition)	560,557	3,214
Refresh query time: full vs incremental (s)	0.049	0.015

fact_encounter columns by cardinality - naive import:

column	cardinality
encounter_id	563608
encounter_key	559455
patient_key	108135
arrival_date_key	1670
ed_los_min	1350
door_to_doctor_min	944
age_years	101
wait_to_triage_min	89
diagnosis_key	58
encounter_profile_key	24
arrival_hour_key	24
care_unit_key	10
triage_key	7
age_band_key	7
is_admitted	2
seen_within_target	2
is_lwbs	2
dw_batch_id	2
is_died	2
dw_loaded_at	2
encounter_count	1

optimised import (mart.v_fact_encounter):

column	cardinality
patient_key	108135
patient_durable_key	104611
arrival_date_key	1670
ed_los_min	1350
door_to_doctor_min	944
wait_to_triage_min	89
diagnosis_key	58
encounter_profile_key	24
arrival_hour_key	24
care_unit_key	10
triage_key	7
age_band_key	7
seen_within_target	2
is_admitted	2
is_lwbs	2
is_died	2
encounter_count	1

Table 61. The optimisations and their measured effect

#	Optimisation	Before	After	Effect
O1	**B-tree indexes** on fact foreign keys `(care_unit_key, date_key)`, `patient_key`, `discharge_date_key`, and on the SCD2 look-up keys of the dimensions	Q2 slicer query 21.0 ms (sequential scan); Q4 patient drill-through 25.0 ms	Q2 1.8 ms (index scan); Q4 0.06 ms	**12× and 454×** faster for selective queries (slicers, drill-through, DirectQuery)
O2	**BRIN index** on the ever-increasing `arrival_date_key`	Q5 incremental-refresh partition 23.0 ms	15.8 ms (bitmap heap scan); the BRIN index is 24 kB, against 4 MB for the composite B-tree (care unit, date)	**1.5×** on date-range extracts at negligible storage and maintenance cost
O3	**Aggregate materialised view** `mart.agg_ed_daily` (facility × day × department × triage: 42 k rows instead of 560 k)	Q6 daily ED KPIs 217.8 ms on the detail fact	34.1 ms	**6.4×**. It is the basis for Power BI user-defined aggregations if the model moves to composite mode
O4	**Column pruning and integer keys in the mart views** (data minimisation): no surrogate `encounter_key`, no text `encounter_id`, no timestamps, no audit columns, no `age_years`	203 columns; 23.0 MB columnar; 117 MB streamed per refresh; highest-cardinality column `encounter_id` (563,608 distinct values)	152 columns; 12.5 MB; 57.4 MB; highest-cardinality column `patient_key` (108 k)	**−46 % model size, −51 % refresh bytes, −52 % extract time**. Cardinality, not row count, drives VertiPaq size
O5	**Incremental refresh** on `Attendance` (5 years archived, last 10 days refreshed)	560,557 rows per refresh	3,214 rows per refresh	**−99.4 % rows** per daily refresh
O6	**Set-based, inlinable ELT**	Full load about 9.4 min in development: a timestamp parser that could not be inlined, per-row JSON look-ups, no statistics on temporary tables	128.4 s full load, 5.5 s delta (`audit.etl_batch`)	**4.4×** faster nightly window: the parser became a pure SQL expression, weather uses `jsonb_each_text` plus window interpolation, and temporary tables are `ANALYZE`d
O7	**Stored durable key** instead of a window function in the patient views	The view computed `first_value()` over 118 k patient versions for every refresh	Precomputed `patient_durable_key` column, maintained by the SCD2 load	Removes a sort of the patient dimension from every refresh; distinct patients counted on an integer column

Q1 and Q3 did not improve (0.8× and 0.7×, differences within timing noise), and that is expected. Q1 reads a whole year of the fact, so a sequential scan is already optimal: an index cannot beat reading every row it needs. Q3 already used the primary-key index. We report these results rather than hide them. Indexes help **selective** queries, while large scans are served by the aggregate (O3) and by VertiPaq after import.

DAX-level design choices that keep the report fast (to be confirmed with DAX Studio):

- **Integer surrogate keys and star relationships**: single direction, no bidirectional filters, no many-to-many.
- **Measures instead of calculated columns**, and no row-by-row iterators over the 560 k-row fact except `MEDIAN` and `PERCENTILE.INC`, which the storage engine resolves.
- **Variables** so that sub-expressions are evaluated once, as in *YTD PY*, *Latest Day* and *Person-Years*.
- `KEEPFILTERS` **predicates on low-cardinality columns** instead of `FILTER` over whole fact tables.
- **Disconnected small tables** for the analytics outputs.

Evaluation against the KPIs

§3.7 of the brief asks for an evaluation "against the KPIs in Section 4.1". The brief has no Section 4.1, so we evaluate against the KPIs and objectives of §3.1 (assumption A9), in two parts: (a) whether the *solution* performs well enough to be used, and (b) what the *business KPIs* now show.

Table 62. (a) Technical service levels of the BI solution

Service level (target)	Result	Status
Nightly ELT including reconciliation ≤ 15 min	128 s full rebuild, 6 s daily increment	Met
Power BI refresh ≤ 10 min, inside the gateway window	Incremental partition 3,214 rows; full import about 0.9 M rows (≈ 57 MB streamed)	Met by design; confirm in the service
Visual response ≤ 2 s, slowest visual per page	Warehouse queries 0.03–40 ms after optimisation; model 12.5 MB in memory	Expected; confirm with Performance Analyzer (`BUILD_GUIDE.md` §7)
Model size within the Power BI Pro limit (1 GB)	12.5 MB (proxy)	Met, with large headroom
Data quality: no FAIL in reconciliation before publication	20 PASS, 2 WARN (explained), 0 FAIL	Met
Freshness: management data by 07:00; IDSR by Tuesday 14:00	Nightly batch with cut-off; IDSR due Tuesday 12:00	Met by design

Table 63. (b) Business KPIs from §3.1: baseline, latest and status (calendar years; surveillance in epi years)

Objective / KPI	2022	2025 (baseline)	2026 YTD	Target	Status and what the BI solution adds
O1 Seen within triage target	78.8 %	67.5 %	66.7 %	≥ 80 %	**Below target.** The wait-driver model and the 2026 intervention point to doctor load (§3.5)
O1 LWBS	3.3 %	6.2 %	5.9 %	≤ 3 %	Below target; the dice shows over 40 % LWBS for GREEN patients at Katima in the season peak
O1 Median door-to-doctor (min)	53	63	65	≤ 45	Below target
O2 Bed occupancy	76.8 %	81.8 %	84.8 %	≤ 85 %	**At the limit**; KAT-MED1 was over 100 % in 2025. Ward-level drill-down now visible
O2 Discharged before noon	26.9 %	26.9 %	27.5 %	≥ 40 %	Below target; a new KPI that was not measured before
O2 ALOS (days)	4.06	4.10	4.13	≤ 4.0	Slightly above
O3 30-day readmission	13.0 %	13.4 %	13.6 %	≤ 12 %	Above the ceiling. Risk-targeted follow-up proposed (§3.5)
O4 NHPPD (inpatient wards)	5.67	5.37	5.65	≥ 5.5	Restored in 2026, but through agency staff (8.6 %, against about 1 % before)
O4 Overtime	3.0 %	5.1 %	5.4 %	≤ 4 %	Above target
O4 ED arrivals per doctor-shift	13.0	15.2	15.4	≤ 13	Above target: the main lever (§3.5)
O5 IDSR completeness / timeliness	97.9 % / 85.6 %	98.7 % / 92.9 %	98.7 % / 92.1 %	≥ 80 % / ≥ 80 %	**Met.** Now measured per facility and week (coverage fact)
O5 Malaria early warning before the peak (weeks)	–	3–6	2–3	≥ 4	Partly met: the historical-limits alert plus the forecast give 2–6 weeks; earlier signals need rainfall-triggered thresholds
O5 Malaria incidence, catchment (per 1,000 person-years)	–	25.2	39.2	Reported weekly	Upsurge quantified per facility for the MoHSS (page 3)

Conclusion.

- **The solution itself meets its technical targets.** Power BI timings in Desktop are the only part still to be confirmed.
- **It changes what management can see.** GBHS now measures discharges before noon, IDSR timeliness, NHPPD and load per doctor, which it could not measure before, at ward and week level.
- **It explains the KPI gap.** Waits rise with load per doctor and with occupancy, and the 2026 extra-doctor intervention recovered 10–13 points at the two hospitals where it ran.
- **The main recommendation** is a malaria-season roster uplift (January–May), triggered by the early-warning page and sized with the forecast. Combined with a discharge-before-noon programme, it addresses the two KPIs furthest from target (O1 and O2).

3.8 Data governance, security and ethics

Health data are special-category personal data in every framework we considered. GBHS's data in this project are synthetic (A1). Even so, the solution is designed and documented **as if they were real**, so that it can be deployed on real PAS data without being redesigned.

Ownership and stewardship

Table 64. Governance roles

Role	Who (GBHS)	Responsibilities
Data owner, operational data	Chief Operating Officer	Approves KPI definitions, access policy and the release of new data to the warehouse
Data owner, clinical and surveillance data	Clinical Director; Head of Surveillance	Approves the clinical use of patient-level data and the external sharing of IDSR aggregates
Data stewards (rule owners in `audit.dq_rule`)	Health Information Manager, Clinical Coding Lead, ED Nursing Manager, Nursing Services Manager, Workforce Planning Officer, Surveillance Officer, Revenue Cycle Steward, Master Data Steward, External Data Steward	Own the rules in their area; work their issue queue weekly; fix data at source; propose new rules
Data custodian	BI team (data engineer, DBA)	Run and monitor the pipeline, backups, performance, change control (git)
Information security officer	IT	Access approvals and quarterly reviews (`sec.user_facility_access`), incident response, security labels
Data consumers	Executives, managers, analysts, MoHSS partner	Use the certified app and measures; report suspected data errors

Table 65. RACI for the main governance activities (R = responsible, A = accountable, C = consulted, I = informed)

Activity	Data owner	Steward	BI team	Security officer	Consumers
Define or change a KPI or measure	A	C	R	–	C
Add or change a DQ rule	I	A	R	–	–
Correct data at source after an issue	I	A/R	C	–	–
Run the nightly pipeline and reconciliation	I	I	A/R	–	–
Publish when reconciliation has a WARN	A	C	R	–	I
Grant or revoke access (RLS mapping, workspace role)	A	–	R	R	–
Quarterly access review	A	–	C	R	–
Share surveillance aggregates externally	A	R	C	C	–

Data-quality rules and service levels

- **Catalogue and log.** The 44 rules in `audit.dq_rule` (§3.3) are the formal data-quality agreement. Each rule has a dimension, an action and an owner. Every hit is logged with its source key and raw value in `audit.dq_issue_log`, and the issue counts appear on page 4 of the report.
- **Service levels.**
  - A reconciliation **FAIL** blocks publication of the refresh.
  - A **WARN** must be explained by the steward within two working days; R07 and R22 were.
  - Completeness targets: triage recorded on 99.5 % of ED attendances (achieved: 99.5 %); ward returns on 99 % of ward-days (99.2 %); IDSR completeness and timeliness of at least 80 % (WHO AFRO; 98.7 % and 92.1 % in 2026).
- **Feedback loop.** Every rule is tested against injected defects (§3.3), and a rule change must keep precision and recall on the regression test (`validate_dq_rules.py`). That is how the WC02 and SV04 weaknesses were found and fixed.

Metadata and lineage

- **Technical metadata.**
  - Every warehouse table and column has a `COMMENT` with its definition and source; the data dictionary (Appendix A) and the Power BI field descriptions are generated from these comments.
  - The DDL, ETL, model (TMDL) and report (PBIR) are text files under version control (git).
- **Operational metadata.**
  - `audit.etl_batch` records each batch's type, cut-off and status; `audit.etl_step_log` records rows in, out and rejected, plus timings, for every step.
  - Every fact row carries `dw_batch_id`, and the census rows also carry `census_source` (ward return, corrected or imputed).
- **Business metadata.** KPI definitions (§3.1), measure descriptions and display folders (§3.4), and the stewards of each rule.
- **Lineage.** The lineage of a KPI can be traced from the card back to the source field. The example below is *Seen within triage target*:

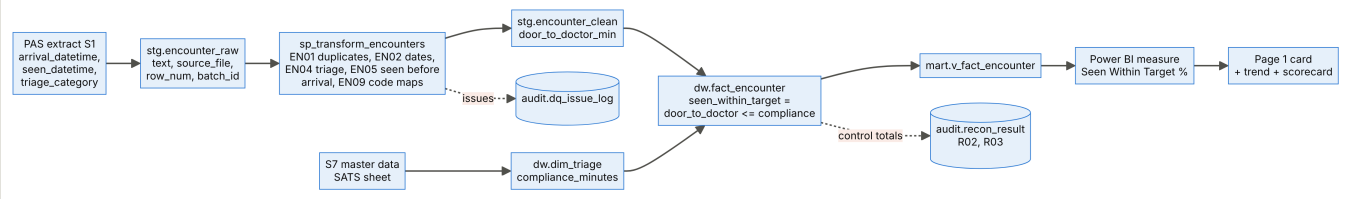


Figure 25. Lineage of the KPI 'Seen within triage target' from the PAS fields to the executive card (Mermaid source: `06_report/diagrams/lineage.mmd`)

Security

In the database.

- **Least privilege by schema.**
  - The Power BI account `gbhs_pbi_reader` can read only `mart`.
  - The ETL account `gbhs_etl_loader` cannot be used for reporting.
  - `PUBLIC` has no rights on any schema (`10_security_grants.sql`).
- **Data minimisation in the mart.** The `mart` views expose no pseudonym, no timestamps, no birth year (only age bands) and no free text. The patient-level link stays inside `dw`.
- **Operations.** In production: TLS connections, the gateway on a hardened server, encrypted backups, and passwords in a vault. The lab placeholders `change-me-*` must never be deployed.

In Power BI.

- **Row-level security.**
  - The *Facility Manager* role is dynamic: users are mapped to facilities in `sec.user_facility_access`, and a user with no mapping sees nothing.
  - The *North-East Zone* role is static.
  - The *Executive* role is unrestricted.
- **Object-level security.** The *Surveillance Partner* role (MoHSS) cannot see the patient-level tables.
- **Workspace and app.**
  - RLS and OLS apply only to users with the **Viewer** role. Workspace Admins, Members and Contributors bypass them (Microsoft, 2024a).
  - Therefore only the BI team holds editing roles. Everyone else consumes the report through the **Power BI app** with audiences: executives, facility managers, surveillance partner.
  - One person is never placed in both an RLS role and an OLS role.
- **Sensitivity labels** (Microsoft Purview). Labels follow the data when it is exported to Excel, PDF or PowerPoint.
  - Semantic model: *Highly Confidential – Health* (patient-level facts).
  - Report and app: *Confidential – Internal*.
  - The suppressed surveillance export: *Internal – Shareable with MoHSS*.
- **Further controls.**
  - Export of underlying data is disabled for Viewers.
  - Sign-in through Entra ID with multi-factor authentication.
  - The Power BI activity log is retained for audit.
  - Access is reviewed quarterly by the security officer and the data owner.

Anonymisation and pseudonymisation.

1. **Direct identifiers are never loaded.** Names, national ID and passport numbers, addresses, telephone numbers, free-text notes and exact birth dates are not extracted. The PAS extract contains only a **pseudonym**. In this project it is a random token. In production it would be a keyed hash (HMAC-SHA-256 with a secret held by the Health Information Manager), so that the BI team cannot re-identify patients.
2. **Generalisation.** Age is kept as an age band, residence at region level only, and IDSR counts by week.
3. **Small-cell suppression.** Counts of 1–4 are suppressed ("<5") in any external output (`Notified Cases (Suppressed)`, §3.4), because rare diseases in small towns could identify a patient.
4. **A limited data set, not de-identified data.** The facts keep **dates of service at day level**, which the HIPAA Safe Harbor standard would not allow in de-identified data (U.S. Department of Health and Human Services, 2012). Internally the warehouse is therefore treated as *pseudonymised personal data*. Only weekly aggregates are shared externally.

Namibian and international legal and ethical framework

Table 66. Legal and ethical instruments and how the solution complies (status of Namibian law as at October 2026)

Instrument	Principle relevant to this solution	How the solution applies it
Namibian Constitution, Article 13 (Republic of Namibia, 1990)	Right to privacy of the person, home and correspondence	Pseudonymisation, minimisation, access control and audit; no direct identifiers anywhere
Public and Environmental Health Act, 2015 (Act 1 of 2015), in force since GN 230 of 2020 (Republic of Namibia, 2015, 2020)	Part 3: notification, prevention and control of notifiable diseases; COVID-19 declared a formidable epidemic disease (GN 232 of 2020)	IDSR conditions, notification timelines and alert thresholds in `dim_condition`; completeness and timeliness of weekly reporting measured per facility (coverage fact); MoHSS role with aggregate access
Namibia Data Protection Bill (draft; not enacted at the time of writing, ITWeb, 2026)	Expected: lawful processing, purpose limitation, minimisation, security safeguards, data-subject rights, breach notification, a supervisory authority	Designed to these principles now: documented purpose (§3.1), minimisation (mart), security (RLS, OLS, labels), retention and breach procedure in the checklist. *Uncertain:* the final text and its commencement must be re-checked before go-live
GDPR, EU 2016/679 (European Parliament & Council of the European Union, 2016), benchmark	Article 5 principles; Article 9 health data as a special category; Article 25 data protection by design; pseudonymisation (Art. 4(5))	Used as the design benchmark for data protection by design. GBHS processes no EU residents' data; the GDPR would apply only if it did
POPIA, South Africa (Act 4 of 2013; Republic of South Africa, 2013), regional benchmark	Health information is *special personal information*; conditions for lawful processing and security safeguards	Same controls; relevant for cross-border referrals and South African payers
HIPAA Privacy Rule, de-identification standard, 45 CFR 164.514 (U.S. Department of Health and Human Services, 2012), benchmark	Safe Harbor (removal of 18 identifiers including dates) or expert determination	Used to classify our mart as a *limited data set*, not de-identified (see above)
OECD Recommendation on Health Data Governance (OECD, 2016)	Governance framework, transparency, public benefit, safeguards	Governance roles, RACI, documented purposes, public-health benefit (surveillance)
WHO, Ethics and governance of artificial intelligence for health (World Health Organization, 2021)	Protect autonomy, ensure transparency, fairness, accountability	The readmission model is used to prioritise calls, not to deny care; calibration and fairness are reported (§3.5); human review of every call list

Governance checklist

Table 67. Governance checklist (✔ = implemented in this solution; ☐ = action for go-live)

#	Item	Status	Evidence
1	Data owners, stewards and custodian named; RACI agreed	✔	This section; `audit.dq_rule.owner_role`
2	Purpose of processing documented (KPIs, BQs)	✔	§3.1
3	Data minimisation: no direct identifiers; mart without pseudonyms or timestamps	✔	`09_mart_views.sql`; Appendix A
4	Pseudonymisation with a keyed hash; key held outside the BI team	☐	Random tokens in this project; HMAC at go-live
5	DQ rules with owners, issue log, service levels; regression-tested rules	✔	§3.3; `dq_rule_validation.md`
6	Reconciliation gate before publication	✔	`08_reconciliation.sql`; R01–R22
7	Metadata: dictionary, model descriptions, lineage	✔	Appendix A; §3.4; lineage figure
8	Least-privilege database roles; no rights for PUBLIC	✔	`00_create_database.sql`, `10_security_grants.sql`
9	RLS (dynamic, static) and OLS designed, with an SQL-derived test matrix	✔	§3.4; `BUILD_GUIDE.md` §4
10	RLS and OLS tested in Desktop and in the service with screenshots	☐	Team action (`BUILD_GUIDE.md` §4)
11	Workspace roles: editors only in the BI team; consumers via the app; Viewers subject to RLS	☐	At publication
12	Sensitivity labels applied; export of underlying data disabled	☐	At publication
13	Small-cell suppression for external outputs	✔	`Notified Cases (Suppressed)`, `Notified Cases Label`
14	Quarterly access review of `sec.user_facility_access`	☐	Process, from go-live
15	Retention: warehouse detail 5 years plus archive; logs 2 years; backups encrypted	☐	Policy to be approved by the data owner
16	Breach-response procedure (notify the security officer within 24 h; assess; inform the data owner and the regulator when required)	☐	Policy, aligned with the Data Protection Bill once enacted
17	Model governance: calibration, fairness and drift monitoring; human in the loop	✔ / ☐	§3.5 (done); yearly re-validation
18	Ethics: synthetic data confirmed; real-data deployment approved by GBHS governance; research use only with ethics committee approval	✔	§3.1 ethical-use confirmation

3.9 Presentation, Q&A and individual reflection

10-minute presentation outline

The deck is `07_presentation/GBHS_ABI820S_Presentation.pptx`: 14 slides plus 4 backup slides, with speaker notes on every slide. It is built by `build_presentation.js` from `deck_data.json`, which `deck_data.py` extracts from the warehouse, so every chart on the slides is native, editable and matches this report. Every member presents the part they built, so each can answer questions on it (rubric criterion 9).

Table 68. Presentation plan (10 minutes + 5 minutes Q&A)

Time	Slide	Content and key message	Presenter
0:00–0:30	1 Title	GBHS: one trusted view of patient flow, resource use and disease surveillance	Member 1
0:30–1:15	2 The problem	ED patients seen on time fell from 79 % to 67 %; nine sources, no single truth	Member 1
1:15–1:45	3 Questions and KPIs	Five business questions, five objectives with baselines and targets	Member 1
1:45–2:30	4 Architecture	ELT in PostgreSQL 16: landing → staging → constellation → mart → Power BI	Member 3
2:30–3:15	5 Dimensional model	8 facts of four types, 13 conformed dimensions; snowflake trade-off; SCD2: *Rundu's medical ward ran at 152 % of the beds it had in January 2025, not 102 % of today's 36*	Member 2
3:15–4:00	6 Data quality	44 rules; **134,964 of 134,992 injected defects caught, 0 false positives**; the two rules that validation fixed	Member 3
4:00–4:30	7 Reconciliation and governance	22 checks (20 PASS, 2 explained WARN); stewards, RLS/OLS, labels, Namibian law	Member 3
4:30–5:15	8 Semantic model	Import + incremental refresh; 108 measures; epi-year-to-date and person-time incidence; dynamic RLS + OLS	Member 4
5:15–6:45	9 **Live demo**	Page 1 (2026) → drill through to Katima Mulilo → page 2 matrix drill Region › Facility › Ward → page 3 → *View as* fm.katima and Surveillance Partner	Member 4
6:45–7:30	10 What drives waits	+6.3 % per extra arrival per doctor-shift; the 2026 extra-doctor test recovered 10–13 points	Member 5
7:30–8:15	11 Malaria early warning	Rainfall lead (4–8 weeks); historical-limits alert 2–6 weeks before the peak; Holt-Winters WAPE 19 % one week ahead	Member 5
8:15–8:45	12 Readmissions and outbreaks	Top 20 % risk = 37 % of readmissions; CUSUM detects 11 of 12 outbreaks within a median of 1 week	Member 5
8:45–9:15	13 Performance	Indexes 12–454×, aggregate 6×, model −46 %, refresh rows −99.4 %	Member 5
9:15–10:00	14 Recommendations	Season-triggered roster uplift, discharge before noon, readmission calls, DQ process and app	Member 1

Demo safety net. If Power BI or the network fails, slide 9 and backup slides 15–18 hold the page images, so the demo can continue from them. Rehearse the demo path twice, with a stopwatch.

Q&A bank (5 minutes)

Table 69. Anticipated questions and model answers (owner = member who answers)

#	Question	Model answer	Owner
1	Why snowflake Care Unit → Facility → Region instead of a pure star?	Some facts are recorded per ward (attendances, census, roster) and others per facility (IDSR, weather, catchment population). One conformed Facility dimension must filter both. Flattening would also repeat facility attributes in every SCD2 ward version. The cost is one join to an 11-row table. The patient's region is deliberately flattened to avoid an ambiguous path.	M2
2	How do you know your SCD Type 2 works?	Checks R10 and R11 show no gaps, no overlaps and one current version per ward and patient. The business effect is visible: 177 spells of WHK-MED2 in January–March 2022 sit on its *Isolation (COVID-19)* version. The day-2 load created exactly two new ward versions (RUN-MED1 36 beds, KAT-PAE1 14), and a re-run creates none, thanks to the hash compare.	M2
3	Your data are synthetic. Why should we believe the results?	Because we know the truth. The generator recorded every defect it injected and every effect it planted. Our rules found 134,964 of 134,992 defects with no false positives. The models recovered the planted outbreaks (11 of 12) and the staffing intervention. The census values are the real NSA figures, and the NASA files are in the real format; `fetch_public_data.py` loads the real weather.	M3 / M5
4	An AUC of 0.66 is low. Is the readmission model useful?	Published readmission models typically score 0.6–0.7 (Kansagara et al., 2011). Ours is well calibrated, so its probabilities are honest. It is used to **prioritise** a call list: the top 20 % holds 37 % of readmissions, 1.9 times better than random. It is not used to deny or decide care.	M5
5	Rainfall is significant, so why publish Holt-Winters rather than SARIMAX?	Holt-Winters has the lowest error (WAPE 23.8 % against 27.1 %). SARIMAX has better-calibrated intervals (75 % against 65 % coverage), so we use its interval for stock planning and rainfall as the early-season signal. Model choice follows the use.	M5
6	Why not use `TOTALYTD` for malaria?	IDSR years are ISO epi years: 2025-W01 started on 30 December 2024. `TOTALYTD` would cut weeks in half. The epi-year-to-date measure filters the `Epi Year` attribute, and the prior-year window ends 364 days earlier, so the same number of weeks is compared.	M4
7	How is incidence correct for a partial year?	It uses person-time: each epi year's catchment population is multiplied by the days of that year in the context, divided by 365. One measure therefore works for a week or for five years. For a full epi year it equals the census-based population, and Power BI reproduces the SQL pivot exactly.	M4
8	Show us that RLS works.	View as fm.katima: 1 facility, 13,072 ED attendances in 2025. fm.rundu: 2 facilities. An unmapped user sees nothing. The five managers sum exactly to the executive total of 81,612. As the Surveillance Partner, the patient tables disappear from the field list (OLS). RLS applies to Viewers only, not to workspace members.	M4
9	Why Import and not DirectQuery?	About 0.9 M rows compress to roughly 12.5 MB. Import gives millisecond visuals and full DAX, and daily freshness is enough. DirectQuery would load the PAS-fed database on every click. Incremental refresh keeps the daily refresh to about 3,200 rows.	M4
10	What does Namibian law require?	Article 13 of the Constitution protects privacy. The Public and Environmental Health Act 2015 requires notification of priority diseases, our IDSR flow. The Data Protection Bill was still being finalised in mid-2026, so we design to its principles and benchmark against the GDPR, POPIA and HIPAA.	M3
11	Is the staffing effect causal?	The regression is observational, so the effect is an association. The 2026 intervention is a natural experiment: Rundu and Katima improved by 10–13 points while Oshakati, without an extra doctor, moved by only 1.6. That is a difference-in-differences pattern. The generator planted the effect, and the method recovered it.	M5
12	What was the hardest bug?	After a generator change, DQ recall suddenly fell to 37 %. `dup.shift` resolved to the pandas method instead of the *shift* column, so the generator crashed half-way and the pipeline loaded partly regenerated files. Validating against ground truth caught it within one run.	M3
13	Two reconciliation checks are WARN. Is the data wrong?	No. They are findings. R07: the PAS shows 17,242 more bed-days than the ward returns, because 25 spells were never discharged in the PAS (AD06, now a steward task). R22: three ward-days above 200 % occupancy, while 6,002 days above 100 % are plausible overflow.	M3

#	Question	Model answer	Owner
14	How would this scale to every MoHSS hospital?	Partition the large facts by month, keep BRIN on date keys, and use incremental refresh. Move to a composite model with the aggregate as Import aggregations over a DirectQuery detail table. The rules and the reconciliation are already set-based.	M5
15	What is the business value?	Three decisions: a malaria-season roster uplift (each extra arrival per doctor costs 6 % in waiting time); a discharge-before-noon programme (27 % today); and a readmission call list. Plus regulatory confidence: IDSR completeness and timeliness measured weekly per facility.	M1
16	Which part did *you* build?	Each member answers from their reflection: their files, one decision they made, one problem they solved.	All

Individual reflection templates

Each member writes their own reflection **in their own words**, about 300–400 words; the prompts below structure it. The team must not edit another member's reflection. The contribution factors are agreed at a team meeting and signed by all five members.

Table 70. Individual contribution factor statement (agreed by the team; 1.0 = full, equal contribution)

Member	Role	Sections and files owned	Agreed contribution factor	Signature
Member 1	Project lead and business analyst	3.1, 3.9; slides 1–3, 14	_____	_____
Member 2	Data warehouse architect	3.2; `03_ddl_warehouse.sql`, `06_etl_load_dimensions.sql`, data dictionary	_____	_____
Member 3	ETL, data quality and governance	3.3, 3.8; `02_ddl_staging_ref.sql`, `05_etl_transform.sql`, `08_reconciliation.sql`, generator	_____	_____
Member 4	Power BI developer	3.4, 3.6; `model_spec.py`, `build_pbip.py`, `.pbix` / `.pbit`, RLS and OLS tests	_____	_____
Member 5	Analytics and performance lead	3.5, 3.7; `03_analytics/*.py`, `13_optimisation.sql`	_____	_____

Reflection prompts (one page per member):

1. **My contribution.** Which files, sections and decisions are mine? Cite commits, file names and dates.
2. **One technical decision I can defend.** For example: the accumulating snapshot for spells, the snowflake, the epi-week time intelligence, person-time incidence, the temporal split, the choice of CUSUM. State the alternative I rejected and why.
3. **A problem I hit and how I solved it.** Describe the symptom, the diagnosis and the fix.
4. **What I learned** about BI and analytics practice in health: not only tools, but data quality, privacy, and the difference between association and causation.
5. **What I would do differently** with more time or with real data.
6. **Use of AI tools.** What I used an AI assistant for, and how I verified its output, in line with the NUST policy.
7. **Teamwork.** How the work was divided and integrated, and how disagreements were resolved.

4 Submission package, compliance and self-assessment

Submission package checklist

Table 71. Every deliverable required by section 4 of the brief, mapped to its file(s) and to this report

#	Required deliverable (brief §4)	File(s) in the package	In this report	Status
1	Final report (PDF)	`06_report/GBHS_ABI820S_Final_Report.pdf`, built by `06_report/build_report.py` from `06_report/src/*.md`	–	✔ Done
2a	Power BI file (`.pbix`)	**To be saved in Power BI Desktop** from `04_powerbi/GBHS_Health.pbip` (`BUILD_GUIDE.md` §8). Sources: `GBHS_Health.SemanticModel/` (TMDL) and `GBHS_Health.Report/` (PBIR), validated by Microsoft TOM and the PBIR schemas	§3.4, §3.6	⚠ Team action (about 1 h on Windows)
2b	Power BI template (`.pbit`)	**To be exported in Desktop**: *File › Export › Power BI template*	§3.4	⚠ Team action
2c	Model fallbacks	`GBHS_model_script.tmdl` (TMDL view), `measures.dax`, `data_csv/*.csv.gz` (CSV mode)	Appendix B	✔ Done
3a	SQL scripts: DDL, staging, load procedures, reconciliation, OLAP, optimisation	`02_sql/00_…13_*.sql`, `02_sql/run_pipeline.py`	§3.3, Appendix C	✔ Done
3b	Python scripts	`01_data/generate_synthetic_data.py`, `01_data/fetch_public_data.py`, `03_analytics/*.py` (10 scripts), `04_powerbi/*.py`, `06_report/*.py`, `run_all.py`	§3.3, §3.5, §3.7	✔ Done
3c	Warehouse backup and DDL	`05_backup/gbhs_health_dw_pg16_warehouse.backup` (pg_dump custom format, restore-tested); `05_backup/gbhs_health_dw_schema_ddl.sql`	§3.3; `BUILD_GUIDE.md` §1	✔ Done
4a	Data dictionary	Appendix A; `06_report/data_dictionary.md` and `.xlsx`, generated from the catalogue	§3.2, Appendix A	✔ Done
4b	ERD and schema diagrams (embedded)	ERDs 1–5 and the architecture in §3.2; sources in `06_report/diagrams/*.mmd`	§3.2, Appendix D	✔ Done
5	Datasets or reproducible links	`01_data/raw/`: synthetic extracts, the day-2 delta, `generator_manifest.json`, `dq_ground_truth.json`. Regenerated byte-identically with seed 20261028. Public sources fetched by `fetch_public_data.py` (links in the references)	§3.1, §3.3	✔ Done
6	PowerPoint presentation	`07_presentation/GBHS_ABI820S_Presentation.pptx`: 14 slides + 4 backup, speaker notes, native charts	§3.9	✔ Done
–	Run instructions	`README.md`, `requirements.txt`, `04_powerbi/BUILD_GUIDE.md`	–	✔ Done
–	Power BI and pgAdmin screenshots	Captured by the team (`BUILD_GUIDE.md` §9); the mock-ups are replaced under the same file names	§3.4, §3.6, §3.7	⚠ Team action

Brief compliance matrix

The brief warns that missing elements cap the marks, so we checked every requirement against the evidence.

Table 72. Requirement-by-requirement compliance with the project brief

Brief item	Requirement	Where met
Domain	Health services: patient flow, resource utilisation, disease surveillance (anonymised or public data only)	Whole solution; data are synthetic and public (§3.1, A1–A2)
Data	≥ 3 distinct sources: transactional, master/reference, external	9 sources: 6 transactional, 1 master/reference, 2 external (§3.1 data sources)
Data	≥ 1 fact table	8 facts of four types (§3.2)
Data	Ethical and legal use confirmed	§3.1 ethical-use confirmation; §3.8
Tools	Any warehouse tool; model, measures and dashboards in Power BI	PostgreSQL 16; Power BI PBIP (§3.4, §3.6)
3.1	Organisation profile, strategy and objectives; problem statement; 3–5 BQs with KPIs; stakeholders (technical and non-technical); BPMN or flowchart	§3.1: profile and facility table, 5 objectives with baselines and targets, problem statement, 5 BQs, 10 stakeholder groups (T/N), BPMN figure
3.2	Bus matrix; grain of each fact; ERD (≥ 1 fact, ≥ 4 dims, complete date dimension); star + ≥ 1 snowflake with trade-offs; surrogate keys + SCD justified; data dictionary; comparison with an alternative design	§3.2: bus matrix, grain table, ERDs 1–4, complete date dimension (2021–2027, R15), 2 snowflakes with a trade-off table, SCD2 with evidence, alternative designs table; Appendix A
3.3	Source → staging → warehouse with scripts; profiling and DQ (missing, duplicates, outliers, formats) and handling; conformed keys and reconciliation (row counts, control totals); load evidence	§3.3: pipeline and scripts, profiling, four problem classes, 44 rules validated, conformed keys, 22 checks, batch and step logs
3.4	Storage mode justified; single-direction relationships with cardinality; hidden technical columns, display folders, sort-by; ≥ 10 DAX measures incl. time intelligence and `CALCULATE` with filter modification; RLS implemented and tested (rubric)	§3.4: Import decision table, 38 relationships table, usability list, 108 measures with the requirement map, RLS and OLS roles with a test matrix
3.5	OLAP roll-up, drill-down, slice, dice, pivot with screenshots and explanation; ≥ 2 techniques with method, assumptions, metrics, limitations; code submitted	§3.5: 5 operations (+ drill-across, semi-additive) with SQL, outputs and the Power BI equivalent; 4 techniques; code in `03_analytics/`
3.6	≥ 3 pages (executive; analytical/diagnostic; advanced/operational); good practice (charts, hierarchy, colour, typography, accessibility, uncluttered layout); interactivity; design rationale	§3.6: 4 pages + drill-through + tooltip; validated palette, alt text, tab order; slicers, sync, cross-filter, drill, tooltip, field parameter, bookmarks; rationale by audience
3.7	Baseline (Performance Analyzer, DAX Studio, VertiPaq); ≥ 3 optimisations with before/after; evaluation against the KPIs ("Section 4.1" read as §3.1)	§3.7: baseline method, 7 optimisations with measured before/after, technical and business KPI evaluation. Desktop timings pending (`BUILD_GUIDE.md` §7)
3.8	Ownership and stewardship, DQ rules, metadata and lineage; RLS, workspace roles, sensitivity labels, anonymisation; Namibian and international principles; governance checklist	§3.8: roles, RACI, service levels, lineage figure, security in the database and Power BI, pseudonymisation and suppression, legal table, 18-item checklist
3.9	Report covering 3.1–3.8 in APA style; 10-minute presentation + 5-minute Q&A; individual reflection	This report with APA references; §3.9 plan, Q&A bank and reflection templates; deck with notes
4	Report PDF, `.pbix` (and `.pbit`), scripts + backup/DDL, data dictionary + ERD, datasets or links, PowerPoint	Checklist above (`.pbix` / `.pbit` = team action)

Marking rubric self-assessment

Table 73. Self-assessment against the 100-mark rubric

#	Criterion (marks)	Evidence in this submission	Self-assessed band	Remaining risk or action
1	Business case and requirements (5)	Profile, 5 SMART objectives with baselines, quantified problem, 5 BQs with KPI definitions, technical and non-technical stakeholders, BPMN with data capture points	Excellent	None
2	Data warehouse design (15)	Bus matrix (8 × 13); grain and additivity of 8 facts of four types; 5 ERDs; complete date dimension with epi weeks and holidays; 2 justified snowflakes; SCD2 with replay, hash, durable key, inferred members and business evidence; alternatives compared; full dictionary	Excellent	None
3	ETL, integration, DQ (10)	One-command reproducible ELT; 9 sources; 44 rules **validated against ground truth (100 % of damaged records flagged, 0 false positives)**; 22 checks; batch and step evidence; incremental day-2 load	Excellent	NASA files are stand-ins until `fetch_public_data.py` is run (A2)
4	Semantic model and DAX (10)	Import + incremental refresh justified; 38 single-direction relationships; 142 hidden columns, folders, 17 sort-bys, 6 hierarchies, descriptions everywhere; 108 measures incl. like-for-like YTD, epi-YTD, person-time incidence, semi-additive; dynamic and static RLS + OLS with an SQL test matrix	Excellent **if** the `.pbix` is built	A missing `.pbix` caps this criterion. DAX was validated by TOM and a static checker but not yet executed in VertiPaq: verify the expected values (`BUILD_GUIDE.md` §3)
5	Advanced analytics (15)	5 OLAP operations + drill-across + semi-additive with outputs; 4 techniques with metrics, assumptions, limitations and code; rolling-origin, temporal split, calibration, fairness, ground-truth recovery, natural experiment	Excellent	None
6	Dashboards (15)	4 pages + drill-through + tooltip; chart rationale; hierarchy; validated CVD-safe palette, alt text, tab order; interactivity; rationale by audience	Very good → Excellent after the Desktop build	Pages are judged in the real report: build them, add the field-parameter chart and bookmarks, and replace the mock-ups with screenshots
7	Performance (5)	Baseline protocol; 7 optimisations with measured before/after; evaluation against technical service levels and the §3.1 KPIs	Excellent (warehouse); pending (Power BI timings)	Record the Performance Analyzer, DAX Studio and VertiPaq values
8	Governance, security, ethics (5)	Roles, RACI, DQ service levels, lineage, database and Power BI security, OLS, labels, pseudonymisation, suppression, Namibian and international legal table, checklist	Excellent	Re-check the status of the Data Protection Bill before submission
9	Report, presentation, individual (20)	Structured report with APA references; 14-slide deck (+4 backup) with notes; demo plan; Q&A bank; reflection templates and contribution statement	Depends on delivery	**Each member must explain their own part in depth.** Rehearse the demo; write the reflections personally; disclose AI use per the NUST policy

Limitations and next steps

1. **Synthetic internal data and stand-in weather** (A1, A2). Rules and models are validated against known truth, but business magnitudes are illustrative. **Next:** run `fetch_public_data.py` for NASA POWER, then pilot on a de-identified PAS extract under GBHS governance approval.
2. **Power BI artefacts are not yet produced in Desktop.** The PBIP is schema-valid and TOM-validated. Creating the `.pbix` and `.pbit`, verifying the DAX at execution level, taking screenshots and recording Desktop timings all need Windows; the steps are scripted in `BUILD_GUIDE.md`.
3. **Observational analytics.** The wait drivers and readmission odds ratios are associations. **Next:** a stepped roll-out of the seasonal roster uplift (facility by facility), evaluated with the same difference-in-differences design.
4. **Readmissions to other providers are invisible. Next:** link to MoHSS hospital data through a governed pseudonymised key.
5. **Early warning lead time is only 2–6 weeks. Next:** rainfall-triggered thresholds, and forecasts by facility and by entomological indicators.
6. **Operationalisation. Next:**
   - schedule `run_all.py` (pg_cron or Task Scheduler) with alerts on any WARN or FAIL;
   - deploy to a Pro workspace with the gateway, app audiences and Purview labels;
   - carry out the quarterly access reviews.

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Appendix A – Data dictionary

`03_analytics/build_data_dictionary.py` generates this appendix from the PostgreSQL catalogue, so it always matches the deployed warehouse. Every column has a `COMMENT ON` definition and a source or lineage note, and the same text becomes the field description in Power BI. The dictionary covers the 21 tables of the `dw` schema and the RLS mapping table in `sec`: 22 tables and 269 columns. It is also submitted as `06_report/data_dictionary.md` and `.xlsx`.

How to read it. *Key* marks the primary key (PK) and the foreign keys (FK → table.column). In the *Null* column, N means NOT NULL. *Source / lineage* names the source field and the data-quality rules (§3.3) that act on it.

dw.fact_admission

ACCUMULATING SNAPSHOT. Grain: one row per inpatient spell (admission to discharge), updated as milestones occur (admission, coding, discharge, 30-day follow-up). Additive: admission_count, los_days, flags.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	admission_key	bigint	N	PK	Surrogate key.	generated
2	admission_id	character varying(12)	N		Degenerate dimension: PAS spell number.	PAS admission_id (AD01)
3	admission_date_key	integer	N	FK → dim_date.date_key	FK to dim_date - admission date (active relationship).	PAS admission_datetime
4	discharge_date_key	integer	N	FK → dim_date.date_key	FK to dim_date - discharge date (role-playing; -2 = still in hospital).	PAS discharge_datetime (AD02, AD03)
5	care_unit_key	integer	N	FK → dim_care_unit.care_unit_key	FK to dim_care_unit version (admitting ward) valid on the admission date.	PAS ward_code (AD04)
6	patient_key	integer	N	FK → dim_patient.patient_key	FK to dim_patient version valid on the admission date.	PAS patient_id
7	diagnosis_key	integer	N	FK → dim_diagnosis.diagnosis_key	FK to dim_diagnosis - principal diagnosis (-1 until coded).	PAS principal_icd10 (AD05)
8	admission_profile_key	smallint	N	FK → dim_admission_profile.admission_profile_key	FK to the admission method x discharge status junk dimension.	PAS admission_method, discharge_status (AD07)
9	age_band_key	smallint	N	FK → dim_age_band.age_band_key	FK to dim_age_band (age at admission).	derived
10	age_years	smallint	Y		Age in years at admission (admission year - birth year).	derived
11	ed_encounter_id	character varying(12)	Y		Degenerate link to the ED attendance that led to the admission (emergency admissions).	PAS ed_encounter_id
12	admission_ts	timestamp without time zone	N		Admission timestamp (milestone 1).	PAS
13	discharge_ts	timestamp without time zone	Y		Discharge timestamp (milestone 2; NULL while in hospital).	PAS
14	admission_count	smallint	N		Always 1; additive count of spells.	ETL
15	los_days	smallint	Y		Length of stay in midnights (discharge date - admission date); NULL while in hospital. Additive (= patient days).	derived
16	is_discharged	smallint	N		1 once the discharge milestone is recorded.	derived
17	is_died	smallint	N		1 for an in-hospital death.	discharge_status = DIED
18	discharged_before_noon	smallint	Y		1 if discharged before 12:00 (flow KPI); NULL while in hospital.	derived
19	is_weekend_discharge	smallint	Y		1 if discharged on Saturday or Sunday; NULL while in hospital.	derived
20	is_readmission_30d	smallint	N		1 if this spell is an emergency admission within 30 days of the same patient's previous live discharge.	derived (window over patient pseudonym)
21	readmission_eligible	smallint	N		1 if this spell counts in the readmission-rate denominator: discharged alive (home or AMA), not maternity, and 30 days of follow-up available.	derived (GBHS KPI definition)
22	readmitted_within_30d	smallint	N		1 if an eligible spell was followed by an emergency readmission within 30 days (milestone 3).	derived
23	prior_admissions_12m	smallint	N		Number of the patient's admissions in the 365 days before this admission (risk feature).	derived
24	is_open_over_60d	smallint	N		1 if the spell is still open more than 60 days after admission at the snapshot date (AD06, probable missing discharge).	derived
25	source_last_updated	timestamp without time zone	N		last_updated of the PAS image applied (newer images overwrite older ones).	PAS last_updated
26	dw_batch_id	integer	N		Batch that inserted or last updated the row.	ETL
27	dw_updated_at	timestamp with time zone	N		Last update timestamp.	ETL

dw.fact_encounter

TRANSACTION FACT. Grain: one row per ED or clinic attendance (encounter). Additive: encounter_count and the 0/1 flags; durations are additive (sum) but reported as averages/medians.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	encounter_key	bigint	N	PK	Surrogate key.	generated
2	encounter_id	character varying(12)	N		Degenerate dimension: PAS attendance number.	PAS encounter_id (EN01)
3	arrival_date_key	integer	N	FK → dim_date.date_key	FK to dim_date (arrival date).	PAS arrival_datetime (EN02, EN12)
4	arrival_hour_key	smallint	N	FK → dim_hour.hour_key	FK to dim_hour (arrival hour).	PAS arrival_datetime
5	care_unit_key	integer	N	FK → dim_care_unit.care_unit_key	FK to dim_care_unit version valid on the arrival date (ED or clinic unit).	PAS facility_code + department (EN03)
6	patient_key	integer	N	FK → dim_patient.patient_key	FK to dim_patient version valid on the arrival date (-1 unidentified).	PAS patient_id (EN10, EN11)
7	diagnosis_key	integer	N	FK → dim_diagnosis.diagnosis_key	FK to dim_diagnosis (primary diagnosis, -1 not coded/invalid).	PAS primary_icd10 (EN07, EN08)
8	triage_key	smallint	N	FK → dim_triage.triage_key	FK to dim_triage (-1 not recorded; 9 clinic attendance).	PAS triage_category (EN04, EN09)
9	encounter_profile_key	smallint	N	FK → dim_encounter_profile.encounter_profile_key	FK to the department x arrival mode x disposition junk dimension.	PAS department, arrival_mode, disposition
10	age_band_key	smallint	N	FK → dim_age_band.age_band_key	FK to dim_age_band (age at arrival).	derived from birth year
11	age_years	smallint	Y		Age in years at arrival, approximated as arrival year - birth year (non-additive).	derived
12	encounter_count	smallint	N		Always 1; additive count of attendances.	ETL
13	wait_to_triage_min	numeric(8,1)	Y		Minutes from arrival to triage (ED only; NULL if invalid).	triage_datetime - arrival_datetime
14	door_to_doctor_min	numeric(8,1)	Y		Minutes from arrival to first assessment by a doctor/clinician (NULL if not seen or invalid, EN05/EN06).	seen_datetime - arrival_datetime
15	ed_los_min	numeric(9,1)	Y		Minutes from arrival to departure (NULL if > 72 h or negative).	departure_datetime - arrival_datetime
16	seen_within_target	smallint	Y		1 if seen within the triage compliance threshold, 0 if not, NULL if not triaged/seen.	derived with dim_triage.compliance_minutes
17	is_lwbs	smallint	N		1 if the patient left without being seen.	disposition = LWBS
18	is_admitted	smallint	N		1 if admitted to a ward from the ED.	disposition = ADM
19	is_died	smallint	N		1 if the patient died in the department (incl. dead on arrival).	disposition = DIED
20	dw_batch_id	integer	N		Loading batch (lineage).	ETL
21	dw_loaded_at	timestamp with time zone	N		Load timestamp.	ETL

dw.fact_idsr_report

COVERAGE (FACTLESS-STYLE) FACT. Grain: one row per facility per epi week that a report was EXPECTED, whether or not it arrived (reporting completeness and timeliness).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	epi_week_date_key	integer	N	PK	FK to dim_date = Monday of the epi week.	scaffold of expected weeks
2	facility_key	integer	N	PK	FK to dim_facility.	scaffold of active facilities
3	report_expected	smallint	N		Always 1 (denominator of completeness).	ETL
4	report_received	smallint	N		1 if any version of the weekly report was received by the extract cut-off.	stg.idsr_submission
5	report_on_time	smallint	N		1 if the first submission arrived by the due time (Tuesday 12:00 after the epi week).	derived (GBHS SOP)
6	hours_after_due	numeric(7,1)	Y		Hours between due time and first submission (negative = early; NULL if not received).	derived
7	first_submitted_at	timestamp without time zone	Y		Timestamp of the first submission.	IDSR submitted_at
8	due_at	timestamp without time zone	N		Due time of the report.	derived (Sunday + 2 days 12:00)
9	dw_batch_id	integer	N		Batch that rebuilt the row.	ETL
10	dw_loaded_at	timestamp with time zone	N		Load timestamp.	ETL

dw.fact_population

ANNUAL SNAPSHOT (external, NSA). Grain: one row per region per year (mid-year, keyed on 1 July). SEMI-ADDITIVE: additive across regions, never summed across years.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	year_date_key	integer	N	PK	FK to dim_date = 1 July of the year (mid-year population).	derived
2	region_key	integer	N	PK	FK to dim_region.	NSA region name mapped through ref.region_alias
3	population	integer	N		Population (semi-additive). 2023 = census count; other years = 2023 count x (1 + national intercensal growth 2011-2023)^(year-2023).	NSA 2023 PHC preliminary results (EX01)
4	is_census_count	boolean	N		True for the census year itself, false for derived estimates.	ETL
5	population_source	character varying(80)	N		Provenance text shown in report tooltips.	ETL
6	dw_batch_id	integer	N		Loading batch.	ETL

dw.fact_staff_roster

PERIODIC SNAPSHOT PER SHIFT. Grain: one row per care unit per date per shift. Hours are additive; head counts on duty are semi-additive (sum across units, average across shifts/days).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	roster_date_key	integer	N	PK	FK to dim_date (date the shift starts).	e-roster roster_date
2	care_unit_key	integer	N	PK	FK to dim_care_unit version valid on the roster date.	e-roster unit_code
3	shift_key	smallint	N	PK	FK to dim_shift.	e-roster shift (RS01)
4	planned_nurse_hours	numeric(7,1)	Y		Nursing hours on the approved roster (additive).	e-roster
5	rn_hours	numeric(7,1)	Y		Registered-nurse hours worked (additive).	e-roster (RS02)
6	en_hours	numeric(7,1)	Y		Enrolled-nurse hours worked (additive).	e-roster
7	nurse_hours	numeric(7,1)	Y		Total nursing hours worked = rn_hours + en_hours (additive; numerator of NHPPD).	derived
8	doctor_hours	numeric(7,1)	Y		Doctor hours worked in the unit (additive).	e-roster
9	overtime_hours	numeric(7,1)	Y		Overtime nursing hours (additive; NULL when inconsistent, RS04).	e-roster
10	agency_hours	numeric(7,1)	Y		Agency (temporary) nursing hours (additive).	e-roster
11	sick_leave_hours	numeric(7,1)	Y		Rostered hours lost to sick leave (additive).	e-roster
12	nurses_on_duty	smallint	Y		Nurses on duty during the shift (semi-additive).	e-roster
13	doctors_on_duty	smallint	Y		Doctors on duty during the shift (semi-additive).	e-roster
14	dw_batch_id	integer	N		Loading batch.	ETL
15	dw_loaded_at	timestamp with time zone	N		Load timestamp.	ETL

dw.fact_surveillance_weekly

PERIODIC SNAPSHOT. Grain: one row per facility per epi week per IDSR priority condition (zero reporting). Counts are fully additive.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	epi_week_date_key	integer	N	PK	FK to dim_date = Monday of the ISO epi week.	IDSR epi_week (SV01)
2	facility_key	integer	N	PK	FK to dim_facility (reporting facility).	IDSR facility_code
3	condition_key	integer	N	PK	FK to dim_condition.	IDSR condition (SV03)
4	cases_under5	integer	Y		Cases in children under 5 (additive).	IDSR return (SV06, SV07)
5	cases_5plus	integer	Y		Cases aged 5 and over (additive).	IDSR return (SV06, SV07)
6	cases_total	integer	Y		All cases = under 5 + 5 and over (additive).	derived
7	deaths_under5	integer	Y		Deaths in children under 5 (additive).	IDSR return (SV04)
8	deaths_5plus	integer	Y		Deaths aged 5 and over (additive).	IDSR return (SV04)
9	deaths_total	integer	Y		All deaths (additive).	derived
10	tests_done	integer	Y		Malaria tests performed (RDT/microscopy); NULL for other conditions (additive).	IDSR return (SV05)
11	tests_positive	integer	Y		Positive malaria tests (additive).	IDSR return (SV05)
12	report_version	smallint	Y		Version of the report applied (2 = corrected resubmission).	IDSR report_version (SV02)
13	dw_batch_id	integer	N		Loading batch.	ETL
14	dw_loaded_at	timestamp with time zone	N		Load timestamp.	ETL

dw.fact_ward_census_daily

PERIODIC SNAPSHOT. Grain: one row per inpatient ward per day (midnight bed state). Stock measures (beds, occupied) are SEMI-ADDITIVE: sum across wards, average or last value across days; summed over days they equal bed-days.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	census_date_key	integer	N	PK	FK to dim_date (census date, midnight).	ward return census_date
2	care_unit_key	integer	N	PK	FK to dim_care_unit version valid on the census date.	ward_code
3	beds_complement	smallint	N		Official beds of the ward version (semi-additive).	dim_care_unit.bed_complement
4	beds_closed	smallint	N		Beds closed (maintenance, staffing) (semi-additive).	ward return
5	beds_available	smallint	N		Beds available = complement - closed (semi-additive; sum over days = available bed-days).	derived
6	occupied_beds	smallint	N		Patients in the ward at midnight (semi-additive; sum over days = patient days).	ward return (WC02, WC04) or PAS imputation (WC01)
7	admissions	smallint	Y		Admissions to the ward during the day (additive flow).	ward return
8	discharges	smallint	Y		Discharges (incl. deaths and transfers out) during the day (additive flow).	ward return
9	deaths	smallint	Y		In-ward deaths during the day (additive flow).	ward return
10	pas_occupied_beds	smallint	Y		Midnight census derived from PAS spells, kept for reconciliation (semi-additive).	stg.pas_ward_census
11	census_source	character varying(20)	N		Ward return, Corrected return or Imputed from PAS.	ETL
12	dw_batch_id	integer	N		Loading batch.	ETL
13	dw_loaded_at	timestamp with time zone	N		Load timestamp.	ETL

dw.fact_weather_daily

PERIODIC SNAPSHOT (external, NASA POWER). Grain: one row per facility location per day. Rainfall is additive over days (not across locations); temperature and humidity are non-additive.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	weather_date_key	integer	N	PK	FK to dim_date.	NASA POWER date (YYYYMMDD)
2	facility_key	integer	N	PK	FK to dim_facility (weather point = facility coordinates).	file name
3	rainfall_mm	numeric(7,2)	Y		Corrected precipitation, mm/day (PRECTOTCORR).	NASA POWER (WE02)
4	tmax_c	numeric(5,2)	Y		Maximum temperature at 2 m, deg C (T2M_MAX).	NASA POWER (WE01)
5	tmin_c	numeric(5,2)	Y		Minimum temperature at 2 m, deg C (T2M_MIN).	NASA POWER (WE01)
6	rh_pct	numeric(5,2)	Y		Relative humidity at 2 m, % (RH2M).	NASA POWER (WE01)
7	is_imputed	boolean	N		True if any value of the day was imputed.	ETL
8	dw_batch_id	integer	N		Loading batch.	ETL

dw.dim_admission_profile

Junk dimension: admission method x discharge status (IN = not yet discharged; updated as the accumulating snapshot progresses).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	admission_profile_key	smallint	N	PK	Surrogate key.	generated (Cartesian product)
2	admission_method_code	character varying(11)	N		EMER-ED, EMER-DIRECT, ELECTIVE, MATERNITY, TRANSFER-IN or UNK.	PAS admission_method via ref.code_map (AD07)
3	admission_method_name	character varying(50)	Y		Admission method description.	ref.code_map
4	is_emergency	boolean	Y		True for unplanned admissions (EMER-ED, EMER-DIRECT, TRANSFER-IN).	derived
5	discharge_status_code	character varying(8)	N		HOME, TRANSFER, DIED, AMA, IN (still in hospital) or UNK.	PAS discharge_status
6	discharge_status_name	character varying(45)	Y		Discharge status description.	ref.code_map

dw.dim_age_band

Age bands (age at the time of care), conformed between attendances and admissions; small-cell friendly groupings.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	age_band_key	smallint	N	PK	Surrogate key = sort order (-1 = Unknown).	generated
2	age_band	character varying(8)	N		Age band label (0-4, 5-14, 15-24, 25-44, 45-64, 65+).	generated
3	age_from	smallint	Y		Lower bound in completed years.	generated
4	age_to	smallint	Y		Upper bound in completed years (NULL = open-ended).	generated
5	idsr_age_group	character varying(12)	Y		IDSR reporting group: Under 5 or 5 and over.	generated
6	life_stage	character varying(12)	Y		Child, Adult or Older adult.	generated

dw.dim_care_unit

Care units (wards, emergency departments, clinic outpatient units). SCD Type 2 on specialty and bed complement; snowflaked to dim_facility.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	care_unit_key	integer	N	PK	Surrogate key of the unit version (-1 = Unknown).	generated
2	unit_code	character varying(10)	N		Unit natural key, - (e.g. RUN-MED1).	master data, Care_Units sheet
3	unit_name	character varying(40)	Y		Unit name (Type 1).	master data
4	unit_type	character varying(25)	Y		Inpatient ward, Emergency department or Outpatient clinic (Type 1).	master data
5	facility_key	integer	N	FK → dim_facility.facility_key	FK to dim_facility (snowflake).	master data facility_code
6	specialty	character varying(25)	Y		Clinical specialty valid for this version (Type 2), e.g. Medical, Paediatric, Isolation (COVID-19).	master data + Unit_History replay
7	bed_complement	smallint	Y		Official (funded) beds valid for this version (Type 2); 0 for ED and clinics.	master data + Unit_History replay
8	scd_effective_from	date	N		First day the version is valid.	Unit_History effective_date
9	scd_effective_to	date	N		Last day the version is valid (9999-12-31 for the current version).	ETL
10	scd_is_current	boolean	N		True for the current version of the unit.	ETL
11	scd_version	smallint	N		Version number per unit (1 = first).	ETL
12	scd_row_hash	character(32)	N		MD5 of the Type 2 attributes, used to detect changes.	ETL
13	dw_batch_id	integer	Y		Batch that created or last expired the version.	ETL
14	dw_updated_at	timestamp with time zone	N		Last update timestamp.	ETL

dw.dim_condition

IDSR priority conditions (grain of the surveillance fact); snowflake parent of dim_diagnosis.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	condition_key	integer	N	PK	Surrogate key (-1 = Unknown, -2 = Not a notifiable condition).	generated
2	condition_code	character varying(5)	N		GBHS IDSR condition code (MAL, COV, MEA ...).	master data, IDSR_Conditions sheet
3	condition_name	character varying(40)	N		Condition name.	master data
4	idsr_category	character varying(50)	Y		IDSR category (epidemic-prone, targeted for elimination, other priority).	master data (IDSR technical guidelines)
5	notification	character varying(30)	Y		Notification timeline (immediate within 24 h or weekly).	master data
6	alert_threshold	character varying(120)	Y		Alert threshold used by GBHS surveillance officers.	master data (GBHS SOP)
7	icd10_codes	character varying(20)	Y		ICD-10 codes mapped to the condition.	master data
8	is_notifiable	boolean	N		False only for the -2 member.	ETL

dw.dim_date

Conformed calendar dimension, one row per day 2021-01-01..2027-12-31 plus -1 Unknown and -2 Not applicable; calendar, ISO epidemiological week, fiscal (Apr-Mar) and rainy-season hierarchies.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	date_key	integer	N	PK	Smart surrogate key YYYYMMDD (-1 = unknown, -2 = not applicable / not yet happened).	generated by etl.sp_load_dim_date
2	full_date	date	Y		Calendar date.	generated
3	day_of_month	smallint	Y		Day number within the month (1-31).	derived
4	day_of_year	smallint	Y		Day number within the calendar year (1-366).	derived
5	day_of_week_iso	smallint	Y		ISO weekday number, Monday = 1 ... Sunday = 7 (sort key for day_name).	derived
6	day_name	character varying(9)	Y		Weekday name (Monday..Sunday).	derived
7	day_short	character(3)	Y		Three-letter weekday abbreviation (sort by day_of_week_iso).	derived
8	is_weekend	boolean	Y		True for Saturday and Sunday.	derived
9	is_public_holiday	boolean	Y		True on Namibian public holidays (Public Holidays Act 26 of 1990, Sunday holidays observed on Monday).	etl.fn_namibian_holiday
10	holiday_name	character varying(40)	Y		Name of the Namibian public holiday, NULL otherwise.	etl.fn_namibian_holiday
11	is_working_day	boolean	Y		Clinic working day: Monday-Saturday and not a public holiday.	derived
12	epi_year	smallint	Y		Epidemiological (ISO-8601) year of the epi week; can differ from the calendar year around 1 January.	derived (IYYY)
13	epi_week	smallint	Y		Epidemiological week number 1-53 (ISO-8601, Monday-Sunday), as used for IDSR weekly reporting.	derived (IW)
14	epi_week_label	character(8)	Y		Epi week label YYYY-Www (e.g. 2026-W39).	derived
15	epi_week_key	integer	Y		Integer YYYYWW (sort key for epi_week_label).	derived
16	epi_week_start	date	Y		Monday of the epi week (key used by the weekly surveillance facts).	derived
17	month_num	smallint	Y		Calendar month number 1-12 (sort key for month_name).	derived
18	month_name	character varying(9)	Y		Calendar month name.	derived
19	month_short	character(3)	Y		Three-letter month abbreviation (sort by month_num).	derived
20	year_month	character(7)	Y		Year-month label YYYY-MM.	derived
21	year_month_key	integer	Y		Integer YYYYMM (sort key for year_month).	derived
22	month_start_date	date	Y		First day of the month.	derived
23	month_end_date	date	Y		Last day of the month.	derived
24	quarter_num	smallint	Y		Calendar quarter 1-4.	derived
25	year_quarter	character(7)	Y		Quarter label YYYY-Qn.	derived
26	calendar_year	smallint	Y		Calendar year (time-intelligence year for YTD and year-on-year measures).	derived
27	fiscal_year_label	character(9)	Y		Fiscal year label FYyyyy/yy; the GBHS financial year runs 1 April - 31 March (aligned with the Namibian public sector).	derived
28	fiscal_year_start	smallint	Y		Calendar year in which the fiscal year starts (sort key for fiscal_year_label).	derived
29	fiscal_quarter_label	character(13)	Y		Fiscal quarter label FYyyyy/yy-Qn.	derived
30	rainy_season_label	character(7)	Y		Rainy season (July-June) label yyyy/yy used for malaria seasonality analysis.	derived
31	is_malaria_season	boolean	Y		True January-May, the usual malaria transmission peak in northern Namibia.	derived (GBHS planning assumption)
32	is_after_cutoff	boolean	Y		True for dates after the latest extract cut-off (future dates used by forecasts).	derived per batch
33	dw_created_at	timestamp with time zone	N		Load timestamp of the row.	ETL

dw.dim_diagnosis

WHO ICD-10 codes used by GBHS with the chapter > block > category hierarchy flattened; snowflaked to dim_condition (IDSR).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	diagnosis_key	integer	N	PK	Surrogate key (-1 = Unknown / not coded).	generated
2	icd10_code	character varying(7)	N		ICD-10 code (e.g. B50.9).	master data, ICD10_Codes sheet (MD06)
3	icd10_category	character varying(3)	Y		Three-character ICD-10 category (e.g. B50).	derived
4	icd10_title	character varying(110)	Y		WHO ICD-10 title.	master data
5	block_range	character varying(7)	Y		ICD-10 block code range (e.g. B50-B64).	master data

#	Column	Data type	Null	Key	Business definition	Source / lineage
6	block_title	character varying(110)	Y		ICD-10 block title.	master data
7	chapter_code	character varying(5)	Y		ICD-10 chapter in Roman numerals (I-XXII).	master data
8	chapter_number	smallint	Y		Chapter number 1-22 (sort key for chapter labels).	derived
9	chapter_range	character varying(7)	Y		Chapter code range (e.g. A00-B99).	master data
10	chapter_title	character varying(120)	Y		ICD-10 chapter title.	master data
11	severity_class	character varying(10)	Y		GBHS severity class (minor, moderate, serious, critical) used for case-mix adjustment.	master data
12	default_specialty	character varying(20)	Y		Default admitting specialty.	master data
13	is_chronic	boolean	Y		True for chronic conditions (readmission risk factor).	master data
14	condition_key	integer	N	FK → dim_condition.condition_key	FK to dim_condition (snowflake); -2 when the code is not notifiable.	master data idsr_condition_code

dw.dim_encounter_profile

Junk dimension: all combinations of department x arrival mode x disposition (low-cardinality attendance flags).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	encounter_profile_key	smallint	N	PK	Surrogate key.	generated (Cartesian product)
2	department_code	character varying(3)	N		ED (emergency department) or OPD (clinic outpatient).	PAS department
3	department_name	character varying(30)	Y		Department name.	generated
4	arrival_mode_code	character varying(4)	N		WALK, AMB, REF or UNK.	PAS arrival_mode via ref.code_map (EN09)
5	arrival_mode_name	character varying(35)	Y		Arrival mode description.	ref.code_map
6	disposition_code	character varying(4)	N		DIS, ADM, TRF, LWBS, DIED, REF or UNK.	PAS disposition via ref.code_map (EN09)
7	disposition_name	character varying(35)	Y		Disposition description.	ref.code_map
8	disposition_group	character varying(30)	Y		Discharged, Admitted or transferred, Referred, Left / died, Unknown.	derived

dw.dim_facility

GBHS hospitals and clinics (10). Conformed across all facts; snowflaked to dim_region; Type 1 (corrections overwrite).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	facility_key	integer	N	PK	Surrogate key (-1 = Unknown).	generated
2	facility_code	character varying(10)	N		Facility natural key (GBH- hospital, GBC- clinic).	master data, Facilities sheet
3	facility_name	character varying(60)	N		Facility name (Type 1).	master data, Facilities sheet
4	facility_short_name	character varying(30)	Y		Town-based short name for compact visuals.	derived
5	facility_type	character varying(30)	Y		Referral hospital, District hospital or Clinic.	master data
6	region_key	integer	N	FK → dim_region.region_key	FK to dim_region (snowflake).	master data region name mapped through ref.region_alias
7	town	character varying(40)	Y		Town where the facility is located.	master data
8	latitude	numeric(9,5)	Y		Latitude (decimal degrees), used for the NASA POWER weather point.	master data
9	longitude	numeric(9,5)	Y		Longitude (decimal degrees).	master data
10	elevation_m	integer	Y		Elevation in metres.	master data
11	has_emergency_department	boolean	Y		True for hospitals with a 24-hour emergency department.	master data
12	catchment_share	numeric(6,4)	Y		Share of the region's population in the facility catchment (0-1), used to derive catchment population.	master data (GBHS planning assumption)
13	opened_date	date	Y		Date the facility opened.	master data
14	ownership	character varying(60)	Y		Operating company.	master data
15	dw_updated_at	timestamp with time zone	N		Last update timestamp.	ETL

dw.dim_hour

Time-of-day dimension at hour grain (24 rows) used for ED arrival patterns and shift alignment.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	hour_key	smallint	N	PK	Hour of day 0-23 (natural and surrogate key).	generated
2	hour_label	character(5)	Y		Hour label HH:00.	generated
3	hour_band	character varying(20)	Y		Part of day: Night 00-06, Morning 07-11, Afternoon 12-16, Evening 17-23.	generated
4	hour_band_sort	smallint	Y		Sort key for hour_band.	generated
5	shift_code	character(1)	Y		Nursing shift covering the hour: D = 07:00-19:00, N = 19:00-07:00.	generated (GBHS rostering policy)
6	shift_name	character varying(12)	Y		Shift name (Day / Night).	generated
7	is_after_hours	boolean	Y		True outside 08:00-16:59 (reduced specialist cover).	generated

dw.dim_patient

Pseudonymised patients (no direct identifiers). SCD Type 2 on residence region and payer class so that facts carry the values valid at the time of care.

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	patient_key	integer	N	PK	Surrogate key of the patient version (-1 = Unknown / unidentified patient).	generated
2	patient_pseudo_id	character varying(13)	N		Pseudonym assigned by the PAS (keyed hash of the medical record number); never shown in reports.	PAS patient index
3	patient_durable_key	integer	Y		Durable (supernatural) key: the surrogate key of the patient's first version, identical on every version - used to count distinct patients across SCD2 versions.	ETL
4	sex	character(1)	Y		F, M or U (unknown) (Type 1).	PAS patient index, mapped via ref.code_map (MD02)
5	birth_year	smallint	Y		Year of birth only (data minimisation; no date of birth) (Type 1).	PAS patient index (MD03)
6	residence_region_code	character varying(5)	Y		ISO code of the region of residence valid for the version (Type 2; flattened, no relationship, to avoid a second filter path to dim_region).	PAS patient index + change log (MD01)
7	residence_region_name	character varying(30)	Y		Region of residence name (Type 2).	dim_region lookup
8	payer_code	character varying(8)	Y		Payer class valid for the version (Type 2): PSEMAS, PMA, SSC-MVA, MOHSS, CASH.	PAS patient index + change log (MD05)
9	payer_name	character varying(60)	Y		Payer class name.	master data, Payer_Classes sheet
10	payer_group	character varying(15)	Y		Medical aid, Statutory, Government or Self-pay.	master data
11	registration_date	date	Y		Date the patient was first registered (Type 1).	PAS patient index
12	is_inferred	boolean	N		True for an inferred member created from a fact before the patient-index record arrived (EN10).	ETL
13	scd_effective_from	date	N		First day the version is valid.	patient change log
14	scd_effective_to	date	N		Last day the version is valid (9999-12-31 = current).	ETL
15	scd_is_current	boolean	N		True for the current version.	ETL
16	scd_version	smallint	N		Version number per patient.	ETL
17	scd_row_hash	character(32)	N		MD5 of the Type 2 attributes.	ETL
18	dw_batch_id	integer	Y		Batch that created or last changed the row.	ETL
19	dw_updated_at	timestamp with time zone	N		Last update timestamp.	ETL

dw.dim_region

Namibian administrative regions (14) - top of the geography snowflake and grain of the population fact (Type 1).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	region_key	integer	N	PK	Surrogate key (-1 = Unknown).	generated
2	region_code	character varying(5)	N		ISO 3166-2:NA region code (e.g. NA-KE).	master data, Regions sheet
3	region_name	character varying(30)	N		Official region name.	master data, Regions sheet
4	zone	character varying(20)	Y		GBHS management zone (North-East, North-Central, Central, West, South, East, North-West).	master data
5	malaria_stratum	character varying(20)	Y		Malaria planning stratum: Endemic - high, Endemic - low, Non-endemic (imported cases only).	master data (GBHS planning assumption based on MoHSS stratification)
6	dw_updated_at	timestamp with time zone	N		Last update timestamp.	ETL

dw.dim_shift

Nursing shifts (Day 07:00-19:00, Night 19:00-07:00).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	shift_key	smallint	N	PK	Surrogate key.	generated
2	shift_code	character(1)	N		D or N.	e-roster shift via ref.code_map (RS01)
3	shift_name	character varying(12)	Y		Shift name.	ref.code_map
4	start_time	time without time zone	Y		Shift start time.	GBHS rostering policy
5	end_time	time without time zone	Y		Shift end time.	GBHS rostering policy
6	shift_hours	numeric(4,1)	Y		Paid hours per shift.	GBHS rostering policy

dw.dim_triage

SATS-style five-colour emergency triage scale plus "Not triaged" (clinic) and "Not recorded" (-1).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	`triage_key`	smallint	N	PK	Surrogate key = priority (-1 = not recorded).	generated
2	`triage_code`	character varying(6)	N		Triage colour code.	master data, Triage_Scale sheet
3	`triage_label`	character varying(40)	Y		Triage label (colour and urgency).	master data
4	`priority`	smallint	Y		Clinical priority 1 (highest) - 9 (sort key).	master data
5	`target_minutes`	integer	Y		Target time from arrival to doctor in minutes (0 = immediate).	master data (SATS-style targets)
6	`compliance_minutes`	integer	Y		Threshold used for the within-target KPI: target, with “immediate” operationalised as 5 minutes.	derived (GBHS KPI definition)
7	`target_text`	character varying(30)	Y		Target in words.	master data
8	`colour_hex`	character(7)	Y		Display colour (used with text labels, never colour alone).	master data

sec.user_facility_access

Dynamic row-level security mapping: which facilities each Power BI user may see (read by the “Facility manager” role).

#	Column	Data type	Null	Key	Business definition	Source / lineage
1	`user_principal_name`	character varying(80)	N	PK	Microsoft Entra ID user principal name (matched with USERPRINCIPALNAME() in DAX).	Information Security Officer
2	`facility_code`	character varying(10)	N	PK	Facility the user may see.	Information Security Officer
3	`access_role`	character varying(30)	N		Business role of the user (documentation only).	Information Security Officer
4	`granted_by`	character varying(40)	N		Approver of the access (data owner).	access request
5	`granted_on`	date	N		Approval date (reviewed quarterly).	access request

Appendix B – DAX measure catalogue

All 108 measures of the semantic model are listed below, grouped by display folder. `04_powerbi/build_pbip.py` writes them from `04_powerbi/model_spec.py` into the TMDL model and into `04_powerbi/measures.dax`. The comment line above each measure is also its description in the Power BI field list. §3.4 explains the design choices. It maps the brief's DAX requirements (time intelligence, `CALCULATE` with filter modification) to these measures and lists the security roles.

Folder 0 Targets & Labels

```
-- KPI target: at least 80 % of triaged ED patients seen within their SATS target.
Target Seen Within Target % =
    IF ( NOT ISBLANK ( [Seen Within Target %] ), 0.8 )

-- KPI ceiling: no more than 3 % of ED patients leave without being seen.
Target LWBS Rate % =
    IF ( NOT ISBLANK ( [LWBS Rate %] ), 0.03 )

-- KPI ceiling: 85 % average bed occupancy (safe-occupancy threshold).
Target Bed Occupancy % =
    IF ( NOT ISBLANK ( [Bed Occupancy %] ), 0.85 )

-- KPI ceiling: 12 % emergency readmissions within 30 days.
Target Readmission Rate 30d % =
    IF ( NOT ISBLANK ( [Readmission Rate 30d %] ), 0.12 )

-- KPI target: 80 % completeness and 80 % timeliness of weekly IDSR reports (WHO AFRO IDSR).
Target IDSR Reporting % =
    IF ( NOT ISBLANK ( [IDSR Completeness %] ), 0.8 )

-- Text status with symbol (never colour alone) for the seen-within-target KPI.
ED Target Status =
    VAR _v = [Seen Within Target %]
    RETURN
        IF ( NOT ISBLANK ( _v ), IF ( _v >= 0.8, "▲ On target (80 %)", "▼ Below target (80 %)" ) )

-- Text status with symbol for bed occupancy against the 85 % ceiling.
Occupancy Status =
    VAR _v = [Bed Occupancy %]
    RETURN
        IF ( NOT ISBLANK ( _v ), IF ( _v <= 0.85, "● Within 85 % ceiling", "▲ Above 85 % ceiling" ) )

-- Freshness label for page headers (patient-flow data).
Data As Of =
    "Data to " & FORMAT ( [Last Data Date], "d mmm yyyy" )

-- Freshness label for the surveillance page (last epi week with IDSR returns).
Surveillance As Of =
    VAR _d = [Last Epi Week Start]
    RETURN
        "IDSR returns to epi week " & CALCULATE ( SELECTEDVALUE ( 'Date'[Epi Week Label] ), REMOVEFILTERS ( 'Date' ), 'Date'[Date] = _d )

-- Dynamic title of the drill-through page.
Facility Detail Title =
    SELECTEDVALUE ( 'Facility'[Facility], "All facilities" ) & " - facility detail"
```

Folder 1 Patient Flow

```
-- All ED and clinic attendances (one row per attendance).
Attendances =
    SUM ( 'Attendance'[Encounter Count] )

-- Emergency-department attendances (CALCULATE with filter modification on the department).
ED Attendances =
    CALCULATE ( [Attendances], KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" ) )

-- Outpatient-clinic attendances.
Clinic Attendances =
    CALCULATE ( [Attendances], KEEPFILTERS ( 'Encounter Profile'[Department Code] = "OPD" ) )

-- Share of triaged ED patients seen by a doctor within the SATS compliance time of their category (RED 5, ORANGE 10, YELLOW 60, GREEN 240 minutes).
Seen Within Target % =
    DIVIDE ( SUM ( 'Attendance'[Seen Within Target] ), COUNT ( 'Attendance'[Seen Within Target] ) )

-- Median minutes from ED arrival to first doctor assessment (patients seen).
Median Door-to-Doctor (min) =
    CALCULATE ( MEDIAN ( 'Attendance'[Door to Doctor (min)] ), KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" ) )

-- 90th percentile of ED door-to-doctor minutes (the long tail that drives complaints and LWBS).
P90 Door-to-Doctor (min) =
    CALCULATE ( PERCENTILE.INC ( 'Attendance'[Door to Doctor (min)], 0.9 ), KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" ) )

-- Median minutes from ED arrival to triage.
Median Wait to Triage (min) =
    CALCULATE ( MEDIAN ( 'Attendance'[Wait to Triage (min)] ), KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" ) )

-- Share of ED attendances that left without being seen.
LWBS Rate % =
    CALCULATE ( DIVIDE ( SUM ( 'Attendance'[Is LWBS] ), SUM ( 'Attendance'[Encounter Count] ) ), KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED"
) )

-- Share of ED attendances admitted to a ward (conversion rate).
ED Admission Rate % =
    CALCULATE ( DIVIDE ( SUM ( 'Attendance'[Is Admitted] ), SUM ( 'Attendance'[Encounter Count] ) ), KEEPFILTERS ( 'Encounter Profile'[Department Code] =
"ED" ) )

-- Share of ED attendances with a valid length of stay that left the department within 4 hours.
ED Stays Within 4 Hours % =
    VAR _within =
        CALCULATE (
            COUNTROWS ( 'Attendance' ),
            KEEPFILTERS ( NOT ISBLANK ( 'Attendance'[ED Length of Stay (min)] ) && 'Attendance'[ED Length of Stay (min)] <= 240 ),
            KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" )
```

```
        )
    VAR _known =
        CALCULATE ( COUNT ( 'Attendance'[ED Length of Stay (min)] ), KEEPFILTERS ( 'Encounter Profile'[Department Code] = "ED" ) )
    RETURN
        DIVIDE ( _within, _known )

-- Identified patients with an attendance (durable key, stable across SCD2 versions; unidentified attendances excluded).
Distinct Patients =
    CALCULATE ( DISTINCTCOUNT ( 'Attendance'[Patient Durable Key] ), KEEPFILTERS ( 'Attendance'[Patient Durable Key] > 0 ) )

-- Identified attendances per distinct patient (frequent-attender signal).
Attendances per Patient =
    DIVIDE ( CALCULATE ( [Attendances], KEEPFILTERS ( 'Attendance'[Patient Durable Key] > 0 ) ), [Distinct Patients] )
```

Folder 2 Inpatient Care

```
-- Inpatient spells by admission date.
Admissions =
    SUM ( 'Admission'[Admission Count] )

-- Share of admissions that were unplanned (via ED, direct or transfer in).
Emergency Admission % =
    DIVIDE ( CALCULATE ( [Admissions], KEEPFILTERS ( 'Admission Profile'[Is Emergency Admission] = TRUE () ) ), [Admissions] )

-- Identified patients admitted (durable key).
Distinct Inpatients =
    CALCULATE ( DISTINCTCOUNT ( 'Admission'[Patient Durable Key] ), KEEPFILTERS ( 'Admission'[Patient Durable Key] > 0 ) )

-- Spells discharged, by DISCHARGE date (activates the inactive role-playing relationship).
Discharges =
    CALCULATE ( SUM ( 'Admission'[Is Discharged] ), USERELATIONSHIP ( 'Admission'[Discharge Date Key], 'Date'[Date Key] ) )

-- Average length of stay (midnights) of spells discharged in the period.
ALOS (days) =
    CALCULATE ( AVERAGE ( 'Admission'[LOS (days)] ), USERELATIONSHIP ( 'Admission'[Discharge Date Key], 'Date'[Date Key] ), KEEPFILTERS ( 'Admission'[Is
Discharged] = 1 ) )

-- Deaths in hospital as a share of discharges (by discharge date).
In-Hospital Mortality % =
    CALCULATE ( DIVIDE ( SUM ( 'Admission'[Is Died] ), SUM ( 'Admission'[Is Discharged] ) ), USERELATIONSHIP ( 'Admission'[Discharge Date Key], 'Date'[Date
Key] ) )

-- Eligible spells (alive home/AMA, not maternity, 30 days follow-up) followed by an emergency readmission within 30 days, by discharge date of the index
spell.
Readmission Rate 30d % =
    CALCULATE ( DIVIDE ( SUM ( 'Admission'[Readmitted Within 30d] ), SUM ( 'Admission'[Readmission Eligible] ) ), USERELATIONSHIP ( 'Admission'[Discharge
Date Key], 'Date'[Date Key] ) )

-- Share of discharges completed before 12:00 (patient-flow KPI: frees beds for the afternoon ED peak).
Discharged Before Noon % =
    CALCULATE ( AVERAGE ( 'Admission'[Discharged Before Noon] ), USERELATIONSHIP ( 'Admission'[Discharge Date Key], 'Date'[Date Key] ), KEEPFILTERS (
'Admission'[Is Discharged] = 1 ) )

-- Share of discharges on Saturday or Sunday.
Weekend Discharge % =
    CALCULATE ( AVERAGE ( 'Admission'[Is Weekend Discharge] ), USERELATIONSHIP ( 'Admission'[Discharge Date Key], 'Date'[Date Key] ), KEEPFILTERS (
'Admission'[Is Discharged] = 1 ) )

-- Spells admitted in the period and not yet discharged at the extract.
Open Spells =
    CALCULATE ( [Admissions], KEEPFILTERS ( 'Admission'[Is Discharged] = 0 ) )
```

Folder 3 Resource Utilisation

```
-- Occupied bed-days: midnight census summed over days.
Patient Days =
    SUM ( 'Ward Census'[Occupied Beds] )

-- Open beds summed over days (complement minus closed beds).
Available Bed Days =
    SUM ( 'Ward Census'[Beds Available] )

-- Patient days / available bed days.
Bed Occupancy % =
    DIVIDE ( [Patient Days], [Available Bed Days] )

-- Average number of occupied beds per day (semi-additive: summed over wards, averaged over days).
Average Daily Census =
    AVERAGEX ( VALUES ( 'Date'[Date] ), CALCULATE ( SUM ( 'Ward Census'[Occupied Beds] ) ) )

-- Occupied beds on the last census date in the period, across the selected wards (semi-additive closing balance).
Occupied Beds (Latest Day) =
    VAR _d =
        CALCULATE (
            MAX ( 'Ward Census'[Census Date Key] ),
            REMOVEFILTERS ( 'Care Unit' ), REMOVEFILTERS ( 'Facility' ), REMOVEFILTERS ( 'Region' )
        )
    RETURN
        CALCULATE ( SUM ( 'Ward Census'[Occupied Beds] ), 'Ward Census'[Census Date Key] = _d )

-- Open beds on the last census date in the period.
Beds Available (Latest Day) =
    VAR _d =
        CALCULATE (
            MAX ( 'Ward Census'[Census Date Key] ),
            REMOVEFILTERS ( 'Care Unit' ), REMOVEFILTERS ( 'Facility' ), REMOVEFILTERS ( 'Region' )
        )
    RETURN
        CALCULATE ( SUM ( 'Ward Census'[Beds Available] ), 'Ward Census'[Census Date Key] = _d )

-- Occupancy on the last census date in the period.
Bed Occupancy % (Latest Day) =
    DIVIDE ( [Occupied Beds (Latest Day)], [Beds Available (Latest Day)] )

-- Nursing hours worked (registered + enrolled).
```

```
Nurse Hours =
    SUM ( 'Staff Roster'[Nurse Hours] )

-- Nursing hours per patient day on inpatient wards (drill-across: e-roster x ward census through Date and Care Unit).
NHPPD =
    DIVIDE ( CALCULATE ( [Nurse Hours], KEEPFILTERS ( 'Care Unit'[Unit Type] = "Inpatient ward" ) ), [Patient Days] )

-- Overtime nursing hours as a share of nursing hours worked.
Overtime % =
    DIVIDE ( SUM ( 'Staff Roster'[Overtime Hours] ), [Nurse Hours] )

-- Agency nursing hours as a share of nursing hours worked.
Agency % =
    DIVIDE ( SUM ( 'Staff Roster'[Agency Hours] ), [Nurse Hours] )

-- Nursing hours worked as a share of the approved roster.
Roster Fill % =
    DIVIDE ( [Nurse Hours], SUM ( 'Staff Roster'[Planned Nurse Hours] ) )

-- Rostered hours lost to sick leave.
Sick Leave % =
    DIVIDE ( SUM ( 'Staff Roster'[Sick Leave Hours] ), SUM ( 'Staff Roster'[Planned Nurse Hours] ) )

-- Doctor-shifts worked in the emergency departments.
ED Doctor Shifts =
    CALCULATE ( SUM ( 'Staff Roster'[Doctors on Duty] ), KEEPFILTERS ( 'Care Unit'[Unit Type] = "Emergency department" ) )

-- ED attendances per doctor-shift (drill-across: attendances x e-roster) - the strongest driver of waiting time.
ED Arrivals per Doctor Shift =
    DIVIDE ( [ED Attendances], [ED Doctor Shifts] )
```

Folder 4 Disease Surveillance

```
-- Cases reported in the weekly IDSR returns.
Notified Cases =
    SUM ( 'Surveillance'[Cases Total] )

-- Deaths reported in the weekly IDSR returns.
Notified Deaths =
    SUM ( 'Surveillance'[Deaths Total] )

-- Notified deaths / notified cases.
Case Fatality Rate % =
    DIVIDE ( [Notified Deaths], [Notified Cases] )

-- Share of notified cases in children under five.
Under-5 Share of Cases % =
    DIVIDE ( SUM ( 'Surveillance'[Cases Under 5] ), [Notified Cases] )

-- Confirmed malaria cases notified (IDSR condition MAL).
Malaria Cases =
    CALCULATE ( [Notified Cases], KEEPFILTERS ( 'Condition'[Condition Code] = "MAL" ) )

-- Malaria deaths notified.
Malaria Deaths =
    CALCULATE ( [Notified Deaths], KEEPFILTERS ( 'Condition'[Condition Code] = "MAL" ) )

-- Positive malaria tests (RDT/microscopy) / tests done.
Malaria Test Positivity % =
    DIVIDE ( SUM ( 'Surveillance'[Tests Positive] ), SUM ( 'Surveillance'[Tests Done] ) )

-- Person-time at risk in the period: catchment population of each epi year x days of that year in context / 365, within the surveillance window.
Catchment Person-Years =
    VAR _first = [First Surveillance Date]
    VAR _last = [Last Epi Week Start] + 6
    RETURN
        SUMX (
            VALUES ( 'Date'[Epi Year] ),
            VAR _y = 'Date'[Epi Year]
            VAR _days =
                CALCULATE ( COUNTROWS ( 'Date' ), KEEPFILTERS ( 'Date'[Date] >= _first && 'Date'[Date] <= _last ) )
            VAR _pop =
                CALCULATE (
                    SUM ( 'Catchment Population'[Catchment Population] ),
                    REMOVEFILTERS ( 'Date' ),
                    'Date'[Calendar Year] = _y
                )
            RETURN
                DIVIDE ( _pop * _days, 365 )
        )

-- Average catchment population over the period (NSA regional population x facility catchment share; time-weighted across epi years).
Catchment Population =
    VAR _first = [First Surveillance Date]
    VAR _last = [Last Epi Week Start] + 6
    VAR _days =
        CALCULATE ( COUNTROWS ( 'Date' ), KEEPFILTERS ( 'Date'[Date] >= _first && 'Date'[Date] <= _last ) )
    RETURN
        DIVIDE ( [Catchment Person-Years] * 365, _days )

-- Malaria cases per 100 000 catchment population in the selected period (period incidence; equals the OLAP pivot by epi year).
Malaria Incidence per 100k =
    DIVIDE ( [Malaria Cases] * 100000, [Catchment Population] )

-- Annualised malaria incidence: cases per 1 000 person-years at risk (comparable across periods of different length).
Malaria Incidence per 1,000 PY =
    DIVIDE ( [Malaria Cases] * 1000, [Catchment Person-Years] )

-- NSA 2023 census count for the selected regions (all regions = 3 022 401).
Census Population 2023 =
    CALCULATE ( SUM ( 'Regional Population'[Population] ), REMOVEFILTERS ( 'Date' ), KEEPFILTERS ( 'Regional Population'[Is Census Count] = TRUE () ) )

-- Small-cell suppression for external sharing: counts of 1-4 are blanked to prevent re-identification.
Notified Cases (Suppressed) =
    VAR _c = [Notified Cases]
    RETURN
```

```
        IF ( _c >= 1 && _c <= 4, BLANK (), _c )

-- Suppressed count as text: "<5" for 1-4 cases.
Notified Cases Label =
    VAR _c = [Notified Cases]
    RETURN
        IF ( NOT ISBLANK ( _c ), IF ( _c >= 1 && _c <= 4, "<5", FORMAT ( _c, "#,0" ) ) )

-- Weekly reports due (facility x epi week).
IDSR Reports Expected =
    SUM ( 'IDSR Report'[Report Expected] )

-- Reports received / reports expected.
IDSR Completeness % =
    DIVIDE ( SUM ( 'IDSR Report'[Report Received] ), [IDSR Reports Expected] )

-- Reports received by Tuesday 12:00 after the epi week / reports expected.
IDSR Timeliness % =
    DIVIDE ( SUM ( 'IDSR Report'[Report On Time] ), [IDSR Reports Expected] )

-- Rainfall total per location, averaged across the selected facility locations.
Rainfall (mm) =
    AVERAGEX ( VALUES ( 'Facility'[Facility Code] ), CALCULATE ( SUM ( 'Weather'[Rainfall (mm)] ) ) )

-- Rainfall in the 28 days ending 4 weeks before the period starts (the lag used by the SARIMAX malaria model).
Rainfall 4-8 Weeks Earlier (mm) =
    VAR _start = MIN ( 'Date'[Date] )
    RETURN
        IF (
            _start >= [First Surveillance Date] && _start <= [Last Epi Week Start] + 6,
            CALCULATE ( [Rainfall (mm)], DATESINPERIOD ( 'Date'[Date], _start - 29, -28, DAY ) )
        )

-- Mean daily maximum temperature.
Avg Max Temperature (C) =
    AVERAGE ( 'Weather'[Max Temp (C)] )
```

Folder 5 Time Intelligence

```
-- Latest attendance date in the model (ignores all filters except RLS).
Last Data Date =
    VAR _k =
        CALCULATE ( MAX ( 'Attendance'[Arrival Date Key] ), REMOVEFILTERS () )
    RETURN
        IF ( NOT ISBLANK ( _k ), DATE ( INT ( _k / 10000 ), MOD ( INT ( _k / 100 ), 100 ), MOD ( _k, 100 ) ) )

-- Monday of the first epi week with IDSR returns.
First Surveillance Date =
    VAR _k =
        CALCULATE ( MIN ( 'Surveillance'[Epi Week Date Key] ), REMOVEFILTERS () )
    RETURN
        IF ( NOT ISBLANK ( _k ), DATE ( INT ( _k / 10000 ), MOD ( INT ( _k / 100 ), 100 ), MOD ( _k, 100 ) ) )

-- Monday of the latest epi week with IDSR returns.
Last Epi Week Start =
    VAR _k =
        CALCULATE ( MAX ( 'Surveillance'[Epi Week Date Key] ), REMOVEFILTERS () )
    RETURN
        IF ( NOT ISBLANK ( _k ), DATE ( INT ( _k / 10000 ), MOD ( INT ( _k / 100 ), 100 ), MOD ( _k, 100 ) ) )

-- Calendar year-to-date ED attendances, no future guard (helper).
ED Attendances YTD Raw =
    TOTALYTD ( [ED Attendances], 'Date'[Date] )

-- Year-to-date ED attendances; blank for periods after the last data date.
ED Attendances YTD =
    IF ( MIN ( 'Date'[Date] ) <= [Last Data Date], [ED Attendances YTD Raw] )

-- Same year-to-date window one year earlier, cut at the same day (like-for-like).
ED Attendances YTD PY =
    VAR _cutoffPY =
        EDATE ( [Last Data Date], -12 )
    RETURN
        CALCULATE (
            [ED Attendances YTD Raw],
            SAMEPERIODLASTYEAR ( 'Date'[Date] ),
            KEEPFILTERS ( 'Date'[Date] <= _cutoffPY )
        )

-- Year-to-date growth versus the like-for-like prior-year window.
ED Attendances YTD YoY % =
    DIVIDE ( [ED Attendances YTD] - [ED Attendances YTD PY], [ED Attendances YTD PY] )

-- Seen-within-target share in the same period last year.
Seen Within Target % PY =
    IF ( MIN ( 'Date'[Date] ) <= [Last Data Date], CALCULATE ( [Seen Within Target %], SAMEPERIODLASTYEAR ( 'Date'[Date] ) ) )

-- Change versus the same period last year, in percentage points.
Seen Within Target Change (pp) =
    VAR _cur = [Seen Within Target %]
    VAR _py = [Seen Within Target % PY]
    RETURN
        IF ( NOT ISBLANK ( _cur ) && NOT ISBLANK ( _py ), ( _cur - _py ) * 100 )

-- Occupancy over the trailing 12 months ending at the context end (capped at the last data date).
Bed Occupancy % Rolling 12M =
    VAR _last = [Last Data Date]
    VAR _end =
        MIN ( MAX ( 'Date'[Date] ), _last )
    RETURN
        IF (
            MIN ( 'Date'[Date] ) <= _last,
            CALCULATE ( [Bed Occupancy %], DATESINPERIOD ( 'Date'[Date], _end, -12, MONTH ) )
        )

-- Malaria cases from epi week 1 of the epi year to the end of the context (capped at the last reported week).
```

```
Malaria Cases Epi-YTD =
    VAR _end =
        MIN ( MAX ( 'Date'[Date] ), [Last Epi Week Start] + 6 )
    VAR _y =
        CALCULATE ( MAX ( 'Date'[Epi Year] ), REMOVEFILTERS ( 'Date' ), 'Date'[Date] = _end )
    RETURN
        IF (
            MIN ( 'Date'[Date] ) <= _end,
            CALCULATE ( [Malaria Cases], REMOVEFILTERS ( 'Date' ), 'Date'[Epi Year] = _y, 'Date'[Date] <= _end )
        )

-- Same number of epi weeks one epi year earlier (like-for-like).
Malaria Cases Epi-YTD PY =
    VAR _end =
        MIN ( MAX ( 'Date'[Date] ), [Last Epi Week Start] + 6 )
    VAR _y =
        CALCULATE ( MAX ( 'Date'[Epi Year] ), REMOVEFILTERS ( 'Date' ), 'Date'[Date] = _end )
    VAR _endPY = _end - 364
    RETURN
        IF (
            MIN ( 'Date'[Date] ) <= _end,
            CALCULATE ( [Malaria Cases], REMOVEFILTERS ( 'Date' ), 'Date'[Epi Year] = _y - 1, 'Date'[Date] <= _endPY )
        )

-- Epi-year-to-date growth versus the like-for-like prior-year window.
Malaria Cases Epi-YTD YoY % =
    DIVIDE ( [Malaria Cases Epi-YTD] - [Malaria Cases Epi-YTD PY], [Malaria Cases Epi-YTD PY] )

-- Malaria cases 52 weeks (364 days) earlier: the same epi week of the previous year.
Malaria Cases Same Week Last Year =
    IF (
        MIN ( 'Date'[Date] ) <= [Last Epi Week Start] + 6,
        CALCULATE ( [Malaria Cases], DATEADD ( 'Date'[Date], -364, DAY ) )
    )

-- Average weekly malaria cases over the four epi weeks ending at the context end.
Malaria Cases 4-Week Average =
    VAR _end =
        MIN ( MAX ( 'Date'[Date] ), [Last Epi Week Start] + 6 )
    RETURN
        DIVIDE ( CALCULATE ( [Malaria Cases], DATESINPERIOD ( 'Date'[Date], _end, -28, DAY ) ), 4 )
```

Folder 6 Advanced Analytics

```
-- 13-week forward forecast of North-East malaria cases (Holt-Winters ETS on log cases).
Forecast Cases =
    CALCULATE ( SUM ( 'Malaria Forecast'[Forecast] ), KEEPFILTERS ( 'Malaria Forecast'[Series] = "Forecast" ) )

-- Lower bound of the 80 % prediction interval.
Forecast Lower 80 =
    CALCULATE ( SUM ( 'Malaria Forecast'[Lower 80] ), KEEPFILTERS ( 'Malaria Forecast'[Series] = "Forecast" ) )

-- Upper bound of the 80 % prediction interval.
Forecast Upper 80 =
    CALCULATE ( SUM ( 'Malaria Forecast'[Upper 80] ), KEEPFILTERS ( 'Malaria Forecast'[Series] = "Forecast" ) )

-- Weekly North-East malaria cases the model was trained and tested on (PAS-coded attendances).
Model Actual Cases =
    CALCULATE ( SUM ( 'Malaria Forecast'[Actual] ), KEEPFILTERS ( 'Malaria Forecast'[Series] = "Actual" ) )

-- One-week-ahead forecasts made at each rolling origin of the back-test.
Backtest Forecast =
    CALCULATE ( SUM ( 'Malaria Forecast'[Forecast] ), KEEPFILTERS ( 'Malaria Forecast'[Series] = "Back-test (1 week ahead)" ) )

-- Weighted absolute percentage error of the one-week-ahead back-test forecasts.
Backtest WAPE % (1 Week Ahead) =
    VAR _rows =
        FILTER ( 'Malaria Forecast', 'Malaria Forecast'[Series] = "Back-test (1 week ahead)" )
    RETURN
        DIVIDE (
            SUMX ( _rows, ABS ( 'Malaria Forecast'[Actual] - 'Malaria Forecast'[Forecast] ) ),
            SUMX ( _rows, ABS ( 'Malaria Forecast'[Actual] ) )
        )

-- Observed 30-day readmission rate in the test set (spell-weighted).
Readmission Observed % =
    DIVIDE ( SUMX ( 'Readmission Deciles', 'Readmission Deciles'[Observed Rate] * 'Readmission Deciles'[Spells] ), SUM ( 'Readmission Deciles'[Spells] ) )

-- Mean predicted readmission probability (spell-weighted) - compare with observed for calibration.
Readmission Predicted % =
    DIVIDE (
        SUMX ( 'Readmission Deciles', 'Readmission Deciles'[Mean Predicted] * 'Readmission Deciles'[Spells] ),
        SUM ( 'Readmission Deciles'[Spells] )
    )

-- Share of all readmissions that fall in the two highest risk deciles (who to call after discharge).
Top-20% Risk Capture % =
    VAR _all = SUMX ( 'Readmission Deciles', 'Readmission Deciles'[Observed Rate] * 'Readmission Deciles'[Spells] )
    VAR _top =
        CALCULATE ( SUMX ( 'Readmission Deciles', 'Readmission Deciles'[Observed Rate] * 'Readmission Deciles'[Spells] ), KEEPFILTERS ( 'Readmission
Deciles'[Risk Decile] >= 9 ) )
    RETURN
        DIVIDE ( _top, _all )

-- Odds ratio of a readmission driver (logistic regression, reference coding).
Odds Ratio =
    AVERAGE ( 'Readmission Drivers'[Odds Ratio Value] )

-- % change in door-to-doctor time associated with the driver (OLS on log waiting time).
Wait Effect % =
    DIVIDE ( AVERAGE ( 'ED Wait Drivers'[Effect %] ), 100 )

-- Wait effect of the drivers management can act on (case-mix controls such as triage category excluded).
Wait Effect % (Operational) =
    CALCULATE ( [Wait Effect %], KEEPFILTERS ( LEFT ( 'ED Wait Drivers'[Driver], 6 ) <> "Triage" ) )
```

```
-- Lower 95 % confidence limit of the effect.
Wait Effect Lower 95% =
    DIVIDE ( AVERAGE ( 'ED Wait Drivers'[CI Low %] ), 100 )

-- Upper 95 % confidence limit of the effect.
Wait Effect Upper 95% =
    DIVIDE ( AVERAGE ( 'ED Wait Drivers'[CI High %] ), 100 )

-- Facility-condition-weeks flagged by the selected detection method(s).
Outbreak Alerts =
    COUNTROWS ( 'Outbreak Alerts' )

-- Share of alerts that fall in weeks of a known (injected) outbreak - a positive-predictive-value proxy.
Alerts in Known Outbreak Weeks % =
    DIVIDE ( CALCULATE ( [Outbreak Alerts], KEEPFILTERS ( 'Outbreak Alerts'[Is Known Outbreak] = TRUE () ) ), [Outbreak Alerts] )
```

Folder 7 Data Quality & Security

```
-- Data-quality issues logged by the ETL rules (all batches).
DQ Issues Logged =
    SUM ( 'DQ Summary'[Issues] )

-- Rules that logged at least one issue.
DQ Rules Triggered =
    CALCULATE ( COUNTROWS ( 'DQ Summary' ), KEEPFILTERS ( 'DQ Summary'[Issues] > 0 ) )

-- Share of reconciliation checks of the latest batch with status PASS.
Reconciliation Pass % =
    DIVIDE ( CALCULATE ( COUNTROWS ( 'Reconciliation' ), KEEPFILTERS ( 'Reconciliation'[Status] = "PASS" ) ), COUNTROWS ( 'Reconciliation' ) )

-- Checks with status WARN (explained differences).
Reconciliation Warnings =
    CALCULATE ( COUNTROWS ( 'Reconciliation' ), KEEPFILTERS ( 'Reconciliation'[Status] = "WARN" ) )

-- Ward-days whose census was imputed from PAS spells (missing ward return, rule WC01).
Imputed Census Days % =
    DIVIDE ( CALCULATE ( COUNTROWS ( 'Ward Census' ), KEEPFILTERS ( 'Ward Census'[Census Source] = "Imputed from PAS" ) ), COUNTROWS ( 'Ward Census' ) )

-- Attendances without a valid ICD-10 code (rules EN07-EN08).
Uncoded Diagnosis % =
    DIVIDE ( CALCULATE ( [Attendances], KEEPFILTERS ( 'Diagnosis'[ICD-10 Code] = "UNK" ) ), [Attendances] )

-- Attendances without a patient identifier (rule EN11).
Unidentified Attendances =
    CALCULATE ( [Attendances], KEEPFILTERS ( 'Attendance'[Patient Durable Key] < 0 ) )

-- Spells still open more than 60 days after admission (probable missing discharge, rule AD06).
Open Spells Over 60 Days =
    SUM ( 'Admission'[Is Open Over 60d] )

-- Facilities visible to the current user (RLS test).
Visible Facilities =
    CALCULATE ( DISTINCTCOUNT ( 'Facility'[Facility Code] ), KEEPFILTERS ( 'Facility'[Facility Key] > 0 ) )

-- User principal name evaluated by row-level security.
Current User =
    USERPRINCIPALNAME ()
```

Appendix C – Script inventory

The table is generated at build time from the files in the package, so the line counts are current. One command, `python run_all.py`, runs every script in dependency order (README and `04_powerbi/BUILD_GUIDE.md`).

Table 74. Submitted scripts, their size and purpose

File	Lines	Purpose
`run_all.py`	86	Rebuilds everything in dependency order: generate, ELT, analytics, evidence, mock-ups, PBIP, backup
`01_data/fetch_public_data.py`	82	Downloads the real NASA POWER files; checks the NSA regional totals
`01_data/generate_synthetic_data.py`	2,102	Seeded generator of all synthetic sources, the day-2 delta, the manifest and the DQ ground truth
`02_sql/00_create_database.sql`	24	Database and least-privilege login roles
`02_sql/01_ddl_schemas_audit.sql`	142	Schemas, audit tables (batches, steps, DQ issues, reconciliation), 44-rule DQ catalogue
`02_sql/02_ddl_staging_ref.sql`	161	Landing (text) and clean (typed) staging tables; reference and alias tables
`02_sql/03_ddl_warehouse.sql`	671	13 dimensions and 8 facts with constraints and a documented comment on every column; RLS mapping
`02_sql/04_etl_functions.sql`	210	Parsing and standardisation functions, holidays, SCD look-ups, batch and step logging
`02_sql/05_etl_transform.sql`	722	Transform procedures per source: DQ rules, issue logging, merge into clean staging
`02_sql/06_etl_load_dimensions.sql`	382	Dimension loads: date, static, Type 1, SCD2 (care unit, patient), inferred members
`02_sql/07_etl_load_facts.sql`	269	Fact loads (as-was key look-ups, accumulating snapshot upsert) and the sp_run_load orchestrator
`02_sql/08_reconciliation.sql`	154	22 reconciliation checks (row counts, control totals, cross-source, integrity, completeness)
`02_sql/09_mart_views.sql`	127	Power BI presentation views (minimised columns), DQ and reconciliation views, RLS mapping view
`02_sql/10_security_grants.sql`	13	Least-privilege grants per schema
`02_sql/11_olap_queries.sql`	119	OLAP operations: roll-up, drill-down, slice, dice, pivot, drill-across, semi-additive
`02_sql/12_evidence_queries.sql`	209	Named evidence queries captured into the report (load evidence, SCD2, KPIs)
`02_sql/13_optimisation.sql`	38	B-tree and BRIN indexes, aggregate materialised view
`02_sql/run_pipeline.py`	271	Extract and land files, call the load and reconciliation per batch (full + day-2 delta)
`03_analytics/build_data_dictionary.py`	98	Data dictionary from the catalogue; column descriptions for the semantic model
`03_analytics/capture_sql_evidence.py`	74	Runs the named SQL queries and writes the Markdown evidence tables
`03_analytics/common.py`	104	Shared helpers: database access, chart style, output paths
`03_analytics/data_profiling.py`	96	Profiling of the raw landed sources before cleansing
`03_analytics/ed_wait_drivers.py`	160	Technique 4: drivers of ED waiting time (OLS with HC3, gradient boosting, permutation importance)
`03_analytics/forecasting.py`	241	Technique 1: weekly malaria forecasting with rolling-origin back-test
`03_analytics/outbreak_detection.py`	211	Technique 3: outbreak detection (EARS C2, CUSUM, historical limits) against injected outbreaks
`03_analytics/performance_benchmark.py`	242	Baseline vs optimised query latency, model-size proxy, incremental refresh
`03_analytics/readmission_model.py`	198	Technique 2: 30-day readmission risk (logistic regression, gradient boosting), calibration, fairness
`03_analytics/validate_dq_rules.py`	62	Scores the DQ rules against the generator's ground truth (precision, recall)
`04_powerbi/build_pbip.py`	741	Generates and validates the PBIP (TMDL via TOM, PBIR via JSON schemas), measures.dax, theme
`04_powerbi/expected_values.py`	179	SQL mirror of the DAX: expected card values and the RLS test matrix
`04_powerbi/export_fallbacks.py`	57	CSV export of the mart views and the single-file TMDL script (fallbacks)
`04_powerbi/model_spec.py`	735	Declarative semantic model: tables, relationships, 108 measures, roles, parameters
`06_report/build_mockups.py`	533	Design mock-ups of the report pages rendered from the warehouse
`06_report/build_report.py`	345	Builds this PDF from Markdown sources, evidence tables, code excerpts and Mermaid diagrams
`06_report/tools/render.js`	65	Headless Chromium: Mermaid to PNG/SVG and HTML to PDF
`07_presentation/build_presentation.js`	343	Builds the PowerPoint deck with native charts and speaker notes
`07_presentation/deck_data.py`	172	Extracts the deck's chart data from the warehouse

Appendix D – Diagram sources

Every diagram in this report is written as Mermaid code inside the report source. `build_report.py` extracts each one into `06_report/diagrams/<name>.mmd` and renders it with headless Chromium, so a diagram can be edited as text and rebuilt. It is never a hand-drawn picture that drifts from the model.

Table 75. Mermaid diagram sources

Source file	Section	What it shows
`bpmn.mmd`	§3.1	Patient-flow and reporting process in four lanes (ED, ward, surveillance, BI), with the data capture points S1–S9
`architecture.mmd`	§3.2	Solution architecture: sources, the PostgreSQL schemas (landing, staging, `dw`, `mart`, `audit`, `sec`), analytics and Power BI
`erd_flow.mmd`	§3.2	ERD 1: `fact_encounter` with its eight dimensions
`erd_spell.mmd`	§3.2	ERD 2: `fact_admission` (accumulating snapshot) with the role-playing date
`erd_resource.mmd`	§3.2	ERD 3: `fact_ward_census_daily` and `fact_staff_roster`, and the Care Unit → Facility → Region snowflake
`erd_surveillance.mmd`	§3.2	ERD 4: `fact_surveillance_weekly` and the factless `fact_idsr_report`, with the Diagnosis → Condition snowflake
`erd_external.mmd`	§3.2	ERD 5: the external facts `fact_population` (NSA) and `fact_weather_daily` (NASA POWER)
`pipeline.mmd`	§3.3	Steps of one ELT batch, from extract to reconciliation
`lineage.mmd`	§3.8	Lineage of the KPI *Seen within triage target*, from the PAS fields to the executive card

The ERDs show keys and the main attributes only. Appendix A is the complete column-level specification.

Appendix E – Synthetic data design

GBHS is a fictitious organisation, and no patient-level health data may be used for a student project without ethics approval (§3.1, §3.8). The internal sources are therefore synthetic. The generator does more than fill tables: it **plants known behaviour and known defects**. This lets us test the warehouse, the data-quality rules and the models against a truth that real data never provides. This appendix documents that truth. All values come from `01_data/generate_synthetic_data.py` and from the manifest it writes (`01_data/raw/generator_manifest.json`).

E.1 Scope and reproducibility

Table 76. Generator settings

Setting	Value
Random seed	20261028 (re-runs are byte-identical: gzip is written with a fixed timestamp)
Software recorded in the manifest	Python 3.13.17, NumPy 2.5.3, pandas 3.0.5
Period of the main extract	1 January 2022 – 30 September 2026. December 2021 is simulated as a warm-up month, so that wards are already occupied on 1 January 2022
Day-2 incremental extract	1 October 2026: 320 attendances, 165 spell rows (78 new admissions and 87 discharges of open spells), 57 new patients, 25 payer changes, 2 ward bed changes and 2 late IDSR reports
Network	10 facilities: 5 hospitals with an ED and 21 wards, and 5 clinics (§3.1). There are 31 care units
Simulated population	149,999 people. 107,172 of them attended by the cut-off, so they appear in the patient index
What is real	The NSA 2023 census totals for the 14 regions. The NASA POWER files have the real layout but stand-in values until `fetch_public_data.py` is run (assumption A2)
Privacy by design	A patient record holds only a random pseudonym, birth year, sex, residence region and payer class. There are no names, ID numbers, dates of birth, addresses or clinical notes

E.2 Behavioural models planted in the data

Table 77. How the generator produces each process (the planted truth)

Process	Model in the generator
ED and clinic arrivals	Poisson counts per facility and day. The base rate is the mean daily arrivals: ED 20 (Katima Mulilo) to 55 (Windhoek); clinics 18–40. It is multiplied by a day-of-week factor (ED: Monday 1.12 to Sunday 0.90; clinics closed on Sundays and public holidays), a yearly trend and noise. Diseases have their own seasonal streams (rainy, winter, summer). Arrivals are spread over the day by hourly profiles: the ED is busy from 09:00 to 19:00, clinics from 07:00 to 15:00
Triage	Category drawn from the diagnosis severity. For example, a minor condition is 0.2 % RED, 3 % ORANGE, 22 % YELLOW and 74.8 % GREEN, while a critical one is 40 % RED and 45 % ORANGE
Admission and death	Admission probability by severity (3 %, 14 %, 48 %, 80 %), multiplied by a triage factor (RED 1.25 to GREEN 0.35). Death probability by severity: 0.1 % to 15 %
ED door-to-doctor wait	Triage median (RED 2, ORANGE 8, YELLOW 42, GREEN 125 minutes) × exp(1.4 × (arrivals per hour ÷ (doctors on duty × 1.6) − 0.8)) × (1 + 1.6 × max(0, hospital occupancy − 0.90)) × 1.10 at night × lognormal(0, 0.55). RED waits do not depend on crowding
Left without being seen	Only patients who would have gone home can leave. The probability rises with the wait: 1 − exp(−(wait ÷ scale)^2.2), with a scale of 560 minutes for GREEN, 900 for YELLOW and 760 in clinics
30-day readmission	Logit = −3.55 + 0.95 (age 65+) + 0.40 (age 45–64) + 0.85 (chronic condition) + 0.04 × length of stay (days, up to 30) + 0.45 (weekend or Friday discharge) + 1.10 (left against advice) + 0.35 (self-pay) + 0.60 (frailty) − 1.50 (maternity). Frailty is latent and never written to any file
Malaria	Daily cases = facility base × season amplitude × (rainfall 21–56 days earlier ÷ January–May mean)^1.3. Imported cases at the non-endemic facilities follow the north-east curve with a 7-day lag. The rainy-season factors make 2023/24 dry (0.62) and 2024/25 (1.22) and 2025/26 (1.32) wet. The amplitudes follow the national pattern: low in 2022–2024, an upsurge in 2025 and a larger one in 2026
COVID-19	Four waves (December 2021, June 2022, December 2022, July 2023), shared between facilities. Windhoek takes 30 %
Outbreaks	12 outbreaks of hepatitis E, measles, acute watery diarrhoea, malaria, typhoid and meningitis, each with a facility, start week, duration and size (listed in §3.5)
IDSR reporting	Each facility has its own probability of a missing report (1–7 %) and of a late one (7–25 %). Both fall by 40 % from 2025, an improvement planted on purpose. A report is due on Tuesday at 12:00 after the epi week
Staffing	Planned nurses per shift follow the beds (1 per 5 beds by day, 1 per 7 at night) or the ED volume. ED doctors per shift = max(1, round(base daily arrivals ÷ 20 by day, ÷ 30 at night)), less random absences. Vacancies (5–15 % by facility) and sickness reduce the hours worked, and overtime rises with demand pressure. Windhoek uses 3 % agency hours. **Intervention:** from 5 January 2026, Rundu, Katima Mulilo and Nkurenkuru receive 25 % agency nurse hours, and the Rundu and Katima EDs get one extra day-shift doctor until 30 June 2026
Slowly changing attributes	7 care-unit changes up to the cut-off, plus 2 in the delta. Two of them make WHK-MED2 a COVID-19 isolation ward from 3 January to 14 March 2022; the other five change bed complements. The simulated population has 16,228 residence-region and payer-class changes; 11,661 of them fall before the cut-off and concern patients in the extract

E.3 Reconciliation of the generator truth with the warehouse

Table 78. Generator truth against the loaded warehouse (after the day-2 load)

Item	Generator truth	Warehouse	Explanation
ED and clinic attendances	560,243 + 320 (delta)	560,557	The 6 rows outside the extract window are rejected by EN12. The 563,209 file rows also contained 2,966 duplicates (EN01)
Inpatient spells	135,551 + 78 (delta)	135,629	✔
Patients (natural keys)	107,172 + 57 (delta)	107,229	✔
Patient SCD2 changes	11,661 + 25 (delta)	11,686 later versions	✔
Care-unit SCD2 changes	7 + 2 (delta)	9 later versions (31 units, 40 versions)	✔
Late-arriving patients	40	40 inferred members, all completed by the delta	✔
Spells never discharged in the PAS	25	25 flagged by AD06	✔ (the source of the R07 WARN, §3.3)
IDSR reports expected / never received	2,470 / 51	2,470 / 51	✔
IDSR reports late	269	264	The generator times a corrected report by its final version. The warehouse judges timeliness on the first version received, as IDSR does. 5 corrected reports had an on-time original
Records with injected defects (record-level truth)	134,992	134,964 flagged by their own rule, 0 false positives	The other 28 were caught by another rule (§3.3)

E.4 Planted effects and what the analysis recovered

The analysis was run without access to these parameters. The table compares the planted effects with the estimates in §3.5.

Table 79. Planted effects against the estimates of §3.5

Effect	Planted	Recovered	Comment
ED load on waiting time	+7.6 % per extra arrival per doctor in a 12-hour shift (1.4 ÷ 19.2 on the log scale), within the unclipped range	+6.3 % (95 % CI 6.2–6.4 %), OLS	Slightly attenuated: the crowding factor is capped at 0.45–4.5, and the roster's doctor counts carry their own noise
Night shift	+10 %	+7.2 %	Same direction, slightly smaller
Access block	0 below 90 % occupancy, then +16 % per 10 points	+3.6 % per 10 points (linear)	A straight line averages an effect that only starts above 90 % occupancy
Malaria rainfall lead	Rainfall 21–56 days earlier, power 1.3	Rainfall 4–8 weeks earlier: SARIMAX coefficient 0.405, $p < 0.001$	The lead is recovered and used as the early-season signal
Readmission drivers	Odds ratios: left against advice 3.0, chronic condition 2.3, weekend or Friday discharge 1.6	2.09, 1.69 and 1.28 (logistic regression)	Same direction and order, but smaller. The observed flag also contains chance readmissions (new, unrelated emergencies), frailty is unobserved, and the feature flags weekends only
Extra ED doctor (Jan–Jun 2026)	Rundu and Katima Mulilo	Seen within target +9.6 and +12.7 points, against +1.6 at Oshakati	A difference-in-differences pattern (§3.5)
Outbreaks	12	CUSUM 11 of 12, EARS C2 9 of 12, median delay 1 week	OB08 is too small to separate from the noise of a busy clinic
IDSR improvement from 2025	Missing and late probabilities × 0.6	Timeliness 82.9–85.6 % in 2022–2024, against 92.9 % in 2025 and 92.1 % in 2026	✔

Limits of the synthetic data. The data reproduce the behaviours we chose to plant, and nothing else. Real data will contain effects nobody planted, such as referral patterns, coding habits and seasonal staff leave. They will also lack a ground truth. The magnitudes in this report are therefore illustrative. The **methods**, the warehouse and the data-quality rules are what transfer to a pilot on de-identified PAS extracts (§4, next steps).